Nadel · payment accounting for garden cooperatives

# board guide

How the chairman, the treasurer and the electrician work in the system: every screen and every button, with examples.

Version %cs

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# Getting started

This page is for a cooperative member, that is, a plot owner. It has everything you need to know first. If you work on the board, see Getting started for the board.

## How to sign in

Open the address of the system given by the board and press **Sign in**. On the sign-in page enter your e-mail and password. There is no self-registration: the board invites you, and a letter "**Set a password**" with a link arrives at your e-mail. The link is valid for 72 hours and works once; you set the password on the set-password page and then sign in with your e-mail. If the link has expired, ask the board to send a new one. If signing in fails or the system does not find you among the members, contact the board: perhaps a different e-mail is in your member card.

## Your home page

After signing in you land on "**My account**". It is your home screen and the first menu item; the logo in the header always returns here. Everything a member needs is on this screen: your plots, charges and payments, meters, invoices and notifications. The list of all the cooperative's plots is not shown to members. You open the card of your plot from the account by pressing its number, and the "**To my account**" link returns. In detail: My account.

## What you can do

- Submit a meter reading: in the **My meters** block press **Submit a reading**. The board checks the reading. Step by step: How to submit a meter reading.
- Get an invoice: in the **Invoices** block press **Download PDF**. How to read an invoice: How to read my invoice. The reconciliation act and other documents: Documents for a member.
- Connect Telegram so that invoices arrive in the messenger: the **Connect Telegram** button in the account. Step by step: How to connect Telegram.
- Install Nadel on your phone and get notifications on the device: Install on your phone and Notifications on your phone.

You see only your own data. The system does not show other people's plots, amounts or owners.

## The help

The "?" button in the header opens the help for the screen you are on. It appears at the side, next to the screen, which stays usable. If you move to another screen, the panel shows the help for that screen. The "**Close**" button or the Esc key closes the panel, and the "**Open in full**" button opens the page on its own. On a phone the help takes the whole screen, and the "**← Back**" button returns to the screen. A member has no "**Help**" item in the menu: the help opens with the "?" button.

## Related pages

- My account
- How to submit a meter reading
- How to read my invoice
- Documents for a member
- How to connect Telegram
- Install on your phone
- Notifications on your phone
- Set a password

# Set a password

This page opens from the link in the letter "**Set a password**" that the board sends when it invites you to the system (page address: `/set-password`). You do not need to be signed in.

## What you see

First the system checks the link: "**Checking the link...**" Then the page shows your login, which is your e-mail: "**Login (e-mail)**". Below are two fields: "**Password**" and "**Repeat the password**".

## Password rules

- "**At least 8 characters**";
- the password is typed twice and both values must match: when they are the same, the system says "**The passwords match**".

Press "**Save password**". When the password is saved, the system says "**Password set**", "**You can sign in now.**", and the "**Sign in**" button leads to the sign-in page. Sign in with your e-mail and the new password.

## If the link does not work

The link is valid for 72 hours and works once. If it has expired, was already used or the board sent a newer one, the system says: "**The link is invalid or has expired. Ask the board to send a new one.**" Ask the board to press "**Send again**" on your member card.

A working link always opens, as often as you like. After 5 wrong or expired links within 15 minutes from one computer, the page answers further such links with: "**Too many attempts. Wait a few minutes and try again.**" Your working link is not spoiled by this and opens as before.

## Related pages

- Getting started for a member
- Member card

# Getting started

## About the system

"**Nadel**" keeps the accounts of your cooperative: who the members are, which plots they own, what is charged and at which tariffs, how much electricity and water was used. Payments are entered at the cash desk and allocated to charges. Invoices, certificates and receipts are printed as PDF. Invoices also reach members by e-mail and in Telegram.

## Who sees what

Everyone signs in with their own account. What a person sees and can do is decided by their roles. A role is made of permissions: separate actions such as "enter payments". All permissions are described on the page Roles and permissions. A cooperative has three base roles.

| Role | What it can do |
|---|---|
| **Chairman** | Everything: members, roles, plots, owners, tariffs, meters, readings, charges, payments, the cash book, import, settings, the change log. The chairman's powers cannot be reduced. The cooperative always keeps at least one chairman. |
| **Treasurer** | Views members, plots, tariffs, meters, charges, reports and the change log. Enters payments, issues and sends invoices, keeps the cash book. Cannot reverse a payment, close a month or change the settings by default: the chairman does that. |
| **Member** | Only their own data: their plots, the tariffs, charges and payments of those plots, their meters, passing readings. |

The chairman can create custom roles. For example, the "Electrician" role has just two permissions: "**View meters**" and "**Enter readings**". A person with this role sees only "**Readings**" and "**Meters**" in the menu. A person can have several roles; their permissions are then added together.

## Signing in

Open the address of the system given by the board and press **Sign in**. Signing in happens on a separate sign-in page: enter your e-mail and password. After signing in you land on the home page. Which one depends on your rights:

- someone who can see plots (has the permission "**View plots**") lands on the plot list;
- an electrician with no plots of their own lands on readings or on the meter register;
- everyone else, including an ordinary member, lands on "**My account**". It is their home screen; there is a separate page about it, Getting started as a member.

If signing in fails, the system says: "**Sign-in failed. Please try again.**". Try again. If it still does not work, contact the chairman.

The system finds you among the members by e-mail. If the member card has no e-mail or a different one, you will see "**Account not linked**". Ask the chairman to correct the e-mail in your member card.

> **IMPORTANT**
> There is no self-registration. A person gets access when someone with the permission "**Invite to the system**" (by default the chairman) presses "**Invite to the system**" on the member card. The person receives a letter with a link, sets a password on the set-password page and signs in with their e-mail. The e-mail in the card is the login.

The **Sign out** button in the header ends your session. If the session expires on its own, the system asks you to sign in again.

## The header and the menu

On the left of the header is the name of your cooperative, on the right your name. The name leads to "**My account**", and the cooperative name with the logo leads to your home screen (the plot list, readings or "**My account**", depending on your rights). Next to it are the language switch, the theme switch, the "?" button and the **Sign out** button.

The menu on the left shows only the sections open to you. The chairman sees everything, the treasurer less. An ordinary member has neither "**Plots**" nor "**Help**" in the menu: their menu starts with "**My account**", and if they open the plot list by its address, it sends them to the account. A member opens their own plots from the account. What is not in the menu is not needed by you: the system does not show a section you have no access to. If you open such a page by a link, the system says: "**You don't have access to this page**".

On a phone the menu hides behind the **Menu** button.

## Language and theme

The language switch in the header changes between Ukrainian and English. The system remembers the choice in this browser. The interface always shows one language only: if English is selected, everything (names, hints, messages) is in English.

The theme has three options: **System**, **Light** and **Dark**. This choice is remembered too.

## The help

The "?" button in the header opens the help for the screen you are on. The help appears at the side, next to the screen: the screen stays usable, you can scroll and work with it while the panel is open. When you move to another screen, the panel shows the help for that screen, or closes if that screen has no help. On a phone the help takes the whole screen, and the "**← Back**" button returns to the screen. The "**Open in full**" button opens the same page on its own. The "**Close**" button or the Esc key closes the panel.

The whole help opens from the "**Help**" item in the menu (it is there only for those who hold at least one permission; a member opens the help with the "?" button). Every help page has "← Back to ...": it returns to the screen you opened it from. It has a table of contents by section and a "**Search the help**" field at the top: search by a word from the text or by a button name; one typo in the word does not matter. Each page explains one part of the system: what every button and field does, when to use it, what happens next and what to do if the system shows an error. There are examples with numbers and screenshots. On a help page you see a list of headings at the right: it helps you jump to the right place.

You see only the pages about the screens you can use. If you open an unavailable page by a link, the system says: "**This section is not available to you**".

The help can also be read without signing in, at the same address.

## Related pages

- Roles and permissions
- Plots
- Members
- Cooperative settings
- Import from Excel
- My account
- Enter a payment

# Import from Excel

## What the screen is for

So that you do not have to type the cooperative in by hand, plots, members, owners, meters, water connections and opening balances can be loaded from one Excel file. The system first checks the file and saves nothing. Saving happens only after you confirm.

## Who sees it

The screen is seen by whoever has the permission "**Run import**". By default this is the chairman. The menu item "**Import**" appears only with this permission.

## Every element

At the top of the screen is the caption: "**Bring plots, members, owners, water connections and opening balances over from an Excel file.**". Then come two panels, then the result of the check and the history.

### The "Template" panel

- The **Download template** button gives an Excel file in the language of the interface. It has the sheets "**Plots**", "**Members**", "**Ownership**", "**Meters**" and "**Opening balances**". Every sheet has a hint and an example. The system accepts sheet and column names in Ukrainian and in English, even mixed.
- Fill in the sheets. Write dates in Excel format or DD.MM.YYYY, areas in sotky or in m², shares as fractions such as 1/2.
- On the "**Members**" sheet you invent the "**Member key**" column yourself: the "**Ownership**" sheet uses it to find the member.
- The "**Opening balances**" sheet carries debts over from your previous records: plot, charge type, amount, date and a reason (the reason is optional). One row per plot, charge type and date. Amounts have at most two decimal places.
- The "**Plots**" sheet has a "**Water**" column. The word "yes" connects the plot to the cooperative's water supply. The name of a common consumer's meter connects it to that source. "No" or an empty cell changes nothing. Import never disconnects a plot from water.

> **IMPORTANT**
>
> You cannot change an owner by import. If a plot already has other owners, the system says so in the check, and the owner is changed on the plot card. Import does not overwrite an existing opening balance: reverse it first.

### The "Upload file" panel

- The **Choose file** button opens the file picker. Next to it you see the name of the chosen file, and until then it says "**No file chosen**". Only a .xlsx file up to 10 MB is accepted, with up to 2,000 plots and 3,000 members.
- The **Check file** button sends the file for checking. Nothing is saved until you confirm.

### The result of the check

The "**Check result**" panel shows the file name and the time until which the file can be imported: a check is valid for 30 minutes. The "**What the import will do**" table counts, for every kind of data, how many records fall into the columns "**New**", "**Updated**" and "**Unchanged**". The kinds of data are: **Plots**, **Members**, **Owners**, **Meters**, **Water connections** and **Opening balances**.

- If there are no problems, the system says: "**The file has no problems. You can import it.**". Then the **Import** button saves everything together: either the whole file or nothing. Importing the same file again duplicates nothing: plots are found by number, members by e-mail (or by surname, first name and phone).
- If the file has no new or changed records, the system says: "**Nothing new or changed in the file**".
- If there are problems, the import button is disabled and the system says: "**Fix the problems and check the file again to enable the import.**". The "**Problems in the file**" table shows every problem: **Sheet**, **Row**, **Column** and the text of the problem. Fix them in the file and check it again.
- The **Choose another file** button resets the check and returns to choosing a file.

### After the import

The "**Import complete**" panel with the "**What was saved**" table shows what was saved. The **Go to plots** and **Go to members** buttons lead to the plot list and the member list. The import is written to the change log. It cannot be undone in one step: correct the records on their cards.

### Import history

Below you see the "**Import history**" table: **Date**, **File**, **By**, **Status**, **What it did** and **Problems**. The status can be "**Checked, not imported**", "**Has errors**", "**Imported**", "**Refused: data changed**" or "**Never imported, expired**". If no files have been uploaded yet, it says "**No files have been uploaded yet.**".

**Error messages**

| What is shown | What to do |
|---|---|
| **The file is too large — the maximum is 10 MB** | Make the file smaller or split it into several. |
| **This is not an Excel file. Choose a file with the .xlsx extension.** | Save the table as an Excel workbook (.xlsx). |
| **The file has too many rows: up to 2,000 plots and 3,000 members.** | Upload the file in parts. |
| **The file has problems. Fix them and check the file again.** | Fix the problems from the table and check the file again. |
| **Nothing to import — everything already matches the cooperative data.** | Nothing to do: everything is already in the system. |
| **The cooperative data changed after the check. Upload the file again.** | Somebody changed the data between the check and the import. Press "**Upload the file again**" and check the file again. |
| **The file check has expired. Upload the file again.** | 30 minutes have passed. Upload the file again. |
| **This file has already been imported.** | This file is already imported; no need to repeat. |

The texts of the problems in the check table explain what is wrong in a row. The most common ones:

| What is shown | What to do |
|---|---|
| **The date is not valid. Use the format DD.MM.YYYY (for example 01.06.2015).** | Fix the date in the row. |
| **The phone number is not valid. Use the format +380501234567.** | Write the number in the format +380501234567. |
| **The chosen delivery channel needs a matching contact in this row.** | Add an e-mail, a Telegram name or a Viber phone depending on the chosen delivery channel, or choose another channel. |
| **There is no such plot in the file or among the active plots.** | Add the plot to the "**Plots**" sheet or fix its number. |
| **The shares of all owners of one plot must add up to exactly 1.** | Fix the shares: together they make one. |
| "The plot ... already has other owners" | Change the owner on the plot card, not by import. |
| **The plot has an ownership history, so this value cannot be changed.** | Change this value on the plot card. |
| "The plot already has an opening balance for this charge type" | Reverse the existing opening balance first. |

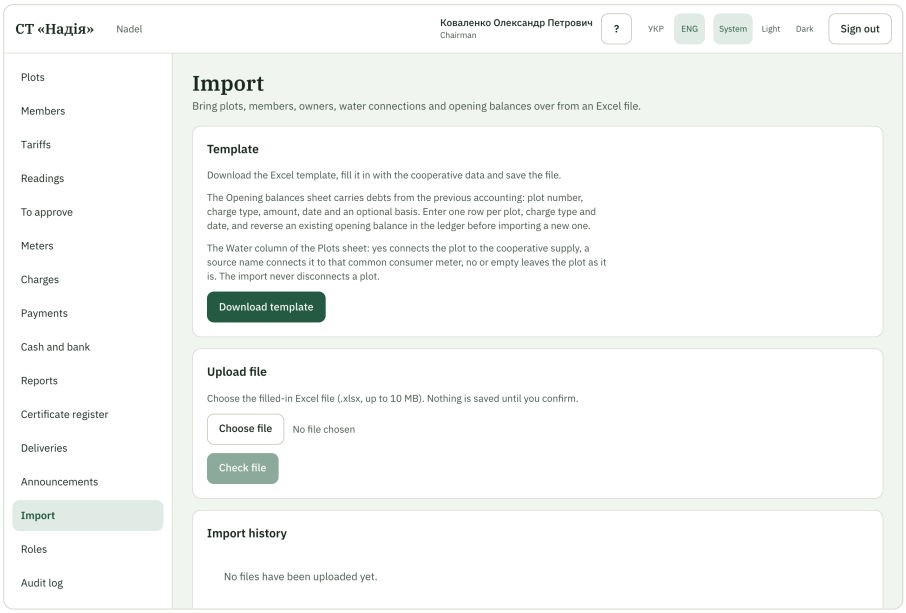
## Worked example

A cooperative moves from Excel to the system. The file must load plot 9 on Sadova street, its owner and the starting debt.

1. Press **Download template** and open the file.
2. On the "**Plots**" sheet write: number 9, street "вул. Садова", area in sotky 15, has a building "yes".
3. On the "**Members**" sheet write the owner: Lytvynenko Svitlana Mykhailivna, e-mail.
4. On the "**Ownership**" sheet write plot 9, the key of this member, share 1 and date 01.06.2025.
5. On the "**Opening balances**" sheet write plot 9, charge type "Membership fee", amount 600.00 and date 01.01.2026.
6. Save the file, press **Choose file** and **Check file**.
7. The system shows: plots new 1, members new 1, owners new 1, opening balances new 1, no problems. Press **Import**.
8. Plot 9 has appeared in the plot list, and its account shows a debt of 600.00.

If you now load the same file again, the check will show "unchanged" in all rows, and there will be nothing to import.

## Screenshots




The import screen: template, file choice and history

## Related pages

- Plots
- Members
- Plot card: changing the owner
- Change log
- Getting started

# Plots

## What the screen is for

This is the register of all plots of the cooperative. Here you search for a plot by number, street or cadastral number, filter by kind and status, and go to the plot card. The status colour shows whether the plot has a debt.

## Who sees it

The full register is seen by those who have the permission "**View plots**": the chairman and the treasurer, and any custom role with this permission. The "**Add plot**" button is seen by whoever also has "**Edit plots**".

A member without the permission "**View plots**" does not have this screen: there is no "**Plots**" item in their menu, and if they open the list address the system sends them to "**My account**". A member opens their own plots from the account, see My account.

> [!note] The setting "**Who sees plot colours**" (in the settings) applies to the plot map that will come later. For now members do not see the plot list, so they do not see other people's plots or colours at all.

## Every element

### The "**Add plot**" button

On the right in the header. Opens the new plot form.

### Search and filters

- The search field: "**Search by number, street or cadastral number**". The list updates as soon as you stop typing. Letter case does not matter.
- The "**Kind**" filter: "**All kinds**", "**Plot**" or "**Common land**". Common land (roads, shared territory) has no owners, tariffs or plot meters.
- The "**Status**" filter: "**All statuses**" or one of the statuses described below.
- The "**Show archived**" switch adds archived plots to the list. They are marked "**Archived**".

Filters are kept in the page address, so a link to a filtered list can be shared. If nothing is found, the system says: "**Nothing found**". When there are no plots at all, it says: "**No plots yet**".

### Table columns

| Column | What is in it |
|---|---|
| **Number** | The plot number. It is a link to the card. The order is natural: 2 before 10, and "12a" after 12. |
| **Street** | The street. A dash if not set. |
| **Area, sotkas** | The area in sotky. |
| **Cadastral number** | The cadastral number, if there is one. |
| **Kind** | "**Plot**" or "**Common land**". |
| **Status** | A colour and a word: the debt status, see below. |
| **Owners** | The owners. Two are shown, the rest as a number such as "+1". |
| **Archived** | The mark "**Archived**" for archived plots. |

At the bottom there is a page switch: a page holds twenty plots.

### Plot status (colour)

| Colour | Meaning |
|---|---|
| Green: "**No debt**" | No debt. The same when the plot has a prepayment. |
| Yellow: "**Has debt**" | There is a debt. |
| Red: "**Debt above threshold**" | The debt has reached the threshold: the amount or age the chairman sets in the settings. |
| Grey: "**No owner**" or "**Common land**" | A plot without an owner, or common land. |

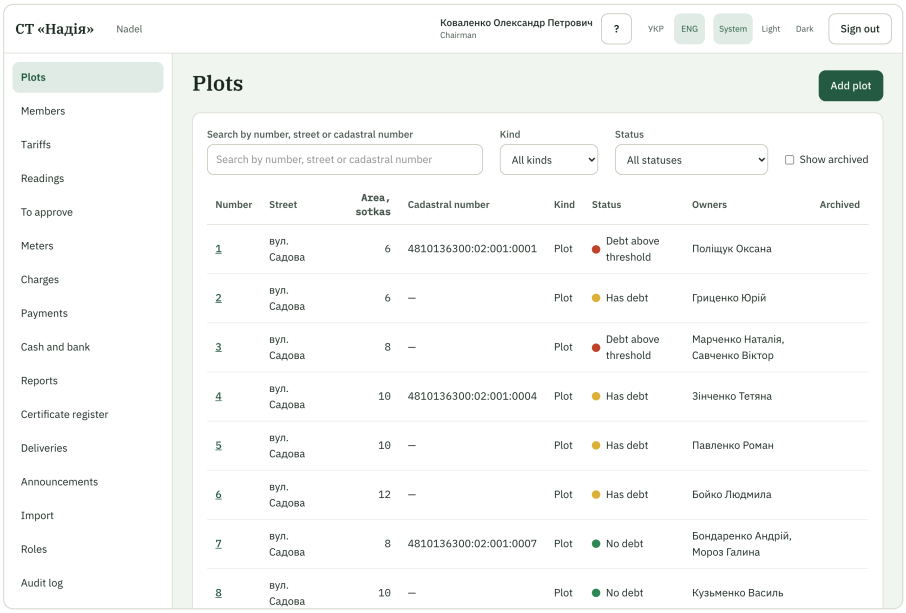
The colour is calculated from the debt after payments: a paid charge is not a debt. More about debt and colour: How the system allocates payments.

## Worked example

The treasurer wants to check the debtors on Sadova street.

1. Open "**Plots**" and type "Садова" in the search.
2. In the "**Status**" filter choose "**Has debt**". The list keeps the plots of Sadova street that have a debt.
3. To see those closer to the threshold, choose "**Debt above threshold**".
4. Click the number of the plot you need, for example 9 (15 sotky, owner Lytvynenko), to open the plot card and see what the debt consists of.

## Screenshots




The plot register with search and filters

## Related pages

- Plot card
- New plot and editing
- Import from Excel
- Cooperative settings
- My account

# Plot card

## What the screen is for

The card gathers everything about one plot: its data, the owners with the ownership history, the account with charges and payments, the tariffs for the chosen month, meters, the water connection, invoices, the certificate and the reconciliation act. Here you also edit and archive the plot, change its owner and enter a payment.

## Who sees it

- Whoever has the permission "**View plots**" sees the card of any plot with all details.
- A plot owner opens the card of their plot from "**My account**": see My account. The owner sees only the card of their own plot, without the "**Notes**" field and without buttons that change data.
- The buttons depend on permissions: "**Edit plots**" gives editing and archiving, "**Edit ownership**" together with "**View members**" gives changing owners, "**Enter payments**" gives the button to enter a payment.
- A member cannot open somebody else's plot: the system does not show its data.

## Every element

At the top are the link "**Back to plots**" and a title with the number, for example "Plot 9" (for common land "Common land R-1"). If the plot is archived, the mark "**Archived**" stands next to it.

### Buttons in the header

- **Enter payment**. Opens the payment form with the plot already chosen. It is not there for common land or archived plots.
- **Edit**. Opens the edit form.
- **Archive**. Hides the plot from the list but keeps all its history. When the plot is already archived, the system shows **Restore** instead. A plot with an open-ended water connection or an active meter cannot be archived or turned into common land: the system refuses and asks you first to end the connection or remove the meter. The refusal texts are: "**Remove the plot's meter first**" and "**The plot has a lasting water connection — disconnect it first**".
- **Delete**. Only for plots that are not archived. The system asks: "**Delete the plot?**", and warns: "**The plot is removed from the registry. This cannot be undone.**".

> **IMPORTANT**
> Only a plot created by mistake can be deleted: one with no owners in its history, no charges and no payments. Otherwise the system refuses: "**This plot has an owner — change the owner first**", "**A plot with ownership history cannot be deleted**", "**The plot has charges, so it cannot be deleted**" or "**The plot has payments, so it cannot be deleted**". For every other plot use the archive.

### Plot data

The table shows: **Number**, **Street**, **Kind**, **Status**, **Area** (in sotky and in m², for example "15 sotky (1,500 m²)"), **Cadastral number**, **Building** ("**Yes**" or "**No**") and **Notes**. The building is for reference only: it does not affect charges. Notes are seen only by those who have "**View plots**".

The status shows the debt colour: see Plots. The owner always sees the status of their own plot.

### The "Owners" panel

Owners are kept here. A plot can have several co-owners with shares, for example 1/2 and 1/3. Together the shares always make one. One member can own several plots, each with its own account. For common land it says: "**Common land has no owners.**"

The panel has two lists. "**Current owners**" shows the owners now: the name (a link to the member card if you can view members), the share, the date from which they own, and the basis. "**Ownership history**" shows previous owners with their periods. Archived members are marked "(**archived**)".

#### The "Change owner" button

Use it when the plot was sold, inherited, or the co-owners changed.

1. Enter the date in the "**Owners from**" field. It cannot be in the future.
2. For every owner choose a member: the "**Member**" field, search by surname ("**Search member by name**"), the **Change** button changes the choice. If the new owner is not among the members yet, press **Add member** right from the form.
3. In the "**Share (like 1/2)**" field write the share as a fraction, for example 1/2. The **Add owner** button adds another owner, and the remove button next to an owner's row removes them. Under the list you see the sum of shares: "**Total: 1**" when all is well. If the sum is not one, the system says why, and does not let you save.
4. Enter the basis in the "**Basis (sale contract, inheritance, court decision)**" field. The basis is optional.
5. Press **Save**. The system says: "**The owners were changed.**"

Previous owners stay in the history until the day before the transfer date. The debt belongs to the plot, not to the person: after a sale it stays on the plot. With no owner at all the plot turns grey from that date: the system explains "**With no owners the plot will have no owner from this date.**"

Possible messages: "**Enter the share like 1/2 (a whole number up to 1000 at the bottom).**", "**Shares must add up to 1**", "**The same member is listed twice.**", "**The date must be later than every current and past owner start and end, and not in the future.**", "**These dates overlap another ownership of the plot.**", "**This member is archived and cannot be an owner.**", "**Restore the plot before changing its owners.**".

#### The "**Correct**" button

It stands next to every record. Use it if a record has a mistake: a wrong share, date or person. The "**Correct an ownership record**" form has the fields "**Member**", "**Share (like 1/2)**", "**Owner from**", "**Owner until (empty for a current owner)**" (empty for the current owner), the basis and the required "**Reason**". If a share changes, the form also asks for the shares of the co-owners in the same period so that the total is one. After saving the system says: "**The record was corrected.**" The old values stay in the change log. Ownership records are never deleted.

#### The "**Account**" tab

This tab is open at first. Its panels:

- "**Account**": the charges and payments of the plot, the debt by type after payments, the prepayment.
  - Payments stand in the list together with the charges, by date; next to each charge are "**Charged**", "**Paid**" and "**Open**". Above the list is the debt after payments by type and in total, and beside the total debt "**Prepayment** X ₴" when the prepayment is above 0.00.
  - The filter "**Payments only**" leaves only payments and their reversals in the list; the debts above the list do not change. Here the opening balance is entered, charges are reversed and the prepayment is allocated (for those who have these permissions). Details: Charges and Payments.
- A link to the report "**Plot balances report for this street**" (for those who see reports).
- "**Tariffs**": the tariffs that apply to the plot in the chosen month and the expected amount. Details: Tariffs.
- "**Meters**": the meters of the plot. Details: Meters.
- "**Water**": whether the plot is connected to water. The **Connect** or **Disconnect** button opens a form with the water source and dates. The buttons are seen by whoever has "**Manage meters**".

#### The "**Invoices**", "**Reconciliation act**" and "**Certificate**" tabs

- "**Invoices**": the invoices of the plot by month with the number, the version and the amount "due". The **Download PDF** button downloads the PDF. The latest version of each month is shown; older ones lie under "**Earlier versions ({count})**". The **Invoice for today** button builds an invoice as of today for the current accounting month and downloads its PDF at once (an indicator is on the button while it is prepared); it is for those who may issue invoices. A new version appears only when something changed since the latest one (a payment, a charge, the owners, the requisites); otherwise the PDF of the latest version is downloaded and nothing new is created. The invoices of a month are built on the month page, sending is described on Delivery. What is in an invoice: How to read my invoice.
- "**Reconciliation act**": the "**Reconciliation act**" form. Choose a period ("**From**" and "**To**") and press **Download the reconciliation act**. You get a PDF in Ukrainian for the plot and the period: the requisites of the cooperative, the plot and its current owners, the period, the opening balance (a debt, an overpayment or nothing), all charges, corrections, reversals and payments by date (columns "**Charged**" and "**Paid**"), the turnover totals, the closing balance and the sentence "According to the cooperative's records the debt as of 31.10.2026 is ..." (or "the overpayment is ...", or "there is no debt"). At the bottom are two signature blocks: "From the cooperative" (with the treasurer's name, or an empty line if there is no treasurer) and "Member of the cooperative". The period cannot start before the first charge of the plot or end after today: otherwise "**Invalid period**". Without the requisites of the cooperative the act is not built, just like invoices. The act is not stored: it is built on request, and the same period always gives the same act. It can be sent like the other documents. For those who have "**View charges**".
- "**Certificate**": the "**Debt certificate**" form. Choose the "**As of**" date and press **Issue certificate**. The system says the certificate is issued and gives its number. The certificate can be downloaded with the **Download PDF** button. All issued certificates are collected in the certificate register. For those who have "**Issue certificates**".

If there is only one tab, the tab bar is not shown.

## Worked example

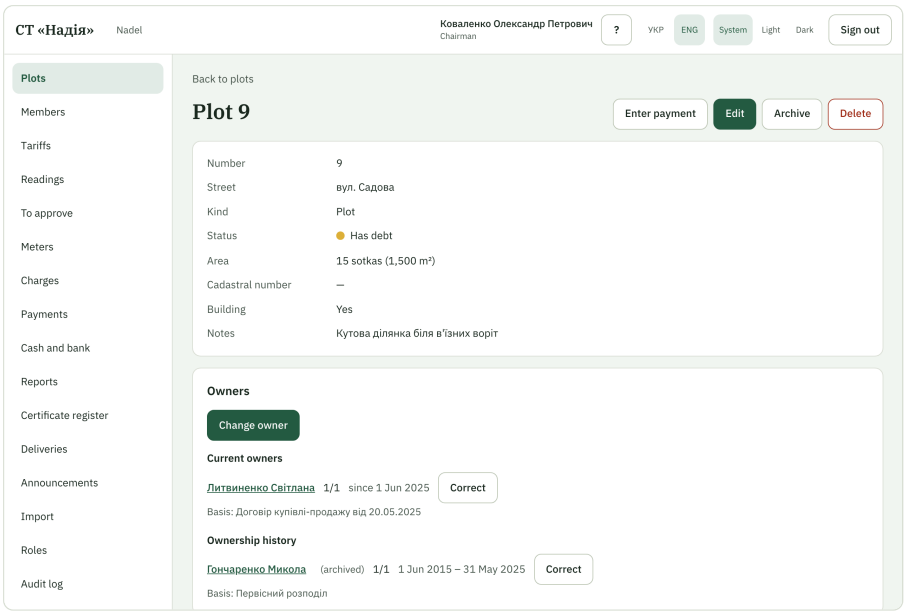
Plot 9, the corner plot by the gate: 15 sotky (1,500 m²), has a building, the note says "Кутова ділянка біля в'їзних воріт" (corner plot by the entrance gate). In 2025 it was sold: until 31.05.2025 it was owned by Honcharenko Mykola (now archived), and since 01.06.2025 by Lytvynenko Svitlana. Basis: "Договір купівлі-продажу від 20.05.2025" (sale contract of 20.05.2025).

How to record such a transfer:

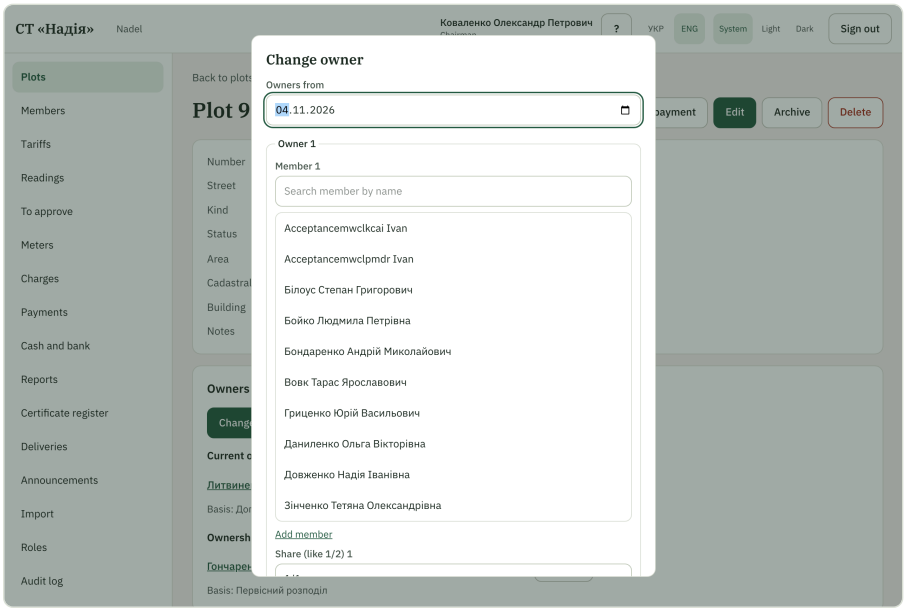
1. Open the card of plot 9 and press **Change owner**.
2. In the "**Owners from**" field enter 01.06.2025.
3. In the "**Member** 1" field find Lytvynenko, share 1.
4. Write the basis and press **Save**.
5. "**Current owners**" now shows Lytvynenko from 01.06.2025, and "**Ownership history**" shows Honcharenko from 01.06.2015 to 31.05.2025. The debts that were on the plot have not gone anywhere.

Plot 7 belongs to two people: Bondarenko 1/2 and Moroz 1/2. Their shares together make 1. If the form said 1/2 and 1/3, the system would show that the sum is not one and would not let you save.

## Screenshots




Card of plot 9: owners, history, tabs




The change owner form

## Related pages

- Plots
- Editing a plot
- Member card
- Enter a payment
- Charges
- Tariffs
- Meters
- Certificate register
- Delivery
- Change log

# New plot and editing

## What the screen is for

One form for two jobs: add a new plot and correct the data of an existing one. The form records only the reference data of the plot. Owners are changed on the plot card, and money and charges are not changed here.

## Who sees it

The form is seen by whoever has the permission "**Edit plots**". By default this is the chairman. The "**Add plot**" button in the list and the "**Edit**" button on the card appear only with this permission. The form title is "**New plot**" for a new plot and "Edit: Plot 9" for an existing one.

## Every element

Above the form is the link "**Back to plots**". The fields come in this order.

| Field | What to enter |
|---|---|
| **Number** | Required, up to 20 characters. Unique among the active plots of the cooperative: 12, 12a, R-1. Lists use natural order: 2 before 10, and "12a" after 12. |
| **Street** | Optional, up to 100 characters. |
| **Kind** | "**Plot**" or "**Common land**". Common land (roads, shared territory) has no owners, tariffs or plot meters. |
| **Area** and **Unit** | Required. Choose the unit "**sotkas**" or "**m²**": 1 sotka is 100 m². When you switch the unit, the number in the field is converted by itself. For a plot the area must be greater than zero, with at most two decimal places in m². |
| **Cadastral number** | Optional. Format: NNNNNNNNNN:NN:NNN:NNNN, for example 4810136300:02:001:0001. |
| **A building stands on the plot** | A tick for reference. It does not affect charges. |
| **Notes** | Optional, up to 2000 characters. Seen only by those who have "**View plots**". |

The **Save** button saves the plot and opens its card. The **Cancel** button goes back without changes.

When you edit an existing plot, a hint appears under the area: "**Past charges have not changed. If needed — Charges → Recalculation.**" Changing the area does not rewrite charges that were already made. To correct them, make a recalculation.

**Error messages**

| What is shown | What to do |
|---|---|
| **Number is required.** | Enter the number. |
| **Area is required.** | Enter the area. |
| **Enter a valid area: at most two decimals of m², greater than zero for a plot.** | Fix the area: a number above zero, at most two decimal places. |
| **Use the form 1234567890:12:345:6789.** | Enter the number in this format or leave the field empty. |
| **Another active plot already has this number.** | Choose another number. An archived plot does not occupy a number. |
| **Another plot of the cooperative already has this cadastral number.** | Check the number: such a plot already exists in the cooperative. |
| **This plot has had owners and cannot become common land** | The kind of such a plot cannot be changed. If this is a mistake, contact the chairman. |
| **The plot has a lasting water connection and cannot become common land — disconnect it first** | First disconnect the plot from water on its card. |

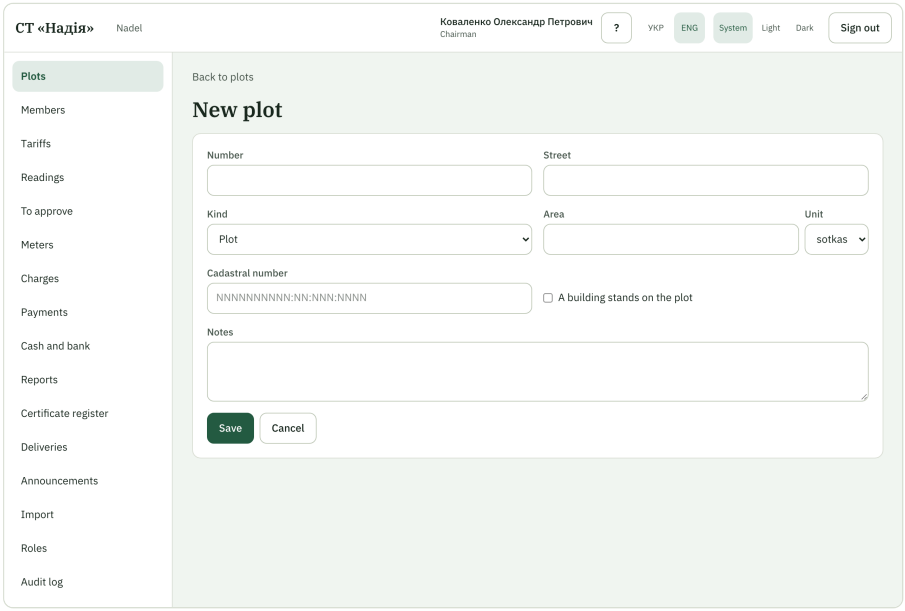
## Worked example

A new plot 31 has appeared on Vyshneva street.

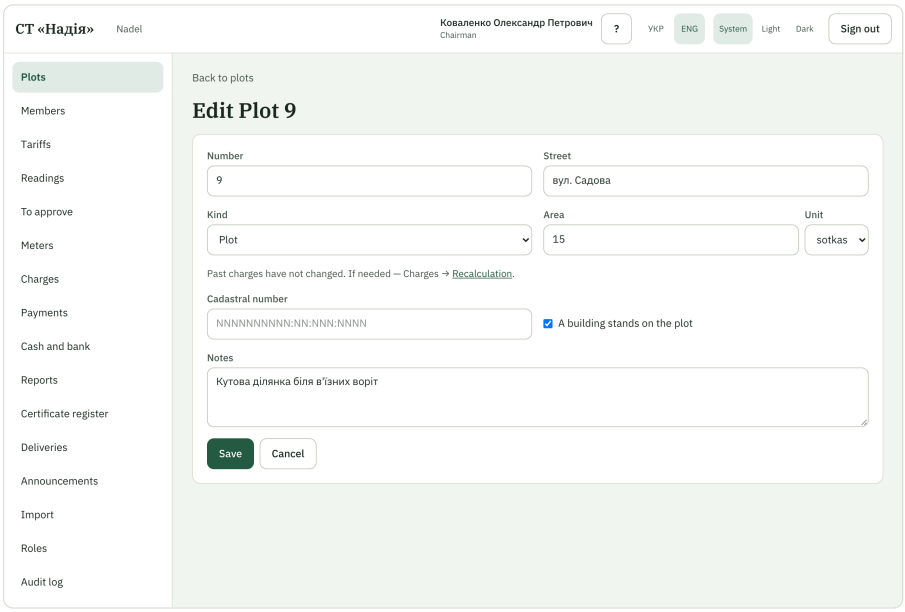
1. Open "**Plots**" and press **Add plot**.
2. In the "**Number**" field enter 31, in the "**Street**" field enter "вул. Вишнева".
3. Leave the kind "**Plot**". In the "**Area**" field enter 12, unit "**sotkas**". If you switch the unit to "**m²**", the field shows 1200.
4. Tick "**A building stands on the plot**" if there is a house on the plot.
5. Press **Save**. The card of the new plot opens.

The plot is grey for now: it has no owner. Add an owner on the card with the "**Change owner**" button, and the plot turns green as long as there is no debt. Existing plots work the same way: plot 9 has 15 sotky, that is 1,500 m².

## Screenshots




An empty new plot form




The edit form of plot 9

## Related pages

- Plots
- Plot card
- Import from Excel
- Tariffs

# Members

## What the screen is for

This is the list of all members of the cooperative. Here you look for a person, see their contacts, the way they receive invoices and their roles, and open the member card. A new member is added from here.

## Who sees it

The list is seen by those who have the permission "**View members**": the chairman and the treasurer. The "**Add member**" button is seen by whoever also has "**Edit members**". An ordinary member does not see this list: they find their own data in "**My account**".

## Every element

**The "Add member" button**

On the right in the header. Opens the new member form.

**Search**

The search field hints: "**Search by name, phone or email**". Under it: "**Name, full phone or last 4 digits, exact email**". The system understands these queries.

- By name: any part of the surname or first name, in any letter case.
- By phone: the full number written in any usual way (+380 50 123 45 67, 0501234567, 380501234567), or exactly its last 4 digits.
- By e-mail: the full address, in any letter case.

> **IMPORTANT**
> Phones, e-mails and addresses are stored encrypted, so a part of an e-mail (for example just what comes before @) or digits from the middle of a phone find nothing. If a member is not found, try the surname or the last 4 digits of the phone.

**The "Show archived" switch**

Archived members are not in the list by default. The switch adds them. They are marked "**Archived**".

**The table**

| Column | What is in it |
|---|---|
| **Name** | Surname, first name, patronymic. It is a link to the card. |
| **Phones** | All phones separated by commas. |
| **Email** | The e-mail. |
| **Preferred channel** | How the member receives invoices: by e-mail, in Telegram, in Viber or on paper. |
| **Roles** | The member's roles. |
| **Status** | "**Active**" or "**Archived**". |

If there are no members at all, it says: "**No members yet**". If the search found nothing: "**Nothing found**". A page holds twenty members, with a page switch at the bottom. The search and the archived switch are kept in the page address.

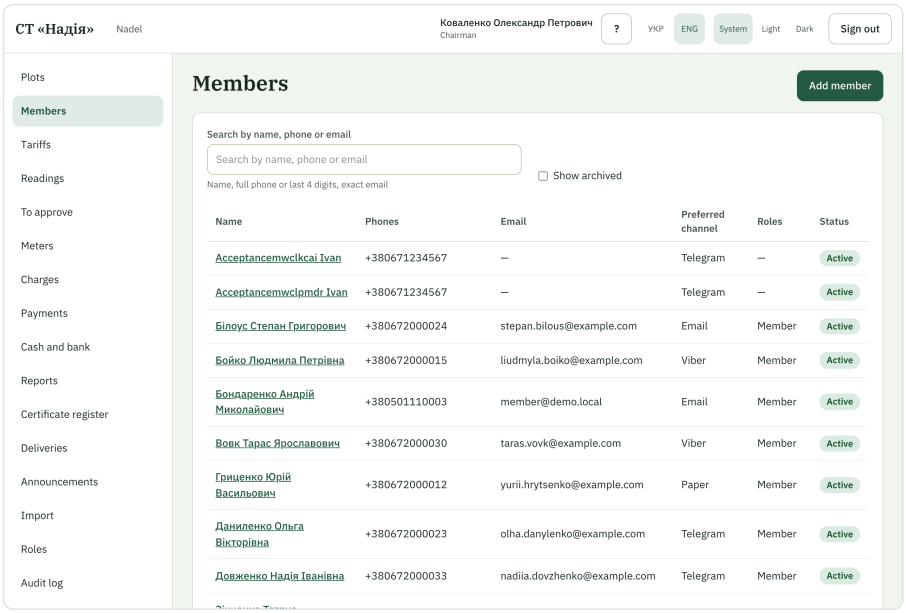
## Worked example

The treasurer got a call from a person who does not give a surname but dictates the number 0501234567.

1. Open "**Members**".
2. Type 0501234567 in the search, or just the last 4 digits.
3. Open the row that was found and see which plots the person owns.

Among the members there is the archived Honcharenko Mykola, the former owner of plot 9. To see him in the list, turn on "**Show archived**".

## Screenshots




The member list with search

## Related pages

- New member
- Member card
- Roles and permissions
- Plot card: owners
- Import from Excel

# New member and editing

## What the screen is for

Here a new cooperative member is recorded. The same form opens on the member card with the "**Edit**" button when the data must be corrected.

## Who sees it

The form is seen by whoever has the permission "**Edit members**": the chairman. The "**Notes**" field appears in it only for those who also have "**View members**". The title: "**New member**".

## Every element

Above the form is the link "**Back to members**". The fields come in this order.

| Field | What to enter |
|---|---|
| **Last name** | Required, up to 100 characters. |
| **First name** | Required, up to 100 characters. |
| **Middle name** | Optional. |
| **Email** | Optional. The system also finds the person by it when they sign in, so the address must match the one they sign in with. One address cannot belong to two active members. |
| **Phones** | Up to ten phones. The **Add phone** button adds another row, and the delete button next to a row removes it. A phone in the format 0XXXXXXXXX is turned by the system into +380XXXXXXXXX. |
| **Postal address** | Optional. The address for paper invoices. |
| **Preferred channel** | How invoices are delivered: "**Email**", "**Telegram**", "**Viber**" or "**Paper**". The chosen channel needs a contact. |
| **Telegram username** | Required if Telegram is chosen. |
| **Viber phone** | Required if Viber is chosen. |
| **Notes** | Up to 2000 characters. Service information: how best to get in touch, agreements, contacts of relatives. Seen only by those who have "**View members**". |

The **Save** button saves the member and opens their card. The **Cancel** button goes back.

> **IMPORTANT**
> The system has no separate fields for tax numbers, passport data or dates of birth: the cooperative does not need them for its accounts. Phones, e-mail, postal address, Telegram, Viber and notes are stored encrypted, so a copy of the database does not reveal them. The system decrypts them for those who may see them. Names stay readable.

### Error messages

| What is shown | What to do |
|---|---|
| **Last name is required.** | Enter the surname. |
| **First name is required.** | Enter the first name. |
| **Enter a phone number like +380671234567 or 0671234567.** | Fix the number in the indicated row. |
| **Email needs an email.** | Enter the e-mail or choose another channel. |
| **Telegram needs a Telegram username.** | Enter the nickname or choose another channel. |
| **Viber needs a Viber phone.** | Enter the Viber phone or choose another channel. |
| **Another active member already uses this email.** | Another member already has this e-mail. Check whether the person is already recorded. |

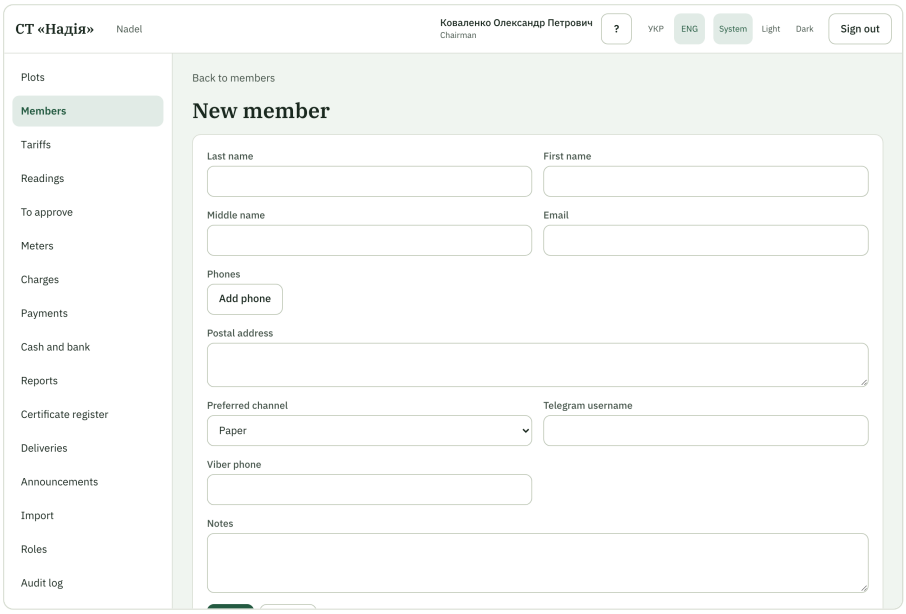
About Viber the system warns: "**Viber is not supported yet — invoices are printed on paper**".

## Worked example

A new owner, Andriienko Maria Petrivna, has joined the cooperative.

1. In the member list press **Add member**.
2. Enter the surname Andriienko, the first name Maria, the patronymic Petrivna.
3. Enter the phone 0671234567 and the e-mail. After saving the phone becomes +380671234567.
4. Choose the delivery channel "**Email**". The system will not save the form if the e-mail is not entered.
5. Press **Save**. Maria's card opens. She has no plots yet: the owner is assigned on the plot card.

## Screenshots




The new member form

## Related pages

- Members
- Member card
- Plot card: changing the owner
- Roles and permissions

# Member card

## What the screen is for

The card shows one person: contacts, the way they receive invoices, roles, the plots they own or used to own, and service notes. Here the member is edited, given roles and archived.

## Who sees it

The card is seen by whoever has the permission "**View members**". The buttons depend on permissions:

- "**Edit**" and "**Archive**" or "**Restore**": "**Edit members**";
- "**Edit roles**": "**Manage roles**";
- "**Change history**": "**View audit log**";
- the buttons of the "**System access**" block: "**Invite to the system**".

Notes are seen only by those who have "**View members**". The "**System access**" block is seen by those who have "**View members**"; the member does not see it on their own card. Every member sees their own data in "**My account**".

## Every element

At the top are the link "**Back to members**" and then the member's name. For an archived member the mark "**Archived**" stands next to it.

### Buttons in the header

- **Change history** opens the change log filtered to this member.
- **Edit roles** replaces the card with a list of roles with tick boxes. Tick the ones you need and press **Save**, or **Cancel**. Permissions add up: when there are several roles, the person has the permissions of all.
- **Edit** opens the form with the current data. After saving the system says "**Saved**".
- **Archive** removes the member from the list but keeps all the history. The system asks: "**Archive this member?**", and explains: "**The member is hidden from the default list and cannot sign in. History is kept.**" The **Restore** button brings them back: "**Restore this member?**", "**The member returns to the list.**"

> **IMPORTANT**
> A member is not deleted but archived. An owner of a plot cannot be archived until the plot passes to another owner: the system says "**This member owns a plot — change the owner first**". The last chairman cannot be archived or stripped of the role: "**The cooperative must keep at least one chairman.**".

### Member data

The table shows: **Name**, **Phones**, **Email**, **Postal address**, **Preferred channel**, the Telegram nickname and the Viber phone (when they exist), the Telegram state, **Notes** and **Roles**. For Telegram the card shows only "**connected**" or "**not connected**", without data of the member's own account. Connecting and disconnecting go to the change log.

### The "System access" block

Shows whether the person can sign in to the system. There are four states:

- "**Not invited**": no invitation has been sent yet;
- "Invited <date>, waiting for a password": the letter was sent, the person has not set a password yet;
- "Has access, last sign-in <date>" or "Has access, has not signed in yet": the password is set;
- "**Access turned off**": signing in is switched off.

The buttons are seen by whoever has the permission "**Invite to the system**":

- **Invite to the system** sends the letter "**Set a password**" to the e-mail on the card, with a link to the set-password page. The letter comes from the cooperative's mailbox; the link is valid for 72 hours and works once. If the member has connected Telegram, the link also arrives from the bot. The system says: "**The invitation was sent.**"
- **Send again** sends a new link; the previous one stops working. The system says: "**The invitation was sent again.**"
- **Turn off access** removes the sign-in, but the card and the data stay. The system asks to confirm: "**Turn off access**". The person can be invited again afterwards: the new invitation removes the old password and ends every session, so the person signs in only after setting a new password with the new link, to the same account and the same card.

When a button is unavailable, the reason is written beside it: "**Add an e-mail — it will be the login**" (the e-mail is the login) or "**Set up the cooperative mailbox**" (see the cooperative settings). When the letter could not be sent, the system says the invitation was not sent and suggests trying again.

Other rules:

- an e-mail that a member of another cooperative already uses is refused;
- archiving a member switches their access off, and restoring does not switch it on: invite the person again;
- nobody can switch off their own access: their own card has no "**Turn off access**" button;
- the access of the last chairman who has access cannot be switched off, and that chairman cannot be archived either: the system says "**This is the last chairman with access. Give another member the Chairman role and access first.**";
- when the e-mail of a member who has access is changed, the old password stops working and every session ends; a new link is sent to the new address, and the member signs in again only after setting a new password with it;
- one member can be sent at most 5 invitations (first and repeated together) within 15 minutes, and one computer at most 30; after that the system says "**Too many attempts. Wait a few minutes and try again.**"

Invitations, resends and switching access off are written to the change log.

#### The "Plots" panel

Shows two lists: "**Owns now**" (plot, share, the date from which) and "**Owned before**" (plot and period). The plot number leads to the plot card if you can view plots. If the person owns nothing, it says: "**This member owns no plots.**" or "**No plots owned before.**".

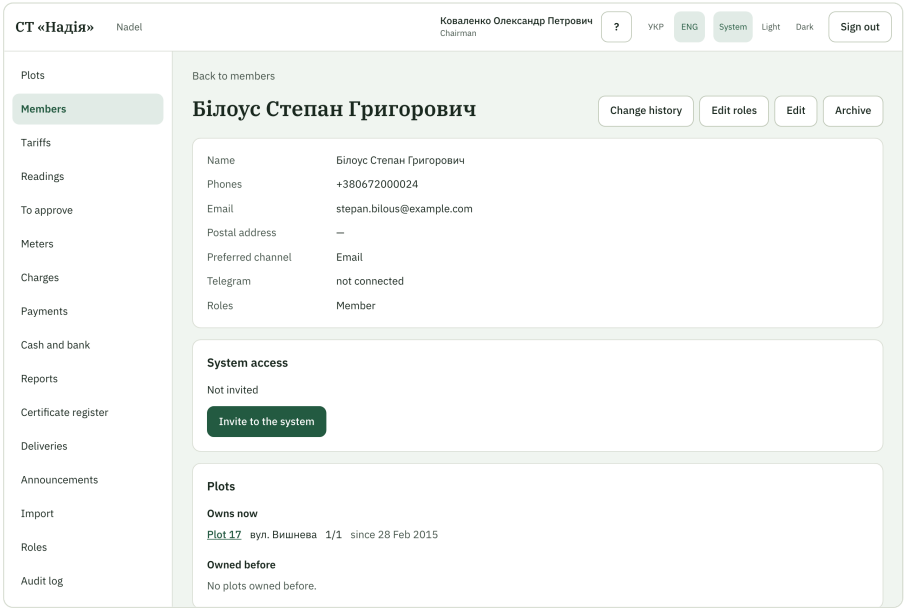
## Worked example

The chairman opens the card of the member Bondarenko Andrii Mykolaiovych.

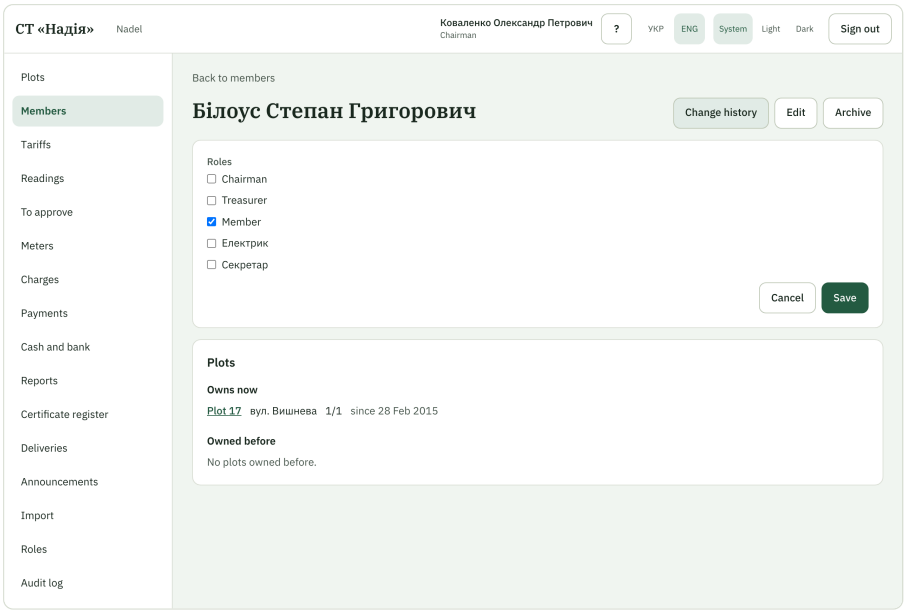
1. "**Owns now**" shows two plots: 7 (share 1/2 since 15.04.2019) and 15 (the whole plot since 25.03.2021).
2. The "**Roles**" field shows only "**Member**". The chairman presses **Edit roles**, ticks one more role and saves: the permissions of the roles add up.
3. The chairman tries to press **Archive**, but the system refuses: Andrii owns plots. First the owners are changed on the plot card, then the member can be archived.

Another example: Polishchuk Oksana, who keeps the member list, has, besides the base role, a custom role "Секретар" (Secretary) with permissions to view and edit members, plots and ownership. In her card the "**Roles**" field shows both roles.

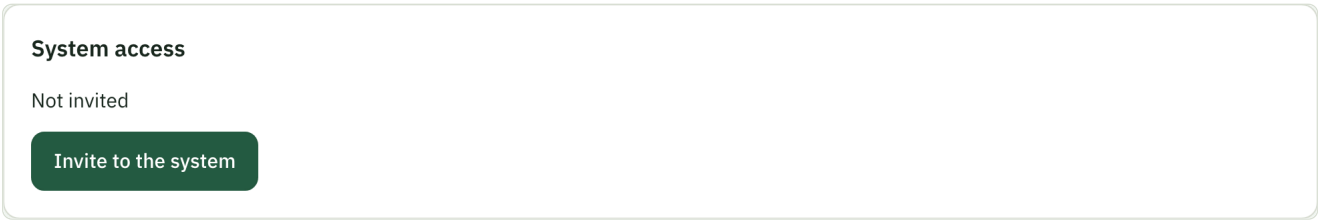
## Screenshots




Member card: contacts, roles, plots




Choosing roles on the member card




The system access block on the member card

## Related pages

- Members
- New member and editing
- Roles and permissions
- Plot card
- Change log
- Set a password

# Tariffs

## What the screen is for

This screen shows what the cooperative charges for and how much. Tariffs are grouped by charge type: membership fee, electricity, land tax, water and others. The charges themselves appear on the "**Charges**" screen. But already here, and on every plot's card, you can see how much a plot will be charged.

## Who sees it

The chairman and the treasurer see this screen: those who have the "**View tariffs**" permission. The create buttons are shown only to those who have "**Manage tariffs**", that is, the chairman. A member does not see this screen: they find their expected amounts in their account, on the page My account.

## Every element

### Buttons at the top

| Button | What it does |
|---|---|
| **Charge types** | Opens the list of charge types: you can add your own type, rename, reorder, archive or switch a type on. |
| **New tariff** | Opens the new tariff form. |

Both buttons are shown to the chairman only.

### The "Active" / "Archived" switch

Above the list there is the switch "**Which tariffs to show**". "**Active**" is shown by default. A tariff that was archived moves to "**Archived**": there it carries the mark "**Archived**" and shows the date until which it applied. If there are no archived tariffs, the screen says "**There are no archived tariffs**".

### Blocks by charge type

Each charge type has its own block with the type name. If the type is switched off, it carries the mark "**Inactive**". Such a type stays in the list, but its tariffs give no expected amount.

If a type has no tariffs yet, the block says "**No tariffs yet**". If there are no types at all, the screen says "**There are no charge types yet.**".

### A tariff row

Each row has:

- the tariff name; click it to open the tariff card;
- the rate, the basis and the frequency, for example "50 ₴ · **Per sotka** · **Yearly**";
- the date the current version takes effect and the number of plots the tariff concerns;
- an overlap warning, if there is one.

The bases are: **Per plot**, **Per sotka**, **Per kWh**, **Per m³** and **Fixed one-time amount**. More about them in the new tariff form.

### The overlap warning

Two tariffs of one charge type cannot apply to the same plot on the same day. If they do, a yellow warning appears under the tariff: "**Overlaps with tariff**" with the name of the other tariff, the number of shared plots and their numbers. Those plots are not charged until you change the plots of one of the two tariffs on the tariff card. The warning reads: "**Tariffs of the same charge type cannot apply to the same plots. These plots are not being charged now. Change the plots of one of the tariffs or choose a differen**" So open the card of one of the tariffs and change its plots, or choose a different charge type. While the overlap stays, those plots land in the queue "Not charged".

## Worked example

The demo cooperative has four tariffs:

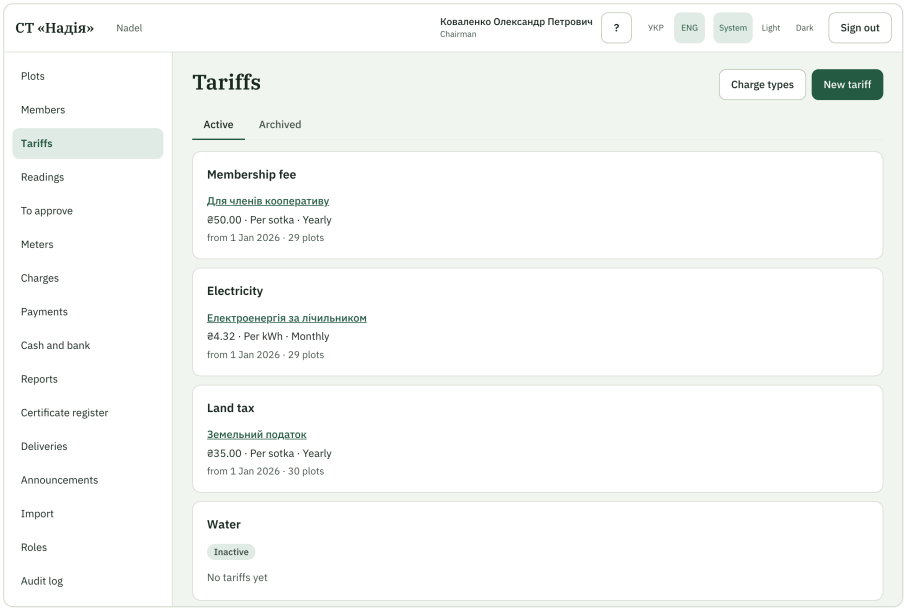
| Type | Tariff | Rate |
|---|---|---|
| Membership fee | Для членів кооперативу | 50.00 ₴ per sotka per year |
| Land tax | Земельний податок | 35.00 ₴ per sotka per year |
| Electricity | Електроенергія за лічильником | 4.32 ₴ per kWh per month |
| Road repair | Збір на ремонт дороги | 1,200.00 ₴ once, pay by 01.12.2026 |

1. Open "**Tariffs**".

2. In the "Membership fee" block find the tariff "Для членів кооперативу" and see how many plots it covers.
3. Click its name to see the versions and the plots.

For a plot of 600 m² (6 sotkas) the membership fee per month is 50 × 6 / 12 = 25.00 ₴, and the land tax is 35 × 6 / 12 = 17.50 ₴.

## Screenshots




The list of tariffs by charge type

## Related pages

- New tariff
- Tariff card: versions and plots
- Charge types
- Not charged
- How the system calculates tariffs and the expected amount

# New tariff

## What the screen is for

Here the chairman creates a new tariff together with its first version. A tariff tells the system what to charge for, how much and how often. After saving, the tariff card opens, where you must set the plots the tariff applies to.

## Who sees it

The chairman sees the form: the one who has the "**Manage tariffs**" permission. The treasurer sees tariffs but cannot create a new one.

> **IMPORTANT**
> Two tariffs of one charge type cannot cover the same plots on the same day. The monthly water fee has its own type "**Water**": a fixed amount per plot, and only connected plots pay. The electricity of the well pump is charged with the type "**Common consumers**" with the split "**Among connected plots**". The type "**Water**" may be switched off: then it is shown as "name (inactive — enable)", and choosing it switches it on (or switch it on in the charge types).

> **IMPORTANT**
> If you created a tariff by mistake and it has no charges yet, you can delete it on the tariff card. If it has charges, it can only be archived.

## Every element

At the top there is the link "**Back to tariffs**".

### Type and name

| Field | What to enter |
|---|---|
| **Charge type** | Required. The list has all types. A switched-off type is shown as "name (inactive — enable)": choosing it switches the type on when you save the tariff (the screen says "**Enabling the charge type...**"), so you do not need to go to the type list. If there are no types, the form is replaced with the message "**There is no active charge type. Add or activate one first.**". |
| **Tariff name** | Required, up to 100 characters, for example "Для членів кооперативу". |

Under the type field there is the hint "**Current tariffs of this type**": the current tariffs of the chosen type with their plots (or "**no plots**", "**none**"). So you see before saving whether the new tariff would overlap an existing one. Next to it there is the link "**Not the type you need?** **Charge types**", which opens the list of charge types.

### The first version

Block heading: "**First version**".

| Field | What to enter |
|---|---|
| **Valid from** | The date the rate takes effect. Required. |
| **Rate, ₴** | The rate with up to four decimal places, for example 4.32 or 1.4532. The amount is rounded to kopecks only at the end, so extra digits do no harm. On screen the rate is shown without extra zeros: 50.00 ₴, not 50.0000 ₴. |
| **Basis** | What the amount is made of. See the table below. |
| **Frequency** | **Monthly**, **Quarterly** or **Yearly**. For the basis "**Fixed one-time amount**" always "**One-off**". |
| **Due date** | Appears only for the basis "**Fixed one-time amount**" and is then required. |
| **Note** | Optional, for example "General meeting decision 14.12.2025". |

If "Valid from" is in the past, a yellow message appears under the fields: "**Charges for past periods will be recalculated**". It is only a warning: a new tariff does not change charges already booked; a recalculation is made separately on the page New recalculation.

### Bases

| Basis | Example |
|---|---|
| **Per plot** | 300 ₴ per plot per quarter |
| **Per sotka** | Membership fee 50 ₴ per sotka per year |
| **Per kWh** | Electricity 4.32 ₴ per kWh, by the meter readings |

| Basis | Example |
|---|---|
| **Per m³** | Water by the water meter readings |
| **Fixed one-time amount** | A targeted fee of 1,200 ₴ with a due date |

The cooperative charges monthly, so a yearly rate is divided by 12 and a quarterly rate by 3. Tariffs per kWh and per m³ apply only to plots that have a meter of that resource.

**Buttons**

- **Create tariff** saves the tariff and opens its card.
- **Cancel** goes back to the previous screen without saving.

**Error messages**

| What is shown | What to do |
|---|---|
| A required field message under the name or the type | Fill it in. |
| An error next to the rate | Enter a number with at most four decimal places. |
| "**A version with this date already exists**" next to "Valid from" | The tariff already has a version from this date. Choose another date or correct the existing version on the card. |
| A red message above the buttons | Read it: it says what exactly is wrong (for example, "**This charge type is archived.**"). |

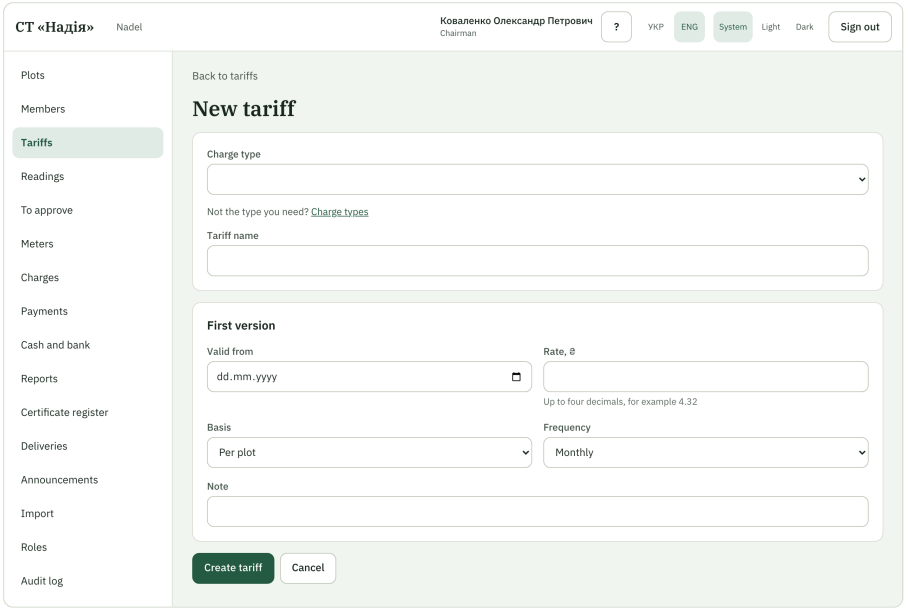
## Worked example

The chairman enters the membership fee, as in the demo data.

1. Click "**New tariff**" on the screen "Tariffs".
2. Choose the type "**Membership fee**", name the tariff "Для членів кооперативу".
3. "Valid from" 01.01.2026, basis "**Per sotka**", frequency "**Yearly**", rate 50.
4. Click "**Create tariff**" and set the plots on the card.

For a plot of 6 sotkas the monthly fee is 50 × 6 / 12 = 25.00 ₴.

## Screenshots




The new tariff form with the first version

## Related pages

- Tariffs
- Tariff card: versions and plots
- Charge types
- How the system calculates tariffs and the expected amount

# Tariff card: versions and plots

## What the screen is for

This screen shows one tariff: what it was and is, by versions, which plots it applies to and which exclusions it has. Here the chairman changes the rate (with a new version), corrects mistakes in versions and decides the plots.

## Who sees it

The chairman and the treasurer see the card: those who have the "**View tariffs**" permission. Only the chairman can change things, that is, the one who also has "**Manage tariffs**". If the charge type is switched off, the buttons for changing versions disappear.

## Every element

At the top there is the link "**Back to tariffs**", then the tariff name, the charge type ("**Charge type**") and the mark "**Inactive**" if the type is switched off.

### The overlap warning

If the tariff overlaps another tariff of the same type, you see a yellow message "**Overlaps with tariff**" with the name of the other tariff, the number of shared plots and their numbers. Those plots are not charged until you change the plots of one of the tariffs in the block "**Plots**". The warning reads: "**Tariffs of the same charge type cannot apply to the same plots. These plots are not being charged now. Change the plots of one of the tariffs or choose a differen**"

### The block "**Plots**"

It shows who the tariff applies to. There are two ways:

- "**All plots**" (except common land) and, if needed, exclusions with a required reason, for example "exempted by the general meeting";
- "**Selected plots**": an explicit list of plots.

If a plot is archived, it carries the mark "**archived**". If there are no exclusions, it says "**No exclusions**", and if no plot is selected, "**No plots selected**".

The button "**Change**" (chairman only) opens the editor:

| Element | What it does |
|---|---|
| **Applies to** | A switch between "**All plots**" and "**Selected plots**". If you switch, the exclusions or the selected plots are deleted on saving, and a yellow message warns about it. |
| **Plot number** | Search by number; plot suggestions below. The button "**Add exclusion**" or "**Add plot**" adds the plot. |
| **Reason** | Required for every exclusion, up to 500 characters. Without it you get "**Enter a reason**". |
| **Remove** | Removes the plot from the list. |
| **Save** | Saves. After that you see "**Scope saved.**". |
| **Cancel** | Closes the editor without changes. |

Editor errors: "**Common land has no tariffs**", "**The plot is archived**", "**There is no such plot**", "**This plot is listed twice**" (the plot is given twice) and "**The reason is too long (500 characters at most)**". Correct the row next to which they appear.

> **IMPORTANT**
> If a tariff was created by mistake and has never produced a charge, you can delete it (see below). If it has produced charges, it can only be archived.

> **IMPORTANT**
> Two tariffs of one type cannot apply to the same plot on the same day. Tariffs per kWh and per m³ apply only to plots that have a meter of that resource: on the plot card, where there is none, you see "**no meter**".

### The block "**Versions**"

The list of versions, the newest on top. A version row says: "from ... to ..." (or "from ..." if it is still valid), the rate, the basis, the frequency and, for a one-off fee, "pay by ...", and the note.

Every row has buttons (chairman only):

| Button | What it does |
|---|---|
| **Edit** | Opens the form "**Edit version**" with the same fields as the new tariff form. The button "**Save**" saves. |

| Button | What it does |
|---|---|
| Delete | Asks for confirmation "Delete the version?". This cannot be undone. |

While a version has no charges, it can be corrected or deleted simply: you see "Version corrected" or "Version deleted". The only version of a tariff cannot be deleted ("A tariff must have at least one version"): add another version or, if the tariff is no longer needed, delete it (while it has no charges) or archive it.

If a version already has charges, a hint "This version is already used in charges — correct it through a recalculation" stands next to it. Then:

- the chairman who can recalculate (the "Recalculate charges" permission), after "Save" or the delete confirmation, sees the recalculation preview step: what will change in past charges. The version change and the recalculation are saved together or not at all. The action is completed by the button "Book the recalculation" after entering the reason. More on the page New recalculation;
- without that permission the system says "This version is already used in charges — correct it through a recalculation" and changes nothing.

#### Deleting a tariff

The "Delete tariff" button is shown only to people with the "Manage tariffs" permission, and only for a tariff that has never produced a charge. Reversed charges count too: if a charge ever existed, the tariff is already used.

1. Click "Delete tariff".
2. Confirm: "Delete tariff "name" with all its versions? This cannot be undone." The name in the question is the name of your tariff.
3. The tariff disappears from the list together with all its versions. The change log gets the entry "Tariff deleted".

If the tariff is already used in charges, the system says: "The tariff is already used in charges — it can only be archived." Such a tariff can only be archived.

#### Archiving a tariff

The "Archive" button is shown only to people with the "Manage tariffs" permission, and only for a tariff that already has charges (instead of "Delete tariff").

1. Click "Archive". The dialog "Archive tariff" opens.
2. In the "Stops applying from" field choose a date. The hint says: "This is the first day the tariff no longer applies. The last day it applies is the one before."
3. Click "Archive".

The date must be after the start of the tariff's last version ("The date must be after the start of the last version.") and after the last day for which the tariff already has charges ("The tariff already has charges on or after this date. Choose a later date.").

From that date the tariff charges nothing: neither in the month run nor in other calculations. Charges before it stay. The tariff leaves the "Active" list and appears under "Archived" on the Tariffs screen. The card shows the mark "Archived" and "Valid until" with a date: the last day it applies, the day before the chosen date. The version history is read-only, and no new versions or plot changes are possible ("The tariff is archived and cannot be changed."). Archiving cannot be undone. The change log gets the entry "Tariff archived".

#### The block "New version"

A tariff is changed not by editing but by a new version with the date it takes effect. The fields are the same as in the first version of a tariff, and the starting values come from the latest one. The button "Add version" adds the version: "Version added".

If the new version takes effect from a past date, the system warns: "Charges for past periods will be recalculated". When you have the right to recalculate, you see the preview step and then "Version added, recalculation booked". Without that right, past charges stay as they were.

The error "A version with this date already exists" means a version from this date already exists: correct it instead of adding.

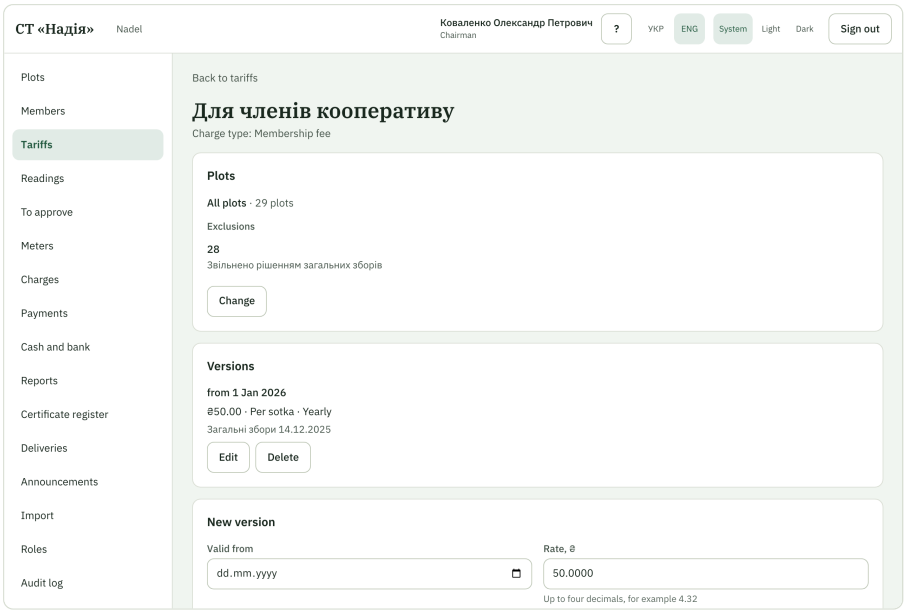
## Worked example

September is closed, October is open. From 1 September 60 ₴/sotka/year was entered by mistake instead of 50, the plot has 6 sotkas (in the demo it is plot 2).

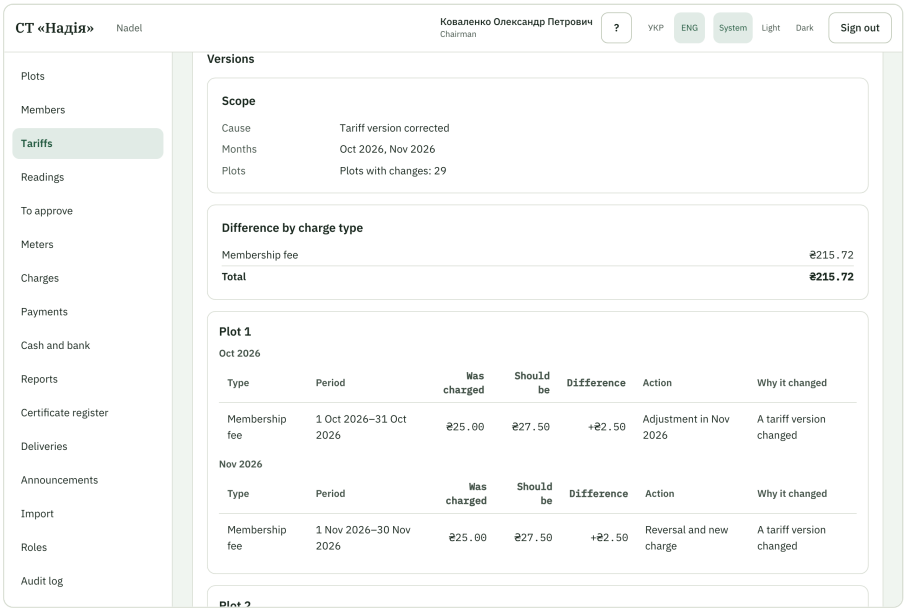
1. Open the tariff "Для членів кооперативу" and click "Edit" next to the version.
2. Change the rate to 50 and click "Save". Because the version is already used, the preview step opens.
3. Look at what will change: in closed September nothing; in October a reversal of 30.00 and a new charge of 25.00; in addition, in October an adjustment of −5.00 "for September".
4. Enter the reason and click "Book the recalculation".

The new rate gives 50 × 6 / 12 = 25.00 ₴ per month, the old one was 60 × 6 / 12 = 30.00 ₴. If the member has already paid at the old rate, the excess becomes a prepayment, not a "negative debt".

## Screenshots




The tariff card with versions and plots




The recalculation preview step after correcting a rate

## Related pages

- Tariffs
- New tariff
- New recalculation
- Not charged
- How the system calculates tariffs and the expected amount
- How the system calculates a recalculation

# Charge types

## What the screen is for

A charge type is "what for" the cooperative takes money: membership fee, electricity, land tax, water, common consumers. Each type has its own tariffs. The order of types in the list matters: when a payment is allocated to debts of the same day, the system follows this order.

## Who sees it

The chairman and the treasurer see the list: those who have the "**View tariffs**" permission. Only the one who has "**Manage tariffs**", that is, the chairman, can add, change, move and archive.

## Every element

### At the top

| Element | What it does |
|---|---|
| **Back to tariffs** | Goes back to the list of tariffs. |
| **Add charge type** | Opens the dialog "**New charge type**". Chairman only. |

The cooperative has these types from the start: **Membership fee**, **Electricity**, **Land tax**, **Water** and **Common consumers**. You can add your own, for example "Ремонт дороги". If there are no types, the screen says "**There are no charge types yet.**".

> **IMPORTANT**
>
> The type "**Water**" may be switched off while the cooperative has no water meters. Then it is shown in the new tariff form as "name (inactive — enable)". You can switch it on there by choosing it, or here: click "**Edit**" next to the type "**Water**", tick "**Active**" and save. You can also do it from the new tariff form: a switched-off type is shown there as "name (inactive — enable)", and choosing it switches it on. Then create the tariff: a fixed amount per plot, and only plots connected to water pay. The electricity of the well pump is charged not with this type but with the type "**Common consumers**".

### A type row

The row shows the name, the mark "**Active**" or "**Inactive**" and the description. The chairman also has buttons:

| Button | What it does |
|---|---|
| **Move up** | Moves the type up in the list. Unavailable for the first one. |
| **Move down** | Moves the type down. Unavailable for the last one. |
| **Edit** | Opens the dialog "**Edit charge type**". |
| **Archive** | Asks for confirmation "**Archive charge type**". |

### The type dialog

| Field | What to enter |
|---|---|
| **Name** | Required. A standard type can be renamed; then your name is shown. |
| **Description** | Optional. |
| **Active** | Only when changing. Hint under the field: "**An inactive type stays in the list, but its tariffs give no expected amount.**" |

Buttons: "**Save**" and "**Cancel**".

### Archiving

The confirmation "**Archive**" hides the type from the lists: it will get no new tariffs. Past charges and old tariffs stay. If a type is just temporarily not needed, it is better to switch it off by clearing "**Active**": the type then stays in the list and can be switched on again.

### Error messages

| What is shown | What to do |
|---|---|
| **A charge type with this name already exists** | Choose another name. |
| **The list of charge types has changed. It was reloaded, try again.** | Someone else changed the list. It has already been refreshed, repeat the move. |

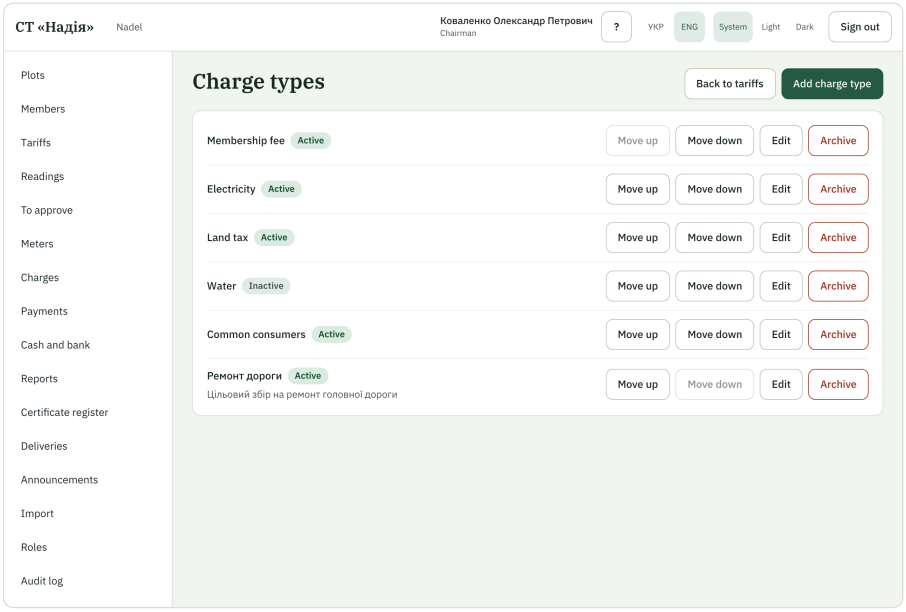
| What is shown | What to do |
|---|---|
| **This charge type is archived.** | The type is archived and can no longer be changed. |

## Worked example

The cooperative decided to collect money for a road repair.

1. Click "**Add charge type**".
2. Name the type "Ремонт дороги", describe it "Цільовий збір на ремонт головної дороги" and save.
3. Move it up with "**Move up**" until it is where you want it: the higher the type, the earlier its debt is paid off by a payment of the same day.
4. Create a tariff for it: a one-off amount of 1,200.00 ₴, pay by 01.12.2026.

## Screenshots




The list of charge types

## Related pages

- Tariffs
- New tariff
- Cooperative settings
- How the system allocates payments

# Charges: months

## What the screen is for

This is the main charges screen. It shows the billing months: the newest on top, with the status and the amount for each type. From here you go to a single month, and to the tabs with the "Not charged" queue, network losses, common consumers and recalculations. A charge is an entry in a plot's account: the type, the amount, the month, the days the service covers and how the amount was obtained.

## Who sees it

The chairman and the treasurer see this screen: those who have the "**View charges**" permission. The treasurer sees months, accounts and lists but has no action buttons. A member sees only the account of their own plots on the page My account, read-only.

## Every element

### Tabs

At the top there is a row of tabs "**Charges sections**":

| Tab | Where it leads |
|---|---|
| **Months** | This screen. |
| **Not charged** | The queue of what the system could not charge: Not charged. |
| **Losses** | The distribution of network losses: Losses. |
| **Common consumers** | Pumps, lighting and other common meters: Common consumers. |
| **Recalculations** | The log of corrections of past charges: Recalculations. |

### The list of months

The list heading is "**Billing months**". Each month is a card:

- the month name, for example "Жовтень 2026": click it to open the month page;
- the mark "**Open**" or "**Closed**";
- the amounts by charge type, and "**Received**", that is, how much money came in;
- the row "**Total**".

If nothing is charged in the month yet, it says "**No charges**". If there are no months at all: "**No billing months yet**".

### Where charges come from

| Source | How it appears |
|---|---|
| Fixed tariffs | The chairman clicks "**Charge the month**" on the month page. Running it again adds only what is missing. |
| Electricity by readings | Every approved reading at once gives a charge for the interval from the previous reading. |
| Network losses | Depending on the setting: a percentage added to every electricity charge, or a separate run for a period. |
| Common consumers | A separate run for a period, a separate line in the account. |
| Opening balance | The plot's debt on the day it started working in the system: by hand on the plot card or with an Excel sheet in the import. |

### Reversing a charge

Charges are never changed or deleted. A mistaken charge is cancelled by a reversal: an entry for the same amount with a minus and a reason. The button "**Reverse**" (the "**Reverse charges**" permission) is in the charge details on the plot card. A charge is reversed once, and a reversal entry is not reversed itself.

> **IMPORTANT**
>
> In a closed month there is no "**Reverse**" button. The details of such a charge say: "**The billing month is closed: the correction is made by an adjustment.**" A charge of a closed month is corrected by a recalculation: nothing in the closed month changes, and an adjustment for the difference appears in the nearest open month.

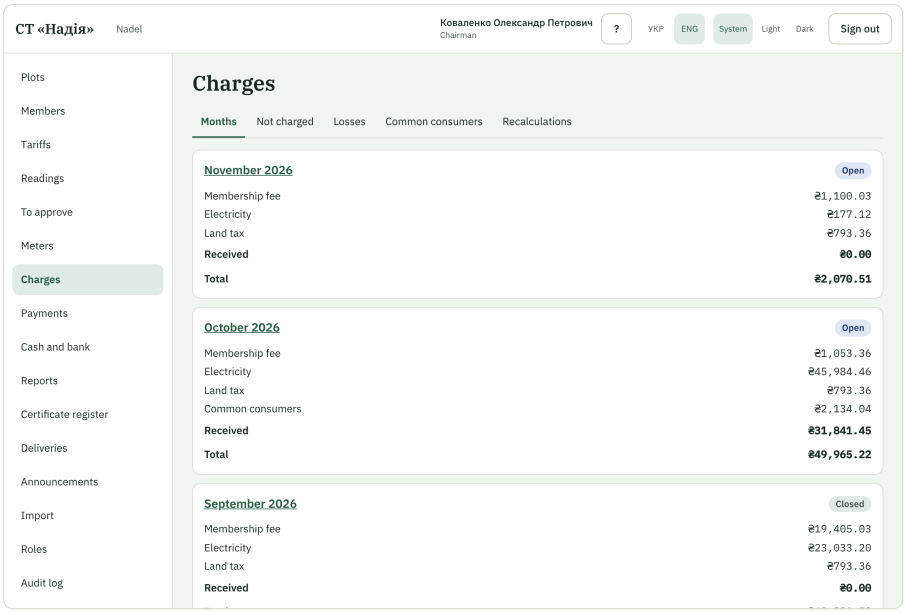
## Worked example

1. Open "**Charges**". On top is the newest month, marked "**Open**".
2. Look at the card of the previous month: it is marked "**Closed**" and has the amounts by type and "**Total**".

3. Click the name of the open month to see the "**Before closing**" list and the buttons "**Charge the month**" and "**Close the month**".
4. Click "**Not charged**" to see whether there are items the system could not charge.

In the demo, plot 14 has a mistaken membership charge in the open month: it is reversed with the reason "Помилкове нарахування". Plot 2 has an adjustment of September of −5.00 ₴ in October.

## Screenshots




The list of billing months

## Related pages

- Month page: charge and close
- Not charged
- Losses
- Common consumers
- Recalculations
- Plot card
- How the system calculates charges and closing of months

# Month page: charge and close

## What the screen is for

This is one billing month: how much is charged by type, how much came in and what is left to do before closing. Here the chairman charges the fixed tariffs for the month and closes the month.

## Who sees it

The chairman and the treasurer see the page (the "**View charges**" permission). The button "**Charge the month**" needs the "**Run charges**" permission, the button "**Close the month**" needs "**Close periods**"; by default only the chairman has them. The treasurer sees months but no actions.

## Every element

At the top are the charges tabs and the link "**All months**". The page title is the month with its status. For a closed month, under the title it says when it was closed and by whom.

### Buttons at the top

| Button | What it does |
|---|---|
| **Report** | Opens the charges-by-month report for this month. Shown if you have the right to reports. |
| **Charge the month** | Open month only. First shows a preview, see below. |
| **Close the month** | Open month only. Asks for confirmation. |

### The block "Charged in the month"

The amounts by charge type, "**Total**" and "**Received**".

### The block "Before closing"

This list helps you decide whether it is time to close the month:

- whether the month's charging is done ("**The month run is done**" or "**The month run has not been done yet**");
- how many items wait in the queue "**Not charged**"; the link leads to it;
- which meters have had no fresh readings for a long time (over 60 days): next to each, the plot number and the date of the last reading, or "**Every meter has a recent reading**".

The list does not stop you from closing: the person decides, seeing everything.

### Invoices of the month

The block "**Invoices of the month**" builds the invoices of the plots for this month and starts sending them. Invoices can be built for an open month too; the board usually does it right after the month's charges. The block is used by those who may build invoices (by default the chairman).

> **IMPORTANT**
>
> Until the IBAN and the ЄДРПОУ code of the cooperative are filled in, invoices are not built. The message "**Invoices cannot be built yet**" names what is missing and leads to the settings: the chairman enters the IBAN in Settings → Requisites (**Fill it in at Settings → Requisites**), the ЄДРПОУ code is set by the platform operator. Details: Cooperative settings.

#### Build the invoices

First the block shows the preview "**What the build will do**": how many plots can get an invoice ("**Plots with an owner: {count}**": active plots, not common land, with a current owner), how many of them already have an invoice for this month ("**Already invoiced for this month: {count}**") and what the build will do. The summary uses the same words as the result: "To be built: 18. New versions: 0. Unchanged: 1. Skipped: 6." A plot that already has an invoice gets a new version only when its figures changed. Plots with no debt are skipped. The switch "**Include invoices with nothing due**" adds plots with no debt; their invoice has no QR code.

Press "**Build invoices for {month}**" and wait: an indicator is shown while the build runs, then the result: how many invoices were built, how many got a new version, how many did not change and how many were skipped. If nobody entered anything between the preview and the build, the result equals the preview. The result stays in place of the preview until you change the month or the switch.

Building again creates nothing while nothing has changed. If somebody else is already building the same month, you will see a message about it, and nothing is created twice.

#### Invoice versions

The system keeps every invoice as it was built: later it opens with the same figures and the same owners, even after a payment or a change of owner. After a payment, a correction or a recalculation the treasurer builds the month again. A plot whose figures changed gets a new version ("version 2") with the same invoice number; a plot where nothing changed gets no new version. Older versions stay available: the lists show the latest one, and pressing "**Earlier versions ({count})**" unfolds the earlier ones. A member sees only the latest version. An invoice is never deleted.

#### The list of the month's invoices

Below is the list "**Invoices of the month**": one row per plot with the latest version (plot, version, "**Due**", build date and a link to the PDF). Older versions of a plot are folded under "**Earlier versions ({count})**". With no invoices yet it says "**There are no invoices for this month yet.**". Each row has a "**Send**" button that sends this one invoice to the owners of the plot.

"**PDF for printing**" downloads one PDF with the latest versions of the invoices, one per A4 sheet, in the order of streets and plots. It is for members who chose paper delivery. The switch "**What to print**" chooses between "**Paper recipients only**" and "**All plots**".

#### Check the QR code before handing out

The code follows the rules of the National Bank of Ukraine. Before you hand out invoices, check one code with a phone: open the QR scanner in a banking app, point it at the code on the screen or on paper and check the recipient, the IBAN, the payment purpose and the amount as described on How to read my invoice. The other fields in the bank are locked; close the form without paying.

#### Send the invoices of the month

The button "**Send invoices for {month}**" does not send anything at once. First the window "**Send invoices for {month}**" shows the preview "**What will be created**": how many members get the invoice in Telegram, how many by email and how many paper copies must be printed. The switch "**Also send invoices with nothing due**" adds plots with nothing to pay: such a member gets "No debt, thank you!" and the PDF. If everything has been sent already, the window says "**Nothing new to send: everything has been sent already.**".

"**Send**" puts the messages in the queue and returns you at once; the progress is on the Delivery screen. In Telegram the member gets "Invoice for October 2026, plot 9: due 413.26 ₴", the PDF and a "Pay" button (it opens the banking app through the QR link of the invoice). The email has the same text, the PDF attached and the signature "The board ...".

Sending the same invoices of a month again duplicates nothing: the same version of an invoice is received by the same member through the same channel once. If an invoice was reissued as a new version, the new one goes out. Paper invoices are marked "to print": print the PDF for printing (every copy has its own QR code for connecting Telegram), then press "**Printed**" on the Delivery screen. Which channel an invoice takes is described on the Delivery page, certificates on the Certificates register page.

### Charge the month

The dialog "**Charge {month}**" first calculates what will be charged and writes nothing. It shows:

- "**Plots that get new charges: {count}**";
- "**New charges**" by type;
- the block "**Not charged**" with a hint: remove the cause and repeat in the "Not charged" section.

If there is nothing new, the dialog says "**Nothing new to charge: every charge of this month already exists.**". The button "**Book**" books exactly what you saw. After that "**Charges created: {count}**" appears. If something changed meanwhile, the system says "**The charges changed since the preview. Preview again before booking.**", and the dialog offers the button "**Preview again**".

Running again creates only what is missing: a plot or a tariff added later gets its charge, and there are no duplicates. A month after the current one cannot be charged: "**A future month cannot be charged**".

### Close the month

The dialog "**Close the month**" shows the list once more and asks for confirmation. A closed month is never reopened. Only the earliest open month can be closed, otherwise you get "**Close the earlier months first**". A charge that belongs to a closed month (for example, a late reading) lands in the earliest open month with a note "for September, charged in October".

> **IMPORTANT**
>
> In a closed month nothing is corrected with the "**Reverse**" button: it is not there. A mistake in a closed month is corrected by a recalculation, which adds an adjustment for the difference in the nearest open month.

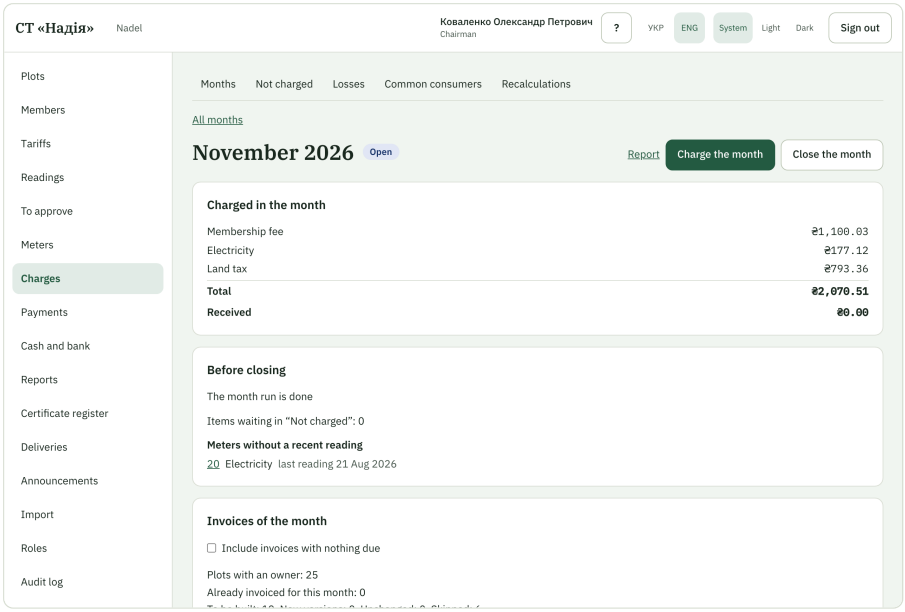
## Worked example

There is an open month in which the chairman wants to charge the fixed tariffs.

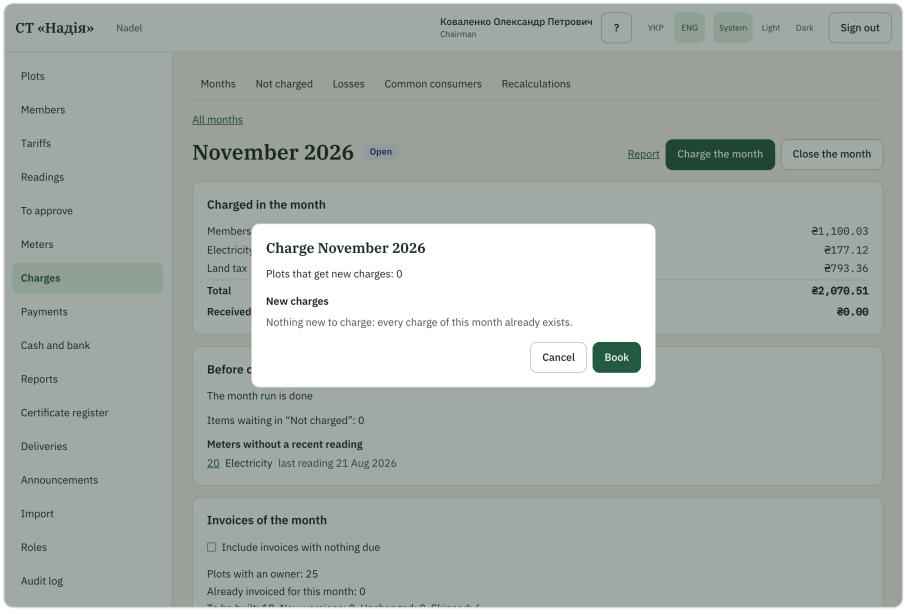
1. Open "**Charges**" and click the month name.
2. Click "**Charge the month**". The dialog shows the number of plots and the new charges by type: a membership fee of 25.00 ₴ and a land tax of 17.50 ₴ for a plot of 6 sotkas.
3. If there is something in the block "**Not charged**", take note, but it will not stop you: click "**Book**".
4. When all items are in order, click "**Close the month**" and confirm.

In the demo the current month has already been charged, so the preview says there is nothing new.

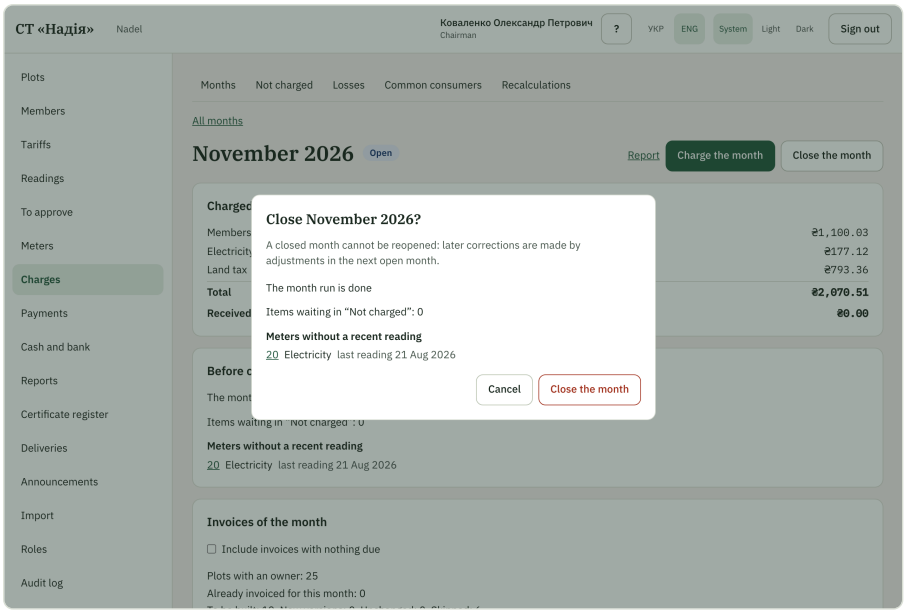
## Screenshots




The month page with the checklist before closing




The preview of charging the month




The confirmation of closing the month

## Related pages

- Charges: months
- Not charged
- New recalculation
- Delivery
- Certificates register
- How the system calculates charges and closing of months

# Not charged

## What the screen is for

Sometimes the system cannot charge because something is missing: there is no tariff for part of the days, two tariffs apply to a plot, a meter reading is unknown. All such items land here, with a reason. You remove the reason and retry the charge. Then the item disappears from the list.

## Who sees it

The chairman and the treasurer see the list (the "**View charges**" permission). The retry button belongs to the one who can run charges (the "**Run charges**" permission), that is, the chairman.

## Every element

At the top are the heading "**Not charged**" with the explanation "**What could not be charged, with the reason. Fix the cause and retry.**" and the charges tabs. If the queue is empty, the screen says "**Nothing is waiting: every charge has been made.**".

### The table

| Column | What it shows |
|---|---|
| **Plot** | The plot number, and for losses or common consumers the node, meter or consumer. |
| **Charge type** | The charge type or the item type: "**Tariff for a month**", "**Reading interval**", "**Losses of a meter tree node**", "**Common consumer**". |
| **Period** | For which month or days. |
| **Reason** | The reason and, where possible, a link to the fix, for example "**Tariffs**". |
| **Actions** | The button "**Retry**" (chairman only). |

### Reasons and what to do

| Reason | What to do |
|---|---|
| **No tariff for every day** | Add a tariff version that covers these days on the tariff card. |
| **Two tariffs apply on the same day; change the plots of one of them** | Change the plots of one of the two tariffs so that they no longer overlap. |
| **The plot is outside the plots of every tariff** | Add the plot to a tariff or create a tariff that covers it. |
| **The plot is excluded from the tariff** | This is a board decision: the plot was excluded on purpose. If the exclusion is a mistake, remove it on the tariff card. |
| **A meter value of the period is unknown** | The node's meter has no reading after the end of the period. Enter a reading on the screen "Readings". |
| **A meter value of the period is unknown** | The same for a common consumer's meter. |

### The button "**Retry**"

Tries to charge one item again. If it works, the item disappears from the list. If not, you see "**Still not charged**" next to the button, and the reason stays. During the attempt the button says "**Retrying...**". The next charging of the month also repeats the tariff and reading records.

Items never block closing a month: they only appear in the checklist before closing.

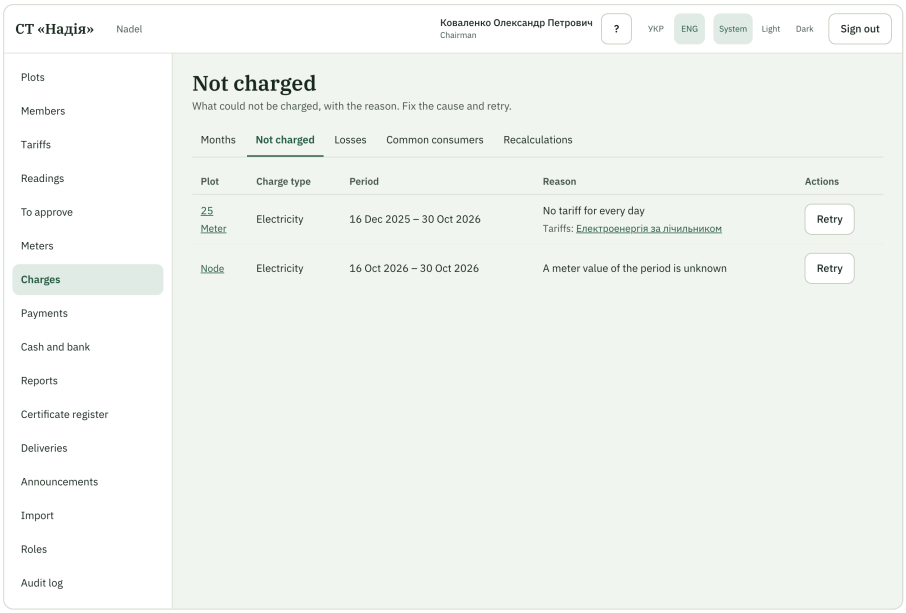
## Worked example

In the demo, plot 25 has a meter installed on 15.12.2025, before the start of the first electricity tariff version (01.01.2026). Its interval cannot be charged: the list has a row with the reason "**No tariff for every day**".

Also, the node "Квартал Б" has an unknown value: plot 20's meter has no reading later than 75 days ago. That row stands in the losses queue.

1. Open "**Not charged**".
2. Look at the reason next to plot 25's row.
3. Add a tariff version valid from 15.12.2025, or leave it as is if those days really are not payable.
4. Click "**Retry**".

## Screenshots




The "Not charged" queue with reasons and the retry button

## Related pages

- Charges: months
- Tariff card
- Readings: the electrician's round
- How the system calculates charges and closing of months

# Network losses

## What the screen is for

The substation shows more electricity than the plot meters under it read: the difference is the loss in the wires. Here the chairman distributes the losses of each node between the plots under it for a period and books them as separate charges. This way the losses are paid by those who consumed.

## Who sees it

The chairman and the treasurer see the screen (the "**View charges**" permission). Whoever has "**Run charges**" can preview and book; whoever has "**Reverse charges**" can reverse a run. Without the right to book, the screen says: "**You can look at past runs here; running needs the permission to run charges.**"

## Every element

At the top are the heading "**Losses in the network**" and the explanation "**Share the losses of every substation and quarter meter among the plots under it for a date range.**".

### The losses mode

The chairman chooses the mode in the settings, the block "**Losses**":

- "**Fixed percentage of the consumption**": a percentage, for example 5 %, is added to every electricity charge; then there is nothing to calculate on this screen, and it says: "**Losses are charged as a fixed percentage of every reading, so there is nothing to run here.**"; whoever has the settings right sees the link "**Open the settings to switch to proportional losses**";
- "**Proportional to the consumption**": the board chooses a period and runs the calculation here.

### Period and preview

| Element | What it does |
|---|---|
| **From** | The first day of the period. |
| **To** | The last day. Hint: "**Not later than today**". |
| **Preview** | Calculates what will be charged, writes nothing. If the dates are wrong: "**Check the dates**". |
| **Preview again** | Refreshes the preview. |
| **Book** | Books exactly what you saw. After that: "**Charges created: {count}**". |
| **Preview without the charged nodes** | Appears when some nodes have already been charged for these days. Books the rest. |

Above the table you see "**Plots charged: {count}**" and the month in which it will be booked: the charge lands in the month of the "To" day or in the earliest open month after it.

### The table "**Losses per node**"

Columns: **Node**, **Node shows, kWh**, **Children, kWh**, **Losses, kWh**, **Losses, %**, **Plots** and **Amount**. A value marked "**calculated**" is calculated in proportion to the days between neighbouring readings.

The result for each node: "**Charged**", "**Negative losses**", "**Unknown value**", "**Skipped**" or "**Already charged**".

### Warnings

The block "**Warnings**" may contain:

| Situation | What it means |
|---|---|
| Negative losses | The plots under the node "consumed" more than the node showed: a reading error, different removal dates or a faulty meter. Nothing is charged, the board sees the numbers. |
| Unknown value | The node or its child has no reading after the end of the period. This node and the nodes above it are skipped, the rest of the tree is charged. The node lands in "Not charged"; when readings appear, "Retry" charges it separately. |
| Already charged | The losses for these days were already calculated for the node. Two runs cannot cover the same days of one node: "**Losses for some of these days have already been calculated for a node. Preview again and book the other nodes.**" |
| No plot consumed | There is nobody to charge. |

**Past runs**

The block "**Past runs**" shows the booked runs: the period, the booking month, the author, the number of plots and nodes. Each has "**Details**" (the list of charges and nodes) and the button "**Reverse the run**". The dialog "**Reverse the run**" requires a reason (up to 500 characters) and reverses all the charges of the run together. The charges themselves are not changed or deleted. If the run's month is already closed, a reversal is impossible: "**The run cannot be reversed: its billing month is closed**".

## Worked example

Quarter A shows 1,100 kWh. Under it are plots P1 = 600 and P2 = 400, 1,000 in all. The losses of 100 kWh are split in proportion to consumption: P1 gets 60 kWh (60 × 4.32 = 259.20 ₴), P2 gets 40 kWh (172.80 ₴).

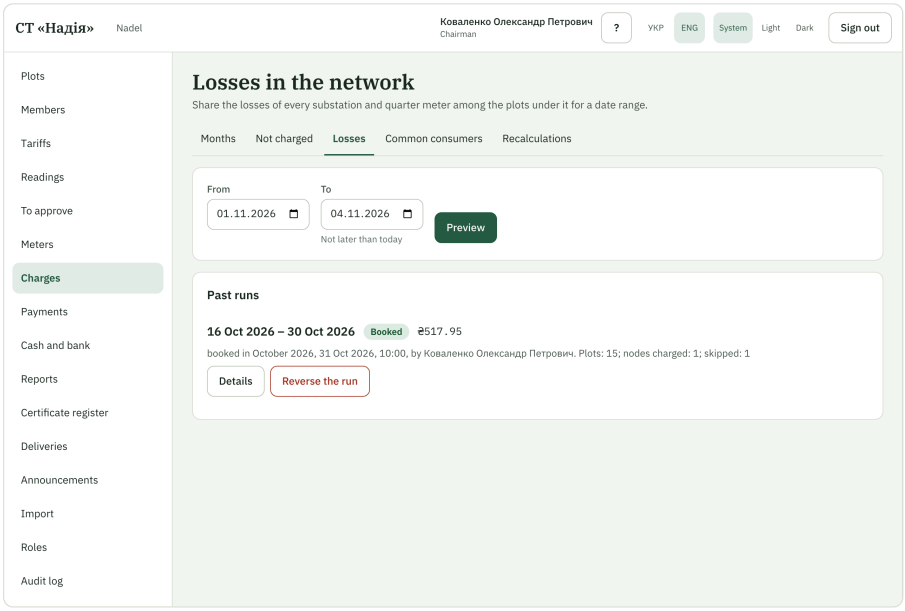
Quarter B has 1,050 kWh and one plot P3 = 1,000, so the losses of 50 kWh go to P3.

The substation shows 2,300. Its children: quarter A 1,100, quarter B 1,050 and the well 50, 2,200 in all. The substation's losses of 100 kWh are split between all plots under it in proportion 600 / 400 / 1,000 out of 2,000: P1 gets 30, P2 20, P3 50 kWh.

For comparison, with a fixed percentage of 5 %: 65 kWh of a plot give losses of 3.25 kWh, that is, 14.04 ₴.

In the demo a run is booked for quarter A, and quarter B is shown with a warning about an unknown value.

## Screenshots




The Losses tab with the run history

## Related pages

- Charges: months
- Common consumers
- Meter tree
- Cooperative settings
- Not charged
- How the system calculates losses and common consumers

# Common consumers

## What the screen is for

Well pumps, street lighting and the office consume electricity for everyone. Their consumption is charged with a separate type "**Common consumers**" with its own tariff per kWh and a separate line in the invoice. Here the chairman previews the distribution for a period and books it.

## Who sees it

The chairman and the treasurer see the screen (the "**View charges**" permission). Whoever has "**Run charges**" can preview and book, and whoever has "**Reverse charges**" can reverse a run. Without the right to book, the screen says: "**You can look at past runs here; running needs the permission to run charges.**"

## Every element

The heading is "**Common consumers**" and the explanation is "**Share the consumption of the well pump, street lighting and other common meters among the plots for a date range.**".

### Split methods

The split method is chosen for each consumer in the settings, the block "**Common consumers**". A consumer without its own choice follows the cooperative default.

| Method | Who is charged |
|---|---|
| **Among connected plots** | Plots connected to this well, by the days of connection. A plot that is not connected does not pay. |
| **Equally per plot** | All active plots, with or without a meter. |
| **Among service users** | Plots from the list in the consumer's settings (for example, lighting). |
| **Not charged** | Nobody: it is included in the membership fee and shown in reports. |

### Period and preview

It works the same way as on the page Losses.

| Element | What it does |
|---|---|
| **From** and **To** | The period; "To" no later than today. |
| **Preview** | Shows what will be charged; writes nothing. |
| **Book** | Books what you saw: "**Charges created: {count}**". The charge lands in the month of the "To" day or in the earliest open month after it. |
| **Preview without the charged consumers** | If some consumers have already been charged for these days, books the rest. |

### The table "Common consumers"

Columns: **Consumer**, **Consumption, kWh**, **Split**, **Plot-days**, **Plots** and **Amount**. A consumer's result: "**Charged**", "**Not charged**", "**No plots**", "**Unknown value**" or "**Already charged**".

### Messages and warnings

| What is shown | What to do |
|---|---|
| **There is no tariff for common consumers, so nothing is priced and the run cannot be booked. Add the tariff first.** | Create a tariff of the type "**Common consumers**" with the basis "**Per kWh**". The button "**Open the tariffs**" leads to the tariffs. Without a tariff the table says "**No tariff**", and nothing can be booked. |
| Unknown value | The consumer's meter has no reading after the end of the period. The consumer lands in "Not charged"; when readings appear, "Retry" charges it separately. |
| **This common consumer has already been charged for these days. Preview again and book the other consumers.** | The consumer has already been charged for these days. Preview again and book the others. |
| No plots | No plot uses the consumer in this period, so there is nobody to charge. |

**Past runs**

The block "**Past runs**" works as on the page Losses: "**Details**" and "**Reverse the run**" with a required reason. You can reverse while the run's month is not closed.

## Worked example

A well pump consumed 60 kWh over the period. Three plots are connected to the well for the whole period, one more plot is not connected. The split method is "**Among connected plots**":

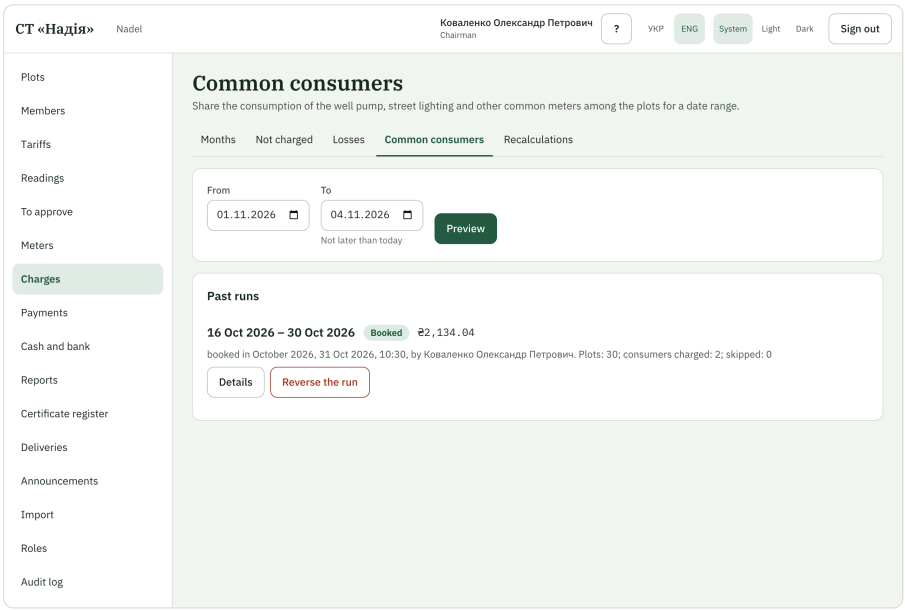
- each of the three gets 60 / 3 = 20 kWh, that is, 20 × 4.32 = 86.40 ₴;
- the plot that is not connected pays nothing.

If a plot was not connected for part of the month, plot-days are split. Pump 60 kWh, plots P1 and P2 connected for the whole month, and P3 only from 16.10: that is 30 + 30 + 15 = 75 plot-days. The shares: P1 24, P2 24, P3 12 kWh.

Lighting of 30 kWh over three plots from the list: 10 kWh each, that is, 43.20 ₴ each.

In the demo a run is booked: the pump among the plots connected to it, and the lighting equally among all plots, at the tariff of 4.32 ₴ per kWh.

## Screenshots




The Common consumers tab

## Related pages

- Charges: months
- Losses
- Cooperative settings
- Plot card: water connection
- How the system calculates losses and common consumers
- Water and connection

# Recalculations: the log

## What the screen is for

Sometimes past charges must be corrected: a tariff rate was entered with a mistake, a meter reading was corrected, a plot's area changed. A recalculation does this. The log here shows all booked recalculations: when, for what reason, who booked them and how much the amounts changed. The log is view-only: a recalculation is never changed or cancelled, and a mistake is corrected by the next recalculation.

## Who sees it

The chairman and the treasurer see the log (the "**View charges**" permission). The new recalculation button is shown only to the one who has the "**Recalculate charges**" permission, that is, the chairman.

## Every element

At the top are the heading "**Recalculations**" with the explanation "**The log of recalculations of past charges: what changed and why. Read only.**" and the charges tabs. If there have been no recalculations yet: "**There have been no recalculations yet**".

### The button

**New recalculation** opens the new recalculation form. The button is shown to the chairman only.

### The table

| Column | What it shows |
|---|---|
| **When** | The date and time of booking. |
| **Cause** | The cause: "**Manual**", "**Tariff version**" or "**Reading**", and what exactly happened (a tariff version added, corrected or deleted; a reading entered, corrected or deleted). |
| **Booked by** | Who booked it; "**System**" if the system made the recalculation. |
| **Reason** | The reason given by the person. |
| **Total** | The difference in the amount and the number of plots. |
| **Actions** | The button "**Open**": opens the recalculation card. |

The list is split into pages; at the bottom are the buttons "**Previous**" and "**Next**".

### A hint on other screens

Screens that change data affecting past charges (a plot's area, the composition of a tariff, a water connection, moving a meter in the tree) recalculate nothing themselves. They remind you: "**Past charges have not changed. If needed — Charges → Recalculation.**" Only those who can recalculate see the hint.

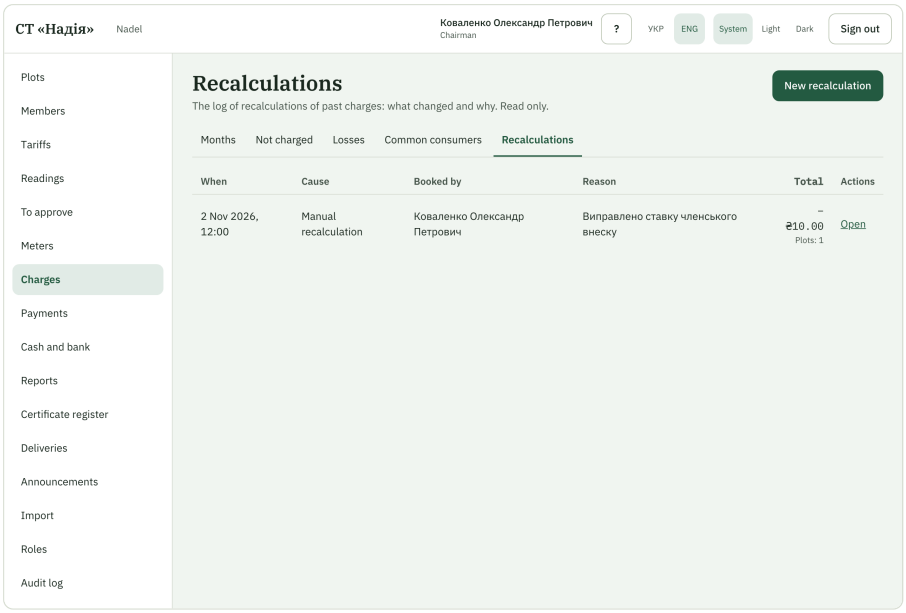
## Worked example

The demo has one booked recalculation: plot 2, the membership fee rate was 60 ₴ per sotka per year instead of 50.

1. Open "**Recalculations**".
2. Find the only row and look at its cause, reason and difference.
3. Click "**Open**" to see exactly what changed: see the Recalculation card.

A plot of 6 sotkas: the old fee is 60 × 6 / 12 = 30.00 ₴ per month, the new one is 50 × 6 / 12 = 25.00 ₴, the difference is −5.00 ₴ for each month.

## Screenshots




The recalculations log

## Related pages

- New recalculation
- Recalculation card
- Charges: months
- How the system calculates a recalculation

# New recalculation

## What the screen is for

Here the chairman recalculates past charges when the data they depend on has changed: a tariff rate, a meter reading, a plot's area, the composition of a tariff, a water connection or the meter tree. The system compares what was charged with what should be by the current data and shows the difference. Nothing is written until you enter a reason and book.

## Who sees it

The screen is seen by the one who has the "**Recalculate charges**" permission: by default the chairman. The treasurer sees the log but cannot recalculate.

An electrician who can enter readings but cannot recalculate, when correcting an already charged reading, sees: "**The reading has already been charged — ask the chairman**". Nothing changes.

## Every element

At the top are the link "**Back to recalculations**", the heading "**New recalculation**" and an explanation.

### Step 1: what to recalculate

| Field | What to enter |
|---|---|
| **From month** | The first month. No later than the current one. |
| **To month** | The last month. No more than 24 months from the first. |
| **Charge types** | "**All types**" or "**Chosen types**" with ticks on the types. At least one type: "**Choose at least one type**". |
| **Plots** | "**All plots**" or "**Chosen plots**": find a plot by number ("**Find a plot by number**") and add it. The button next to a number removes the plot. At least one: "**Add at least one plot**". |

The button "**Preview**" calculates what will change; it writes nothing. Form errors: "**Enter the months**", "**The first month cannot be later than the last**", "**A future month cannot be recalculated**", "**A recalculation covers at most 24 months**".

### Step 2: what will change

The block "**What would change**" shows the scope (the months and the number of plots with changes), the difference by type ("**Difference by charge type**") and the rows "was charged → should be → difference". Next to each row is the reason for the change, for example "**A tariff version changed**" or "**The plot area changed**". The action for a row:

| Action | When |
|---|---|
| **Reversal and new charge** | The month is open: the old charge is reversed, a new one is created. |
| An adjustment in an open month | The month is closed: nothing changes in it, and an adjustment for the difference, referring to the original month, appears in the nearest open month. |
| **New charge** | It was not charged before. |
| **Reversal** | The charge no longer belongs. |

Then:

- the field "**Reason**": required, up to 500 characters, with the counter "**{count} / 500**"; the hint "**Stored on every reversal and adjustment.**"; without a reason: "**Enter a reason**";
- the button "**Book the recalculation**" books exactly the preview you saw;
- the button "**Back**" goes back to the selection.

If charges changed meanwhile, the preview is refreshed and says "**The preview was refreshed.**": confirm again. When the charges already match the data, the preview says "**No changes**", and there is nothing to book.

> **IMPORTANT**
> A closed month is never changed. The correction lands in the nearest open month as an adjustment. If a debt decreased and has already been paid, the excess becomes a prepayment, not a "negative debt".

### Other ways into a recalculation

The same preview step opens without this form when you:

- correct or delete a tariff version that is already used, or add a new version with a past date: on the tariff card;

- correct or delete an already charged reading: on the meter card, the heading "**Recalculation for the reading**".

In "**To approve**" there is no recalculation step. If a reading has already been approved and charged, correct it on the meter card.

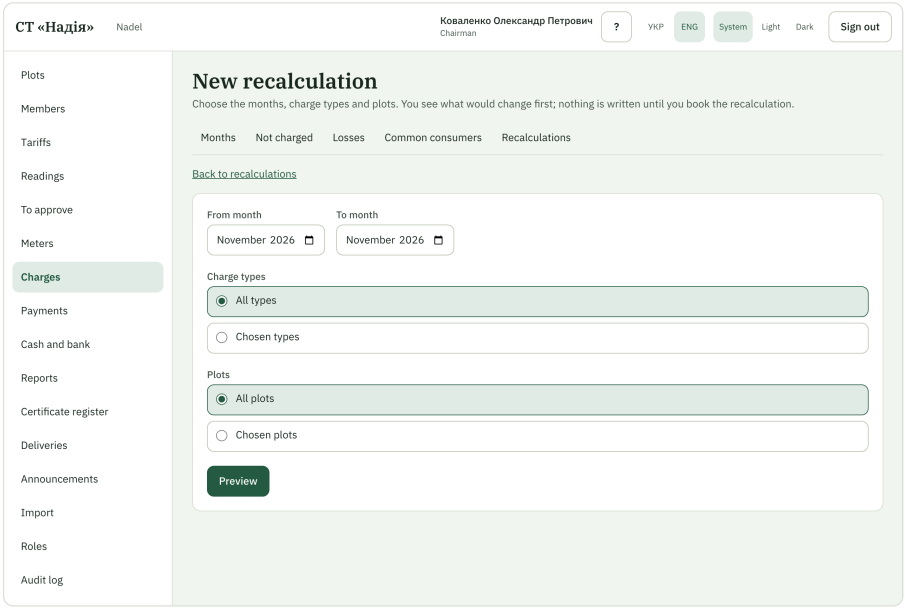
## Worked example

From 1 September 60 ₴/sotka/year was entered by mistake instead of 50, the plot has 6 sotkas. September is closed, October is open.

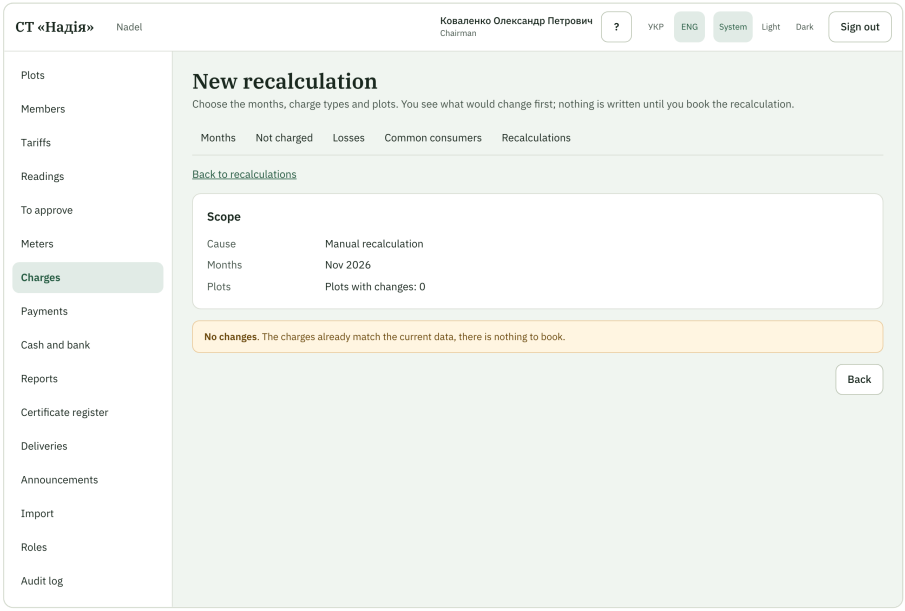
| | Was | After the correction |
|---|---|---|
| September (closed) | 30.00 | Does not change; in October an adjustment of −5.00 "for September" |
| October (open) | 30.00 | A reversal of −30.00 and a new charge of 25.00 |
| Total | 60.00 | 50.00; if 60.00 was paid, a prepayment of 10.00 |

Steps: choose the months September and October, the type "Membership fee", plot 2, click "**Preview**", check the rows, enter the reason "Wrong rate" and click "**Book the recalculation**". The demo is already consistent, so the preview of all plots of the open month says "**No changes**".

## Screenshots




The new recalculation form




The preview when there are no changes

## Related pages

- Recalculations: the log
- Recalculation card
- Tariff card
- Meter card
- How the system calculates a recalculation

# Recalculation card

## What the screen is for

Here you see one booked recalculation in detail: what was recalculated, why, who booked it and how every plot changed. It is view-only. A recalculation is never changed or cancelled, and a mistake is corrected by the next recalculation.

## Who sees it

The chairman and the treasurer see the card (the "**View charges**" permission). A member sees only the adjustments of their own plots in "My charges": the reason, but not the recalculation itself.

## Every element

At the top is the link "**Back to recalculations**". If the recalculation is not found: "**The recalculation was not found**".

#### General information

Fields: **Cause**, **Months**, **Plots**, **Reason**, **Booked**, **Booked by**. The block "**Scope**" shows the months and plots.

#### The block "**Totals by charge type**"

The difference for each charge type and "**Total**".

#### The rows of the recalculation

Each plot has its own block with the heading "**Plot {number}**" and the link "**Plot account**". Rows are grouped by month. Columns:

| Column | What it shows |
|---|---|
| **Type** | The charge type. |
| **Period** | For which period. |
| **Was charged** | How much was charged. |
| **Should be** | How much should be. |
| **Difference** | The difference. |
| **Action** | The action: "**Reversal and new charge**", "**New charge**", "**Reversal**" or an adjustment in an open month. |
| **Why it changed** | The reasons for the change, for example "**A tariff version changed**", "**A reading changed**", "**The water connection changed**", "**The meter tree changed**". |
| **Entries** | The created entries: "**Original**", "**Reversal**", "**New charge**", "**Adjustment**"; each links to the plot account. |

If there are no rows, it says "**This recalculation has no lines**".

#### What it looks like on the plot account

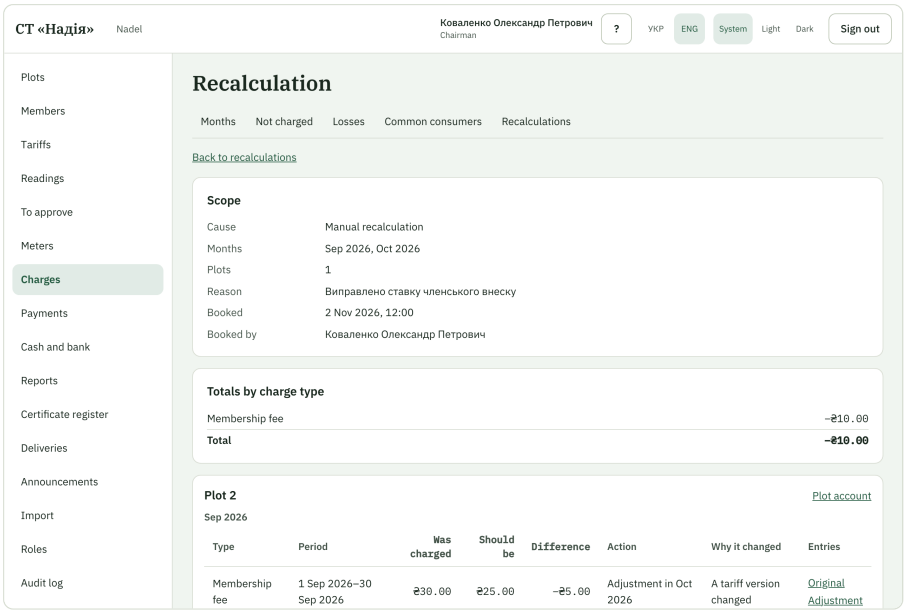
An adjustment of a closed month stands on the plot account as a separate row, for example "Adjustment for 09.2026: −5.00 ₴", with a reason, the links "**Original charge**" and "**Recalculation**" and the mark "**Adjustment**". The tick "**Adjustments only**" leaves only such rows.

## Worked example

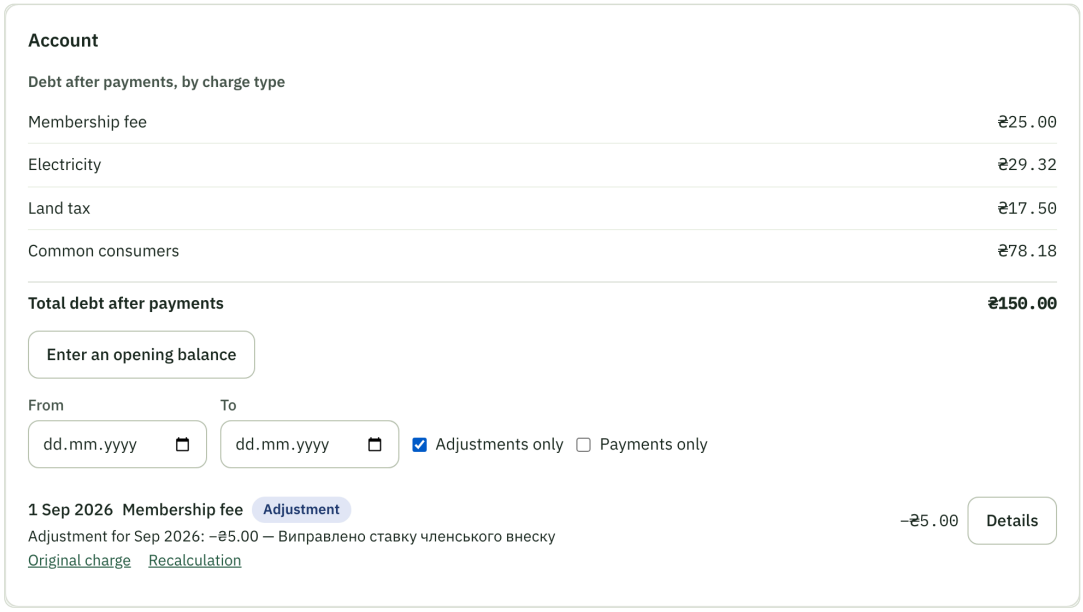
The recalculation of plot 2 in the demo: the fee rate 60 → 50 ₴ per sotka per year.

1. In the log click "**Open**".
2. In the block "**Plot {number}**" find the two months. In closed September: was 30.00, should be 25.00, difference −5.00, the action is an adjustment in October.
3. In open October: a reversal of −30.00 and a new charge of 25.00.
4. Click "**Plot account**" to see the "Adjustment" row on the plot account.

## Screenshots




The recalculation card with rows and links to entries




The September adjustment on plot 2's account

## Related pages

- Recalculations: the log
- New recalculation
- Plot card
- How the system calculates a recalculation

# Meters

## What the screen is for

This is the register of all the cooperative's meters. Each has a resource icon: a lightning bolt for electricity, a drop for water. From here the chairman registers a new meter, and by opening a meter sees its card with the reading history.

## Who sees it

The chairman and the treasurer see the register: those who have the "**View meters**" permission. Only the one who has both permissions, "**View meters**" and "**Manage meters**", that is, the chairman, can register meters. An electrician with "**Enter readings**" enters readings on the screen "Readings".

## Every element

### The button

**Register a meter** opens the dialog "**Register a meter**" (chairman only).

### Meter kinds

| Kind | What it is |
|---|---|
| **Plot meter** | Belongs to a plot. A plot has at most one working meter of each resource. |
| **Substation** | The meter on the cooperative's transformer, the root of the meter tree. |
| **Quarter meter** | The meter of a group of plots: a street or a quarter. |
| **Common consumer** | A well pump, street lighting, the office. |

The number of substations, quarter meters and wells is not limited.

### Filters

| Filter | What it does |
|---|---|
| **Resource** | "**All resources**", "**Electricity**" or "**Water**". |
| **Kind** | "**All kinds**" or one of the kinds above. |
| **Street** | Search by street; the list updates by itself. |
| **Show removed meters** | Adds removed meters to the list. |

The chosen filters are kept in the page address, so the link can be shared. If nothing matches the filters, the screen says "**Nothing found**", and if there are no meters at all: "**No meters yet**".

### Columns

**Plot**, **Name** (a link to the card), **Serial number**, **Kind**, **Resource**, **Last reading** (value and date) and **Status**: "**In service**" or "**Removed**".

### The dialog "**Register a meter**"

| Field | What to enter |
|---|---|
| **Kind of meter** | The meter kind. |
| **Resource** | Electricity or water; for substations, quarters and consumers only electricity. |
| **Plot** | For a plot meter: find the plot by number. The button "**Change plot**" lets you choose another. Without a plot: "**Pick the plot of the meter.**". |
| **Name** | For non-plot meters, for example "Свердловина №1". |
| **Serial number** | Required. |
| **Installation date** | The installation date. |
| **Initial reading** | The initial reading, at most three decimal places: "**Up to 3 decimal places**". |
| **Note** | Optional. |

Buttons: "**Save**" and "**Cancel**".

**Error messages**

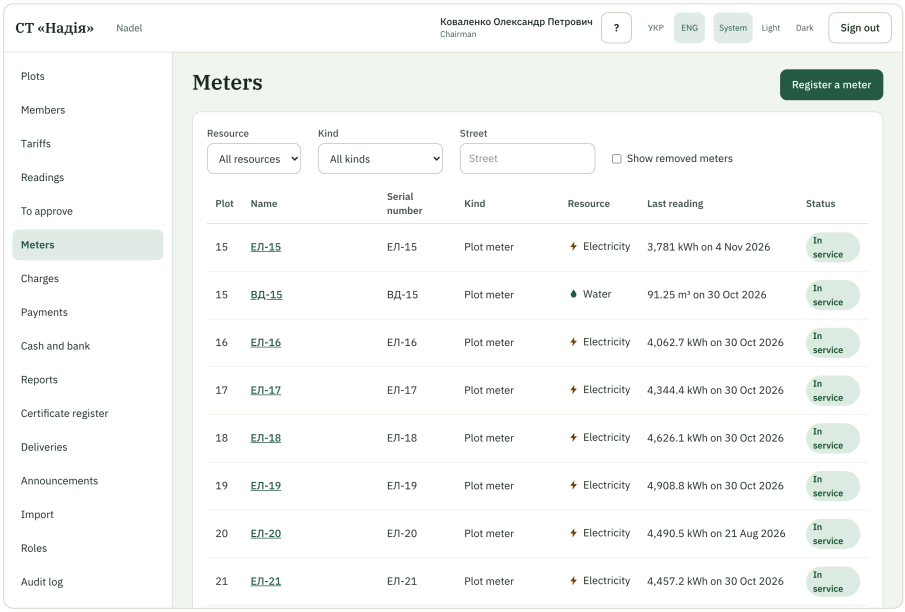
| What is shown | What to do |
|---|---|
| **The plot already has an active meter of this resource** | Do not add a second one: replace the current one on its card. |
| **A reading must be a non-negative number with at most 3 decimal places** | Correct the initial reading. |
| **Pick a date.** | Choose the installation date. |

## Worked example

The demo has 38 meters: the substation "Підстанція №1", quarters "Квартал А" and "Квартал Б", "Свердловина №1" with a pump, "Освітлення вул. Вишнева", plot meters and four water meters.

1. Open "**Meters**" and choose the resource "**Water**" to see the water meters.
2. Clear the filter and tick "**Show removed meters**": the meter ЕЛ-9, removed 30 days ago, appears.
3. Click ЕЛ-9/2 to open its card with the replacement history.

## Screenshots




The meter register

## Related pages

- Meter card
- Meter tree
- Readings: the electrician's round
- How the system calculates meters and the tree

# Meter card

## What the screen is for

This is one meter: its data, the link to the previous or next meter in case of a replacement, and the history of all readings. Here the chairman edits, replaces, removes and deletes the meter and corrects readings.

## Who sees it

The chairman and the treasurer see the card (the "**View meters**" permission). The buttons for changing a meter belong to the one who also has "**Manage meters**", that is, the chairman. A reading can be corrected or deleted with the "**Enter readings**" permission. But if the reading has already been charged, only the one who can recalculate ("**Recalculate charges**") does it. An electrician without that permission sees: "**The reading has already been charged — ask the chairman**", and nothing changes.

## Every element

At the top are the link "**Back to meters**", the title "Meter ..." and the mark "**In service**" or "**Removed**".

### The meter data

Kind, resource, plot (a link to the plot card), serial number, name, installation date, initial reading, and for a removed meter the removal date and the final reading. Also the last reading and the note.

If the meter was replaced, the link is shown: "**Replaced meter {serial}: removed {date}, final reading {value}**" (on the new one) and "**Replaced by {serial}: removed {date}, final reading {value}**" (on the old one). Both meters keep their history.

### Buttons

| Button | What it does |
|---|---|
| **Edit** | The dialog "**Edit meter**": change the serial number, the name, the installation date or the note. |
| **Replace meter** | The dialog "**Replace meter**", see below. Working meters only. |
| **Remove** | The dialog "**Remove the meter**": the removal date and the final reading. The meter stops working from that date, the readings stay. |
| **Delete** | The question "**Delete the meter?**". Only a meter without readings, created by mistake, can be deleted. This cannot be undone. |

### Replacing a meter

One action: the replacement date, the final reading of the old meter, the serial number and the initial reading of the new one. The old meter is removed on that day, and the new one is installed the same day. Fields: **Replacement date**, **Final reading of the old meter**, **Serial number of the new meter**, **Initial reading of the new meter**. The button "**Replace**". Without a serial number: "**Enter the serial number.**".

Did the meter roll over zero (for example, from 99 999 to 00 012)? This is entered as replacing the meter with itself. The button "**Fill in as a wrap-around**" fills in the same serial number, the final reading before the rollover and 0 as the initial reading. The hint in the dialog: "**Did the counter wrap around (for example from 99 999 to 00 000)? Enter it as a replacement by itself: the same serial number, the final reading before the wrap**"

### The block "Readings"

The table of readings: **Date**, **Reading**, **Status**, **Source**, consumption for the interval and actions. A reading's state: "**counted**", "**pending**", "**rejected**" (with the rejection reason) or "**superseded**". The source: "**Entered**", "**Import**" or "**From the owner**". A reading for the same date replaces the previous one; the old one stays in the history marked "replaced". An empty history: "**There are no readings yet**".

Each reading has actions:

- "**Correct**" opens the dialog "**Correct the reading**", the button "**Save the correction**";
- "**Delete reading**" asks for confirmation "**Delete the reading?**": the reading is deleted from the history, and the consumption for the intervals is recalculated.

While a reading has no charge, correcting it is simple. After a charge:

- the chairman sees the preview step "**Recalculation for the reading**" with a reason: what exactly will change in the charges, and after booking "**Reading corrected, recalculation booked**";
- an electrician without the right to recalculate sees "**The reading has already been charged — ask the chairman**".

Reading errors: "**The reading is lower than the previous one or higher than the next one**", "**The reading date cannot be in the future**", "**The date is outside the meter's working period**", "**A reading must be a non-negative number with at most 3 decimal places**".

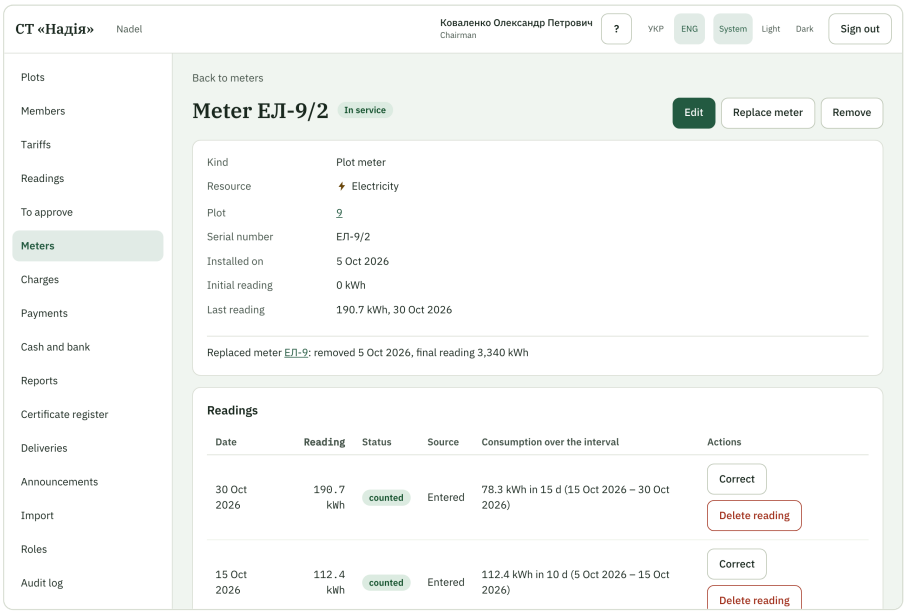
## Worked example

Plot 9 in the demo: the old meter ЕЛ-9 was removed 30 days ago with a final reading of 3,340, and the new ЕЛ-9/2 was installed the same day from zero.

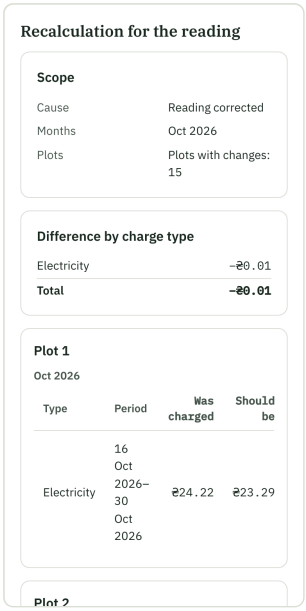
1. Open ЕЛ-9/2 from the register. In the data block you see "Replaced meter ЕЛ-9...".
2. In the history find a reading and click "**Correct**". Change the value to 195 and click "**Save the correction**".
3. Both readings of this meter are already charged, so the recalculation preview step appears. Nothing is saved until you enter a reason and click "**Book the recalculation**".

A replacement splits the consumption into separate intervals, and each is charged at the tariff of 4.32 ₴ per kWh. In the demo: the old meter from its last reading of 3,250 to the final 3,340 gives 90 kWh, that is, 90 × 4.32 = 388.80 ₴. The new meter from 0 to the reading of 112.4 gives 112.4 × 4.32 = 485.57 ₴, and from 112.4 to 190.7 gives 78.3 kWh, that is, 78.3 × 4.32 = 338.26 ₴.

## Screenshots




The meter card with the reading history




The preview step after correcting a charged reading

## Related pages

- Meters
- Readings: the electrician's round
- Readings for approval
- New recalculation

- How the system calculates readings and consumption

# Meter tree

## What the screen is for

The tree shows how electricity flows through the cooperative: a substation, under it quarter meters, common consumers and plot meters. The system calculates losses by it: the difference between what a node showed and what the meters under it read. Here the chairman builds the tree and sees consumption for a period.

## Who sees it

The screen opens from the cooperative settings, the item "**Meter tree**". It is seen by those who have the "**View meters**" or "**Manage meters**" permission. The button "**Move**" is seen by the one who has both, that is, the chairman. The tree has electricity meters only: there are no water meters in it.

## Every element

At the top is the link "**Back to settings**".

### The period

The fields "**From**" and "**To**" set the period for which consumption is shown. By default it is from the start of the current month to today. If the end is earlier than the start: "**The end of the period cannot be before its start.**".

### The tree

Each node is a frame with:

- an electricity icon and the meter name;
- the kind ("**Substation**", "**Quarter meter**", "**Plot meter**", "**Common consumer**");
- the plot number for a plot meter;
- "**removed {date}**" if the meter is removed;
- consumption for the period: "**Consumption: {value} {unit}**";
- the button "**Move**" (for a working meter).

Under a node, indented, are its children. If a meter changed its place in the tree during the period, you see the consumption for each segment under it.

If there is no reading on the border of the period, the system calculates the value in proportion to the days between neighbouring readings and marks it "**calculated**". If there is still no reading after the end of the period: "**no reading after {date}**"; if there are no readings at all: "**no reading**".

An empty tree: "**There is no tree yet. Move a meter under a substation to build it.**"

### The dialog "Move meter {label}"

The button "**Move**" moves the meter under another node from the given date: past periods are calculated by the old tree.

| Field | What to enter |
|---|---|
| **Parent meter** | The node the meter is moved under, or "**Outside the tree**". |
| **Effective from** | The date from which the new place applies. Without a date: "**Pick a date.**" |

Buttons: "**Move**" and "**Cancel**".

### Errors

| What is shown | What to do |
|---|---|
| **A quarter meter cannot be under another quarter meter** | Choose a substation. |
| **This kind of meter cannot be under the chosen parent** | Choose another parent. |
| **The parent meter is already removed** | This parent is already removed, choose a working one. |
| **The date is before the meter's latest change in the tree** | Choose a later date. |
| **The meter has other meters under it — move them first** | Move the children before removing. |

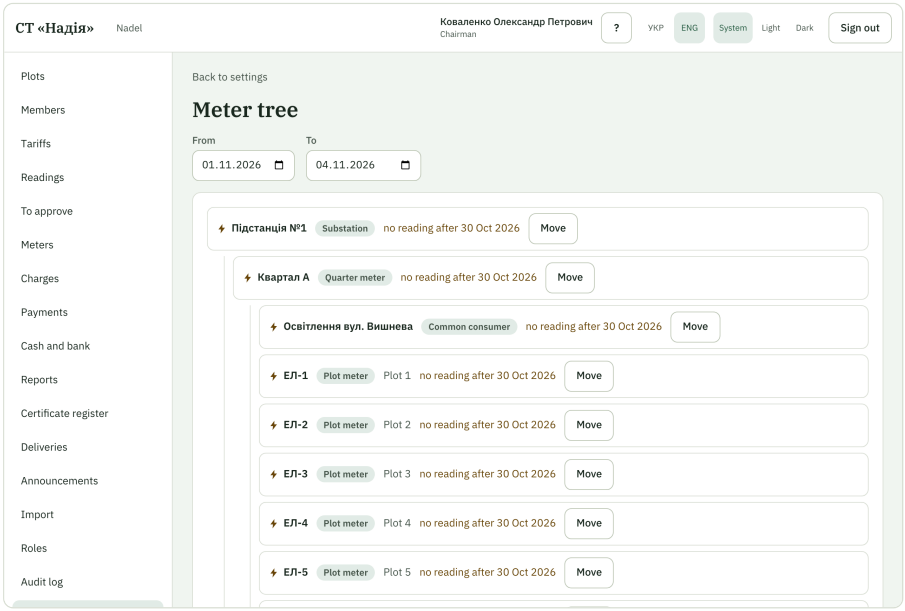
Moving recalculates nothing. For past charges a separate recalculation is needed; the screen reminds you about it.

## Worked example

In the demo the substation "Підстанція №1" has the children "Квартал А", "Квартал Б" and "Свердловина №1". The lighting "Освітлення вул. Вишнева" hangs under quarter A, and the plot meters under their quarters.

1. Open "Cooperative settings" and the item "**Meter tree**".
2. Choose the period "From" 21 days ago, "To" 7 days ago: there are readings in this range.
3. Look at the consumption of the substation and its children. The difference is the losses.
4. To move a plot meter under another quarter, click "**Move**", choose the new node and the date.

## Screenshots




The meter tree with consumption for a period

## Related pages

- Meters
- Network losses
- Cooperative settings
- How the system calculates meters and the tree

# Readings: the electrician's round

## What the screen is for

Here meter readings are entered. Readings are not tied to months: the electrician may come in a month, in three or in a few days. A reading is the meter's value on a date, and the system calculates consumption between neighbouring readings. The screen is convenient on a phone: a card per plot, with a row for each meter.

## Who sees it

The screen is seen by the one who has the "**Enter readings**" permission: the chairman and the electrician (the special role "Electrician"). The treasurer does not see this screen. If a reading has already been charged, only the one who also has "**Recalculate charges**", that is, the chairman, can correct it.

## Every element

### Filters at the top

| Element | What it does |
|---|---|
| **Round date** | The date for all readings of the round. It already holds today's date, with the hint under the field: "**Reading date is today, {date}. Change it if the meter was read on another day**". For a single meter the date can be changed in its row. A future date cannot be entered. |
| **Resource** | "**All resources**", "**Electricity**" or "**Water**". The choice is remembered in this browser. |
| **Street** | Search by street. |
| **No reading since** | Shows the meters that have not been read for a long time. After switching it on, the field "**Date since which there is no reading**" appears. |

If nothing matches: "**No meters match these filters**", and if there are no meters for the round at all: "**There are no meters to read**".

### Cards and rows

Each card is a plot with its meters ("**Plot {number}**") or a single non-plot meter. In a meter's row:

- the resource icon, the serial number (a link to the meter card) and the mark "**Saved**" after saving;
- "**Last reading: {value} {unit}, {date}**": the last reading and its date;
- the "Reading" field with the unit (kWh or m³);
- the field "**Reading date for this meter**";
- the button "**Save**" saves this row's reading; it says "**Saving...**" while saving.

Next to the date field there is the **Add a photo** button: it adds a photo of the meter to this reading (on a phone the camera opens). A thumbnail shows above the button, **Replace** changes the photo, **Remove** removes it. The rule "**Meter photo required for members**" applies to members only, so the electrician and the board add a photo if they wish.

After saving, the consumption for the interval appears at once: "**Interval {from} – {to}: {consumption} {unit} in {days} d**". The Enter key in a saved row moves the cursor to the next card.

### Warnings and errors

| What is shown | What to do |
|---|---|
| **Check the reading: consumption is much higher than usual** | The reading is saved, but the consumption is much higher than usual. Check the digits; if wrong, correct the reading on the meter card. |
| **The reading is lower than the previous one or higher than the next one** | The reading is not accepted. Check the value: it cannot be less than the previous reading or more than the next one. |
| **The reading date cannot be in the future** | Choose a date no later than today. |
| **The date is outside the meter's working period** | The date is outside the period the meter worked. |
| **A reading must be a non-negative number with at most 3 decimal places** | Correct the number. |
| **A reading for this date already existed — it is marked as superseded.** | Not an error: a reading for the same date replaces the previous one, and the old one stays in the history marked "replaced". |
| **The reading has already been charged — ask the chairman** | The reading has already been used in a charge. Ask the chairman to correct it. |

**IMPORTANT**

This screen does not take photos. But if you attach photos from a phone elsewhere (for example, a receipt for an expense), note that HEIC photos from an iPhone are accepted in Safari only. In another browser on the iPhone open "Settings → Camera → Formats" and choose "Most Compatible", then photos will be saved as JPEG.

**Water and electricity**

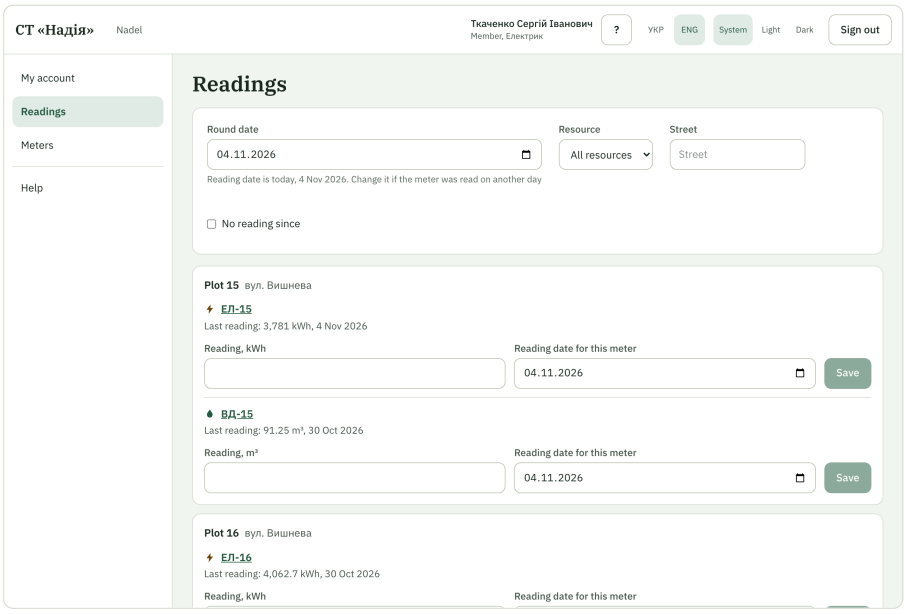
For water, each plot pays only for the days it is connected to the well. A tariff per cubic metre also needs a water meter. The connection is entered on the plot card.

## Worked example

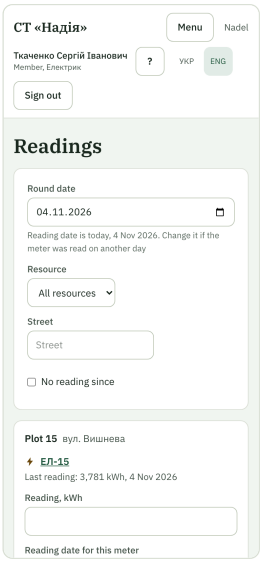
1. Open "**Readings**". The round date is already today.
2. Choose the resource "**Electricity**" and, if needed, a street.
3. On the card of plot 7 you see the last reading and its date. Enter the new reading, for example 205.7, and click "**Save**" in this row.
4. If the previous reading was 112.4 of 22.09 and the new one is 205.7 of 04.10, the consumption of 93.3 kWh over 12 days (23.09–04.10) appears under the row. The electricity charge: 93.3 × 4.32 = 403.06 ₴.
5. If the tariff changed inside the interval, the system splits the kWh by days. Readings 1,000 (01.10) and 1,300 (31.10), tariff 4.32 until 16.10 and 5.00 from 17.10: 15 days each, 150 × 4.32 + 150 × 5.00 = 648.00 + 750.00 = 1,398.00 ₴.

A reading lower than the previous one is not accepted by the system: "**The reading is lower than the previous one or higher than the next one**".

## Screenshots




The Readings screen: a card per plot, a row per meter




The same screen on a phone

## Related pages

- Meters
- Meter card
- Readings for approval
- How the system calculates readings and consumption
- Water and connection

# Readings for approval

## What the screen is for

A cooperative member can submit a reading of their own meter from their account. Such a reading waits here. The chairman approves it or rejects it with a reason that the owner will see. Until a reading is approved, it is not used in calculations.

## Who sees it

The screen is seen by the one who has the "**Approve readings**" permission: by default the chairman. A member submits a reading from their account, with the button "**Submit a reading**" in the block "**My meters**": see How to submit a meter reading.

## Every element

The heading is "**Readings to approve**". If there is nothing to wait for: "**No readings are waiting for approval**".

### A reading card

Each submitted reading is a card:

- the plot ("**Plot {number}**"), the resource and the serial number of the meter; the serial number leads to the meter card;
- "**Reading {value} {unit} on {date}**": what the owner submitted;
- "**Once approved: {consumption} {unit} over {days} days ({from} – {to})**": what the consumption will be if approved;
- a thumbnail of the meter photo, if the owner added one: press it to open the photo in a large window and check the number; the photo stays with the reading and cannot be deleted;
- warnings, for example "**Check the reading: consumption is much higher than usual**".

If the reading does not fit between the accepted readings, the card says: "**The reading does not fit between the counted readings, so it cannot be approved as it is**". Then it cannot be approved without a correction, so reject it with a reason.

### Buttons

| Button | What it does |
|---|---|
| **Approve** | Approves the reading: it becomes accepted, and a charge for the interval from the previous reading appears at once. |
| **Reject** | Opens the dialog "**Reject the reading**". |

In the rejection dialog the field "**Reason**" is required, up to 500 characters. The hint: "**The owner will see the reason. At most 500 characters**". Without a reason you get "**Enter a reason**". The button "**Reject reading**" rejects the reading. The owner sees the reason, and in the meter history the reading has the state "**rejected**".

### Errors

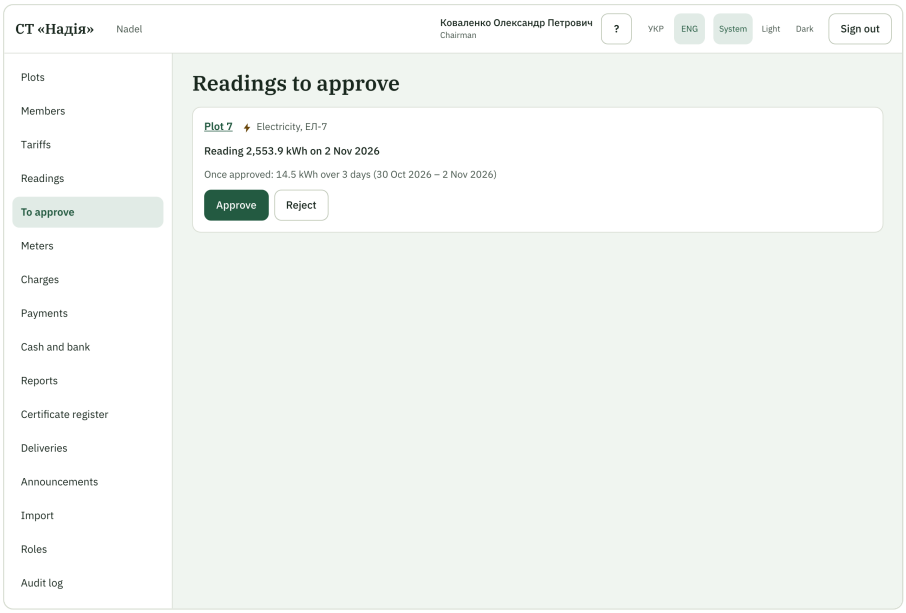
| What is shown | What to do |
|---|---|
| **The reading is lower than the previous one or higher than the next one** | The reading does not fit the sequence. Reject it with a reason. |
| **The reading has already been charged — ask the chairman** | The reading is already used. Seen by someone who cannot recalculate: contact the chairman. |

> **IMPORTANT**
> This screen has no recalculation step. If a reading has already been approved and charged, correct it on the meter card: the preview step is there.

## Worked example

1. A plot owner submitted a meter reading from their account.
2. The chairman opens "**To approve**" and sees a card: "Reading 205.7 kWh on 04.10", under it "After approval: 93.3 kWh over 12 days" (the previous reading is 112.4 of 22.09).
3. Everything looks plausible, so the chairman clicks "**Approve**". The consumption of 93.3 kWh × 4.32 = 403.06 ₴ goes into the charges.
4. If the value were clearly wrong, the chairman would click "**Reject**", give the reason "Check the last digit" and confirm. The owner will see the reason and submit the reading again.

## Screenshots




Readings for approval: approve or reject

## Related pages

- Readings: the electrician's round
- Meter card
- How to submit a meter reading
- How the system calculates readings and consumption

# Payments

## What the screen is for

Here you see all the payments entered in the cooperative: cash at the till and bank transfers. Payments go from newest to oldest. From here you can enter a new payment or open the payment card.

A payment is never deleted. If it is reversed, it stays in the list with the mark "**Reversed**".

## Who sees it

The screen is seen by those who have the right "**View payments**": the chairman and the treasurer. The "**Enter payment**" button is seen by those who also have "**Enter payments**" (the chairman and the treasurer). Only a person with "**Reverse payments**" can reverse a payment; by default that is the chairman.

A member does not see this screen. A member sees their own payments in the account, and only for their own plots.

## Every element

### The "**Enter payment**" button

On the right in the header. It opens the payment entry form. Without the right to enter payments the button is not there.

### Filters

The filters are above the table. You can combine them. Whatever you choose stays in the page address, so you can send a link to a filtered list to a colleague.

| Filter | What it does |
|---|---|
| **From** and **To** | Show payments by the date the money arrived. You can enter just one of the dates. |
| **Billing month** | Shows the payments of one accounting month. The accounting month may differ from the receipt date if the money arrived in an already closed month. The option "**All months**" removes the restriction. |
| **Source** | "**Cash**", "**Bank**" or "**All sources**". |
| **Plot** | The field "**Plot number**": start typing a number and choose the plot. The button "**Clear plot**" clears the choice. |

When no payments match, the screen says "**No payments match these filters.**". If there are no payments at all, it says "**There are no payments yet.**".

### The payment table

| Column | What is in it |
|---|---|
| **Receipt / document** | The receipt number for cash ("**№ {number}**") or the bank document number. If there is neither, the date is shown. It is a link to the payment card. |
| **Date received** | The day the money arrived. |
| **Plot** | The plot number. The link opens the plot card. |
| **Payer** | Who paid, if entered. |
| **Source** | **Cash** or **Bank**. |
| **Amount** | The payment amount. |
| **Allocated** | How much of this amount is allocated to charges. |
| **Prepayment** | How much is left over the debts. |

Marks may stand by the number.

- "**Reversed**": the payment is reversed, its money is taken off the charges.
- "**One payment, plots: {count}**": the payer brought one sum for several of their plots. Each plot has its own row, and the card shows all the parts.

At the bottom there is a page switch: 20 payments per page.

## Worked example

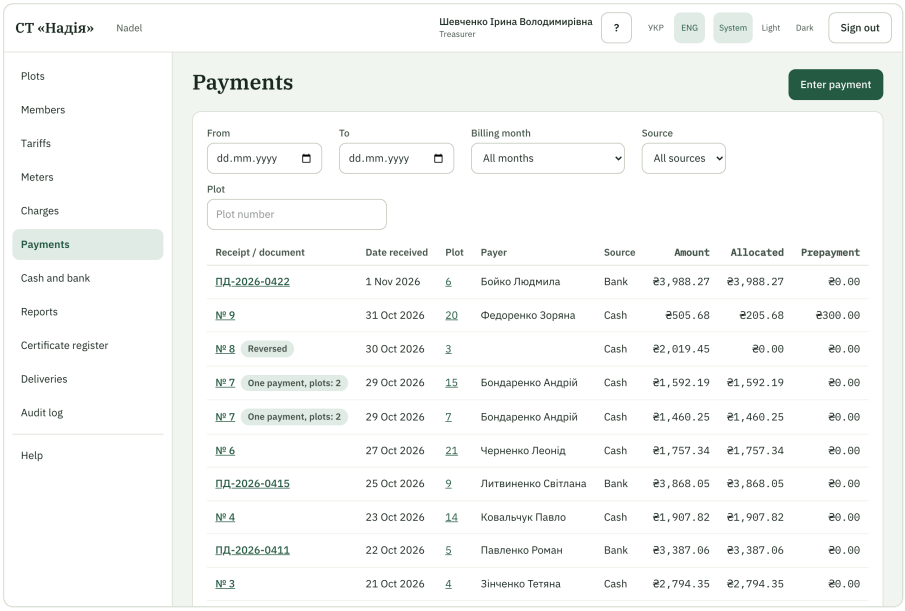
The treasurer wants to find all cash payments of plot 9 for the current month.

1. Open "Payments".

2. In the filter "**Source**" choose "**Cash**".
3. In the filter "**Plot**" type 9 and choose the plot.
4. In the filter "**Billing month**" choose the current month.
5. Press the receipt number in the row you need: the payment card opens.

The payment of plot 9 in the demo data is a bank payment and is paid only in part, so with the source "**Bank**" this payment is in the list, and with "**Cash**" it is not.

## Screenshots




The payment list with filters

## Related pages

- Enter a payment
- Payment card: reversal and reallocation
- How the system allocates payments
- Cash and bank

# Enter a payment

## What the screen is for

The treasurer records money received here: cash at the cooperative's till or a bank transfer. The system allocates the payment to the plot's debts. After that, the debt, the plot colour on the map and the reports take this payment into account.

## Who sees it

The chairman and the treasurer see it: they have the right "Enter payments". Other roles have no entry for entering payments. If you are not among them and need to enter a payment, ask the chairman to give you this right on the Roles page.

## Every element

Open the screen with the **Enter payment** button in the payment list or on a plot card. From a plot card the plot is already chosen.

### Form fields

| Field | What to enter |
|---|---|
| **Plot** | The plot number. Common land and an archived plot do not accept payments. The **Change plot** button lets you choose another plot. |
| **Source** | **Cash** or **Bank**. |
| **Date received** | Today by default. You can enter another date, but the warning "**The date is not today**" appears. A future date is not accepted. |
| **Amount** | More than zero, at most two decimals, for example 1250.50. |
| **Payer** | Optional. The plot owners come first (**Owners of the plot**), then **Another member**. The option **Not specified** leaves the field empty. If the plot has exactly one current owner, that owner is preselected as the payer; with several owners, choose the payer yourself (owners are listed first). |
| **Note about the payer** | Optional, up to 200 characters, for example the name of a relative who brought the money. This is personal data, so the note is stored encrypted and only a person who also has the right "View members" can see it. |
| **Bank document number** | Only for the source "**Bank**". Optional, up to 64 characters. |

### The proposed allocation

As soon as the plot and the amount are known, the table "**Allocation of the payment to charges**" appears under the form. It lists the plot's unpaid charges: **Charge**, **Period**, **Open** and the **Allocate** field. The **Prepayment** row shows what is left over the debts.

The system pays the oldest debt first. If debts are from the same day, it follows the order of charge types set in the cooperative.

- You can change the amount in any row.
- The **Add a charge** button adds another open charge of this plot.
- At the bottom you see how much is allocated: "**Allocated** X **of** Y" (X allocated out of the payment amount Y).
- If the amount in a row is larger than the charge balance, the line "More than the remaining {amount}" appears above the amount field with the balance.

> **IMPORTANT**
> When automatic allocation of the prepayment is on in the settings, rows you leave empty do not become a prepayment: the free part of the payment goes straight to the oldest debts of the plot. The note "**The remainder will be placed automatically on the oldest debts**" then stands under the "**Prepayment**" row. A prepayment remains only when the plot has no open debts.

### The mode "Several plots of one payer"

If a member brings one sum for several of their plots, choose the mode "**Several plots of one payer**", then enter the payer and the amount. The form shows the payer's plots with their debts in the table "**Parts of the payment by plot**" and proposes parts in proportion to the debts. Cents that do not divide go to the plots with the largest balances. You can change any part, but together they must equal the payment amount. Each part becomes a separate payment of its plot; a cash group has one receipt number.

### Saving

The **Save** button records the payment. A payment is never deleted or edited: a mistake is corrected by a reversal on the payment card.

- For cash the screen shows "**Payment saved**" and the receipt number, for example "Receipt № 42", and the **Another payment** button. For a bank payment it shows the saved payment.
- Receipt numbers run through the whole cooperative, with no gaps and no repeats, even when two treasurers enter cash at the same time. The receipt is internal; it is not a fiscal receipt.
- The **Print receipt** button downloads the receipt PDF (cash only). It is also on the payment card. A bank payment has no receipt.

- The **Open the payment card** button opens the payment card.
- If payment notifications are on in the settings, the plot owners who get messages in Telegram or by e-mail learn about the payment and the remaining debt. More on the Delivery page.

**Error messages**

| What is shown | What to do |
|---|---|
| **No payment is taken for common land** | Payments for common land are not entered. Choose a plot with an owner. |
| **The plot is archived** | The plot is archived. A payment cannot be entered for it. |
| **Enter the amount as a number, for example 1250.50 (at most two decimals)** | Correct the amount: a number with at most two decimals. |
| **The data changed after the preview. Review it again before saving.** | Someone changed the plot account between the preview and saving. Review the allocation again and save. |
| The amount in a row is more than the charge balance | The line "More than the remaining {amount}" stands above the amount field. Reduce the amount to the balance. |

If any allocation row is wrong, the table stays on the screen, a message stands by the row, and you can save as soon as there are no errors. The total allocated cannot be more than the payment.

**A payment for a past month**

If money arrived on 31 January and you enter it on 3 February, and January is already closed, the payment keeps the date 31.01 but counts in February. January totals do not change.

## Worked example

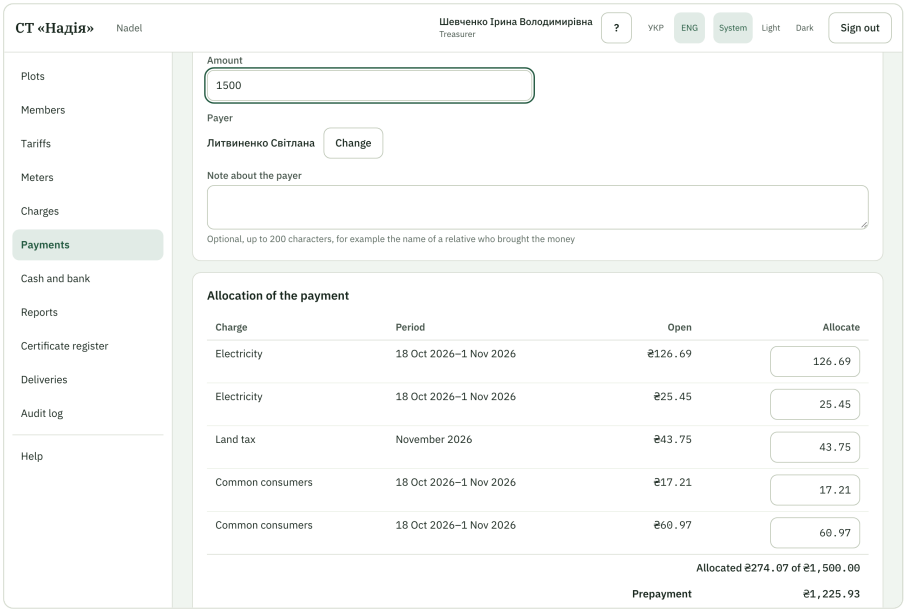
A plot has two debts: the membership fee for January 600.00 and electricity for February 250.00. The order of types in the cooperative: membership fee, then electricity.

1. Open the plot card and press **Enter payment**.
2. Choose the source "**Cash**" and enter the amount 700.00.
3. Look at the proposal: fee 600.00 and electricity 100.00, prepayment 0.00. The oldest debt is paid first.
4. Press **Save**. The screen shows the receipt number.

Afterwards the electricity balance is 150.00, and this is the whole debt of the plot. Had the payment been 1,000.00, both debts would be paid in full and 150.00 would remain as a prepayment: the plot is green even though a prepayment exists.

The same screen on demo plot 9: open its card, press **Enter payment**, enter the amount 100.00. All hundred hryvnias go to the oldest charge if it is more than 100.00; in the table "**Allocation of the payment to charges**" the charge row has 100.00 and "**Prepayment**" is 0.00. If you enter more than the whole debt of the plot, the excess becomes a prepayment.

## Screenshots




The payment form with the proposed allocation of 1,500.00

## Related pages

- Payment list
- Payment card: reversal and reallocation
- Plot card
- How the system allocates payments

# Payment card: reversal and reallocation

## What the screen is for

The card shows one payment: where the money came from, what it is allocated to and who entered it. This is where a payment is corrected: **reverse a payment** that was entered by mistake, or reallocate the money between charges. The payment record itself is never deleted or edited.

## Who sees it

The card is seen by those who have the right "**View payments**". The buttons "**Reallocate**" and "**Allocate prepayment**" are seen by those who have "**Enter payments**" (the chairman and the treasurer). The button "**Reverse**" is seen only by those who have "**Reverse payments**": by default the chairman. A treasurer who made a mistake asks the chairman.

The payer note is seen only by those who also have the right "**View members**".

## Every element

### The title and buttons

The title is "**Receipt № {number}**" for cash, or the bank document number (or "**Bank payment**") for a bank payment. If the payment is reversed, the mark "**Reversed**" stands next to it.

| Button | When it is there | What it does |
|---|---|---|
| **Print receipt** | Only for cash with a receipt number | Downloads the receipt PDF. For a payment split between plots the receipt is one, with all the plots. The receipt (A5, half of A4) holds the requisites of the cooperative, the number, the date, the plot, the payer, the amount in figures and in words, what the money was allocated to ("Allocation as of" the print date) and the name of the person who accepted the money, with a line for a signature. The note about the payer is printed only for users who may see member data. The receipt of a reversed payment carries the mark "REVERSED" with the date. It is an internal receipt, not a fiscal one. |
| **Reallocate** | The payment is not reversed and can be reallocated | Moves money from some charges to others. |
| **Allocate prepayment** | Automatic allocation of the prepayment is off, the payment has a prepayment, and the plot has an open charge | Allocates the prepayment to open charges. |
| **Reverse** | The payment can be reversed | Opens the reversal window. |
| **Back to payments** | Always | Returns to the payment list. |

### Payment data

A table with the fields: **Plot**, **Date received**, **Billing month**, **Source**, **Bank document** (for bank), **Payer**, **Note about the payer**, **Amount**, **Allocated** and **Prepayment**.

If the payment is reversed, the rows **Reversed**, **Reversal month** and **Reversal reason** are added. At the bottom of the page you see who entered the payment and when.

### Allocation

The allocation has two parts.

- "**Allocated now**": which charges the payment is allocated to now (charge, period, amount). If nowhere yet, it says "**The payment is not allocated to any charge yet**".
- "**Allocation history**" (expands): every allocation and every return from a charge with date, amount, author and reason. Reasons for a return: "**Charge reversal**", "**Adjustment**", "**Payment reversal**", "**Reallocation**". Where an allocation came from: "**Entry**", "**Reallocation**", "**Prepayment allocation**", "**Automatically from prepayment**". Records of the system are signed "**System**".

### Other parts of the payment

If the payer brought one sum for several plots, there is a table "**Other parts of this payment**": the plot and the amount of each part. Each part is reversed separately, the rest stay.

### Reverse a payment

Press "**Reverse**". In the window "**Reverse the payment**" enter the "**Reason**" (up to 500 characters, for example "Wrong plot") and press "**Reverse**". Without a reason you see "**Enter a reason**". The button "**Cancel**" closes the window without changes.

What happens next:

- The system removes the whole allocation of the payment, and the charges it paid become debts again.
- The payment record stays; in the list it has the mark "**Reversed**". A reversed cash payment keeps its receipt number.
- A line of the opposite amount appears in the cash or bank journal, dated the day of the reversal. The cash may become negative: then the cash card asks you to do a cash count.
- Owners who receive messages learn about the reversal if "notify about payments" is on in the settings.
- A payment of a plot in the archive can be reversed.
- A payment is reversed only once.

> **IMPORTANT**
> A reversal cannot be undone. If you reversed a payment by mistake, enter it again as a new payment. A closed month does not change because of this: the reversal goes to the nearest open month.

#### Reallocate

The button "**Reallocate**" opens the window "**Reallocate the payment**". The window shows what is allocated now ("**Allocated now**"), and below it a new proposal ("**New allocation**") that you can change the same way as when entering. The button "**Reallocate**" removes the old allocation and records the new one in one step. It goes to the change log. This is how money is moved between charges.

When automatic allocation of the prepayment is on in the settings, empty rows do not create a prepayment while the plot has open debts: the free sum goes to the oldest debts.

#### Allocate a prepayment

A plot's prepayment is money entered for the plot but not allocated to any charge. If automatic allocation of the prepayment is on (it is by default), the prepayment goes to new charges by itself. If it is off, the prepayment waits until the treasurer allocates it. The button "**Allocate prepayment**" opens the window "**Allocate the prepayment**" with a proposal over the open charges; you can change it and confirm with "**Allocate**". Turning automatic allocation on does not allocate the old prepayment at once: it goes to the plot's next charges.

#### Error messages

| What is shown | What to do |
|---|---|
| **Enter a reason** | Write the reason for the reversal. |
| **The payment has already been reversed** | Refresh the page: the payment is already corrected. |
| **The data changed after the preview. Review it again before saving.** | Someone changed the plot account. Review the new allocation and save. |

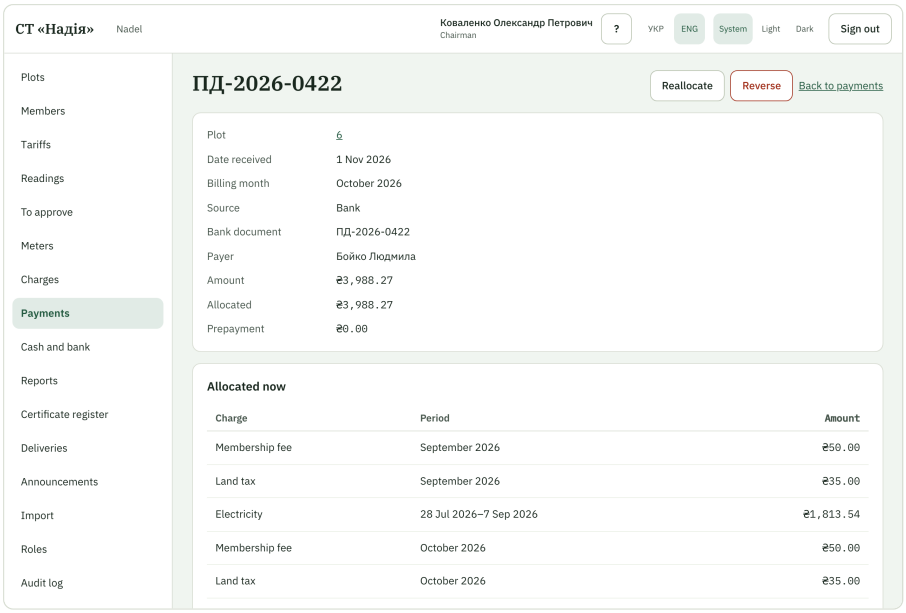
## Worked example

The treasurer mistakenly entered 100.00 ₴ in cash for plot 9 instead of plot 19. He asks the chairman.

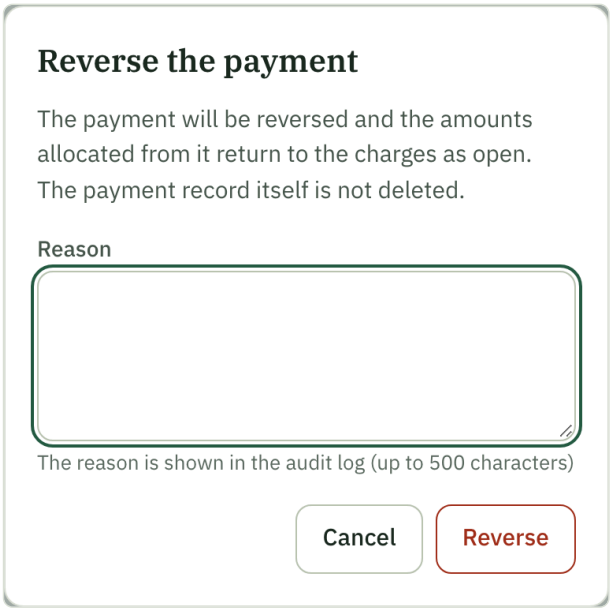
1. The chairman opens "Payments" and the card of this payment (the receipt is, say, № 42).
2. He presses "**Reverse**", writes the reason "Wrong plot" and confirms.
3. The charges of plot 9 that this payment closed are a debt of 100.00 ₴ again. The payment stays in the list with the mark "**Reversed**".
4. The treasurer enters 100.00 ₴ for plot 19 as a new payment. The new receipt has the next number, for example № 43.

Had the money been entered for the right plot but against the wrong charge, it would be enough to press "**Reallocate**": the payment stays, only what it closes changes.

## Screenshots




The payment card with the allocation and buttons




The payment reversal window with a reason

## Related pages

- Payment list
- Enter a payment
- Cash and bank
- How the system allocates payments
- Delivery

# Cash and bank

## What the screen is for

The system knows not only how much members owe and pay, but also where that money is: in the treasurer's till or in the bank account, and what it was spent on. This screen shows both accounts, their balances and the movement journal. It does not change members' debts: it shows where their money went.

The cooperative has exactly two accounts: "**Cash**" (cash held by the treasurer) and "**Bank account**". There is no second till or second bank account.

## Who sees it

The screen is seen by those who have the right "**View cash and bank**": the chairman and the treasurer. The buttons for entering records are seen by those who have "**Enter cash book entries**" (the treasurer and the chairman). Opening balances, settling differences and reversing records are available only to those who have "**Approve cash book operations**": by default the chairman. Without the right to enter you see only balances and journals.

A member does not see this menu item, and gets a message that there is no access at a direct address. Each of the three rights can be given to a custom role on the Roles page.

## Every element

### Header buttons

| Button | What it does |
|---|---|
| **Expenses** | Opens the expense list. |
| **Cash count** | Opens the cash count. Needs the right to enter. |
| **Hand over to the bank** | The window for handing cash to the bank (below). Needs the right to enter. |
| **Other income** | The window for other income (below). Needs the right to enter. |
| **Expense** | Opens the expense form. Needs the right to enter. |

On a phone the cards stand one under another, and everything works the same.

### An account card

Each account has its own card.

- **Balance.** The large line "**Balance on {date}**" shows the balance for today. Under it you see the opening balance and from which date.
- **Opening balance.** While the account has none, the sum is replaced by "**Enter the opening balance**". The chairman enters it once: the amount and the date "**As of**" (the balance at the start of that day) and presses "**Save the opening balance**". After saving it cannot be changed: a mistake is corrected by a cash count (for cash) or a bank difference (for the bank). A treasurer without the approval right sees a hint but no button.
- **Last cash count**: date and result ("**matches**" or "**difference {amount}**"). The link leads to the count. If a difference is not yet settled, the warning "**Unresolved difference {amount}**" is shown.
- **Months of bank reconciliation** (only in the bank card): a list of months with the mark "**Not reconciled**", "**Reconciled**" or "**Needs a new check**". Press a month to open the bank reconciliation.
- **Warnings.** If the cash went negative, you see "**The cash balance is negative — do a cash count**". For the bank it is "**The bank balance is negative**": an overdraft is the bank's business, the system only warns.

While there is no opening balance, expenses, handing over to the bank, other income, counts and reconciliations for this account are rejected with the message "**Enter the opening balance of the account first.**". Member payments are recorded anyway. Payments dated before the opening balance stay in the journal with the mark "**before the opening balance**" and are not part of the balance.

The balance for a day is the opening balance plus all receipts and minus all expenses from the day of the opening balance to that day inclusive.

### The movement journal

Under each card there is the journal "**Movements**". The filters "**From**" and "**To**" set the period; without them the current month up to today is shown. Above the table you see "**Opening balance of the period**" and "**Closing balance of the period**".

Columns: **Date**, **Type**, **Basis** (receipt number, document, category, reason), **Amount**, **Entered by** and **Actions**.

Kinds of entries: "**Member payment**", "**Expense**", "**Handover to the bank**", "**Other income**", "**Cash count difference**" and "**Bank difference**". A member payment appears here by itself at the moment the payment is saved. A reversed entry and its reversal both stay in the journal: the reversal has the mark "**Reversal**" and the reversed entry has "**Reversed**".

In the actions column:

- by an expense there is the link "**Open the expense**": an expense is reversed on its card;

- the button "**Reverse**" (only with the approval right) is by a handover to the bank, other income, a count difference and a bank difference;
- a member payment is reversed on the payment card, not here. Its reversal appears in the journal by itself.

A reversal is a new line of the opposite amount dated the day you reverse. The entry itself is not deleted. An entry is reversed only once, and a reason is required (up to 500 characters, otherwise "**Enter the reason for the reversal.**"). A reversal of a cash receipt after which the cash would go negative is rejected.

> **IMPORTANT**
> An entry dated in an already closed accounting month keeps its date but counts in the nearest open month, as payments do. Reports by day use the entry date, reports by accounting month use the accounting month.

**The window "Hand cash over to the bank"**

The button "**Hand over to the bank**". Cash goes from the till to the bank account, and two lines appear in the journals: an expense in the cash and a receipt in the bank of the same date and amount.

| Field | What to enter |
|---|---|
| **Date** | Not later than today: otherwise "**The date cannot be later than today.**". |
| **Amount** | More than 0.00: otherwise "**The amount must be greater than 0.00.**". |
| **Bank slip number** | Required, for example "KB-123" (error: "**Enter the slip number.**"). |

The button "**Record the handover**" records the handover, "**Cancel**" closes the window. A handover after which the cash would go negative is rejected with the message "**Not enough cash in the till: balance {balance} on {date}**".

**The window "Other income"**

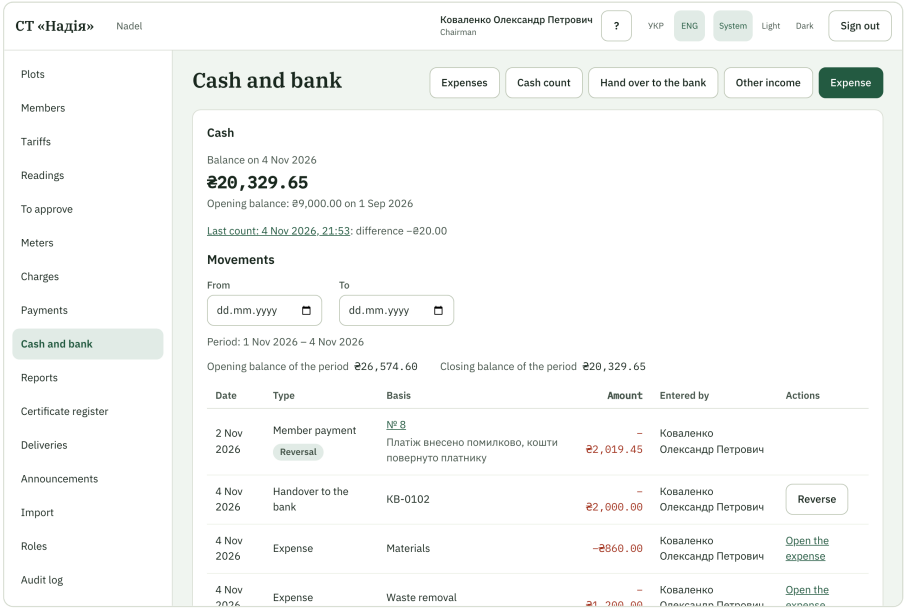
Receipts that are not from members: bank interest, a grant, sale of scrap metal. Fields: "**Account**", "**Date**", "**Amount**", "**Income category**" and "**Description**" (up to 500 characters). The button "**Record the income**". If the category list is empty, the window says "**No income categories. Add them in the settings.**": the chairman adds categories in the settings.

## Worked example

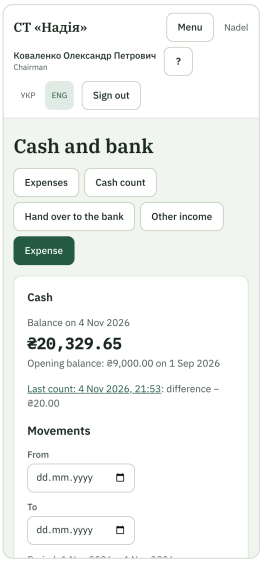
Suppose the till holds 9,000.00 ₴ and the bank account 48,200.00 ₴ (these are the opening balances in the demo cooperative).

1. The treasurer enters a cash payment of 500.00 ₴. A line "**Member payment**" of +500.00 appears in the cash journal, and the cash balance becomes 9,500.00 ₴.
2. He presses "**Hand over to the bank**", enters 3,000.00 and the number "KB-123". The cash gets a line of −3,000.00, the bank +3,000.00. Cash: 6,500.00 ₴, bank: 51,200.00 ₴.
3. He got a digit wrong. The chairman presses "**Reverse**" by the handover and writes a reason. Both lines are reversed, the balances are back to 9,500.00 ₴ and 48,200.00 ₴.

## Screenshots




Cash and bank: two cards with balances and journals




Cash and bank on a phone

## Related pages

- Expenses
- Cash count
- Bank reconciliation
- Payments
- Cash reports
- How the system calculates cash, bank and reconciliation

# Expenses

## What the screen is for

All the cooperative's expenses are listed here: what was paid and to whom, from which account, and how much. From here you open an expense card with receipts and enter a new expense.

## Who sees it

The list is seen by those who have the right "**View cash and bank**": the chairman and the treasurer. The button "**Expense**" is seen by those who also have "**Enter cash book entries**". A member does not see this screen.

## Every element

### Header buttons

- "**Back to cash and bank**" returns to the Cash and bank screen.
- "**Expense**" opens the new expense form.

### Filters

| Filter | What it does |
|---|---|
| **From** and **To** | Show expenses by the expense date. |
| **Account** | "**Cash**", "**Bank account**" or "**All**". |
| **Category** | One expense category or "**All**". |

When no expenses match, the screen says "**No expenses in this period.**".

### The table

Columns: **Date**, **Account**, **Category**, **Paid to**, **Amount** and **Files** (how many receipts are attached). The date is a link to the expense card. A reversed expense is marked "**Reversed**".

An expense is not edited: a mistake is corrected by a reversal on the card.

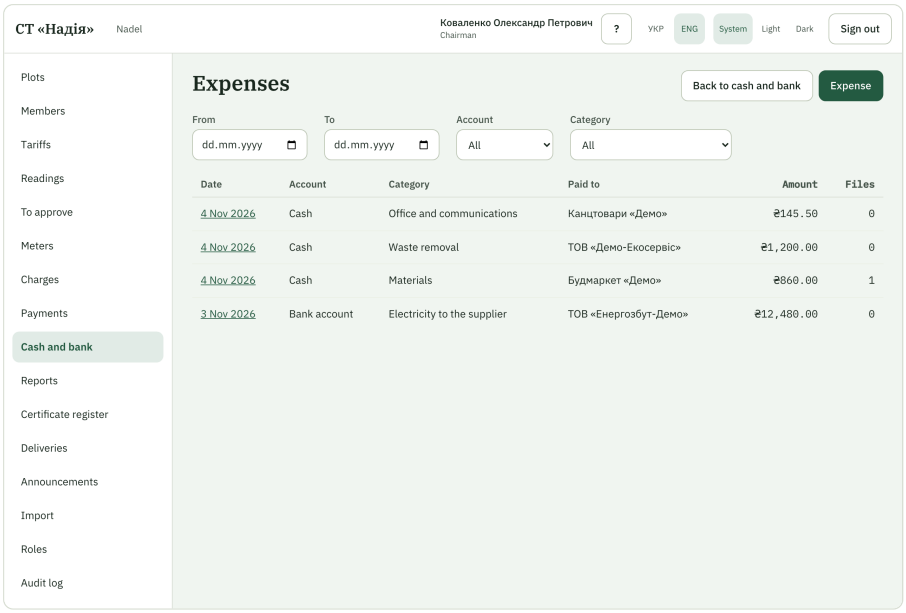
The field "**Paid to**" may contain the name of a private person, so it is personal data: it is seen only by those who have the right "**View cash and bank**".

## Worked example

The demo cooperative has expenses in four categories: "Electricity to the supplier" 12,480.00 ₴ from the bank, "Materials" 860.00 ₴ in cash (cable and terminals for a lamp, a photo of the receipt attached), "Waste removal" 1,200.00 ₴ and "Office and communications" 145.50 ₴.

1. Open "Expenses".
2. In the filter "**Account**" choose "**Cash**": three cash expenses remain.
3. Press the date of the 860.00 ₴ expense: a card with the photo of the receipt opens.

## Screenshots




The expense list with filters

## Related pages

- Enter an expense
- Expense card
- Cash and bank
- Cash reports

# Enter an expense

## What the screen is for

Here you record the money the cooperative spent: payment of electricity to the supplier, materials, waste removal. You can attach a photo of the receipt or an invoice to the expense. After saving, the expense reduces the balance of the cash or the bank.

## Who sees it

The form is seen by those who have the right "**Enter cash book entries**": the treasurer and the chairman. After saving, an expense is not edited. It can be reversed only by a person with "**Approve cash book operations**" (the chairman), on the expense card.

## Every element

Open the form with the button "**Expense**" on the Cash and bank and Expenses screens. The title is "**New expense**". If you came from a reversed expense, "**The data comes from a reversed expense. Correct it and record it again.**" stands at the top, and the fields and files are already filled.

### Fields

| Field | What to enter |
|---|---|
| **Account** | "**Cash**" or "**Bank account**". |
| **Expense date** | Not later than today: otherwise "**The date cannot be later than today.**". |
| **Amount** | More than 0.00: otherwise "**The amount must be greater than 0.00.**". |
| **Expense category** | Required, from the cooperative's list of categories. If the list is empty, the form says "**There are no expense categories. Add them in the settings.**": the chairman adds categories in the settings. Otherwise "**Choose an expense category.**". |
| **Paid to** | Optional, up to 200 characters. |
| **Purpose** | Optional, up to 500 characters. |
| **Document number** and **Document date** | Optional: the number and date of the supplier's invoice or receipt. |
| **Receipt or invoice** | From 0 to 5 files (more below). |

The fields "**Paid to**" and "**Purpose**" may contain the name of a private person, so they are personal data. They are stored protected and visible only to those who have the right "**View cash and bank**".

The button "**Record expense**" records the expense.

### Files: receipts and invoices

The hint under the buttons: "**Up to 5 files, each up to 5 MB. Photos (JPEG, PNG) and PDF are allowed.**".

- "**Add a file**" adds a file from the device.
- "**Take a photo**" (on a phone) opens the camera: the shot goes into the expense as an attachment.
- Files added before saving are shown as thumbnails. The button "**Remove**" removes a file. The same file (same name, size and content) cannot be attached twice: the form says "**This file is already attached**".
- Location (GPS) and camera data are removed from photos on upload. The system determines the file type from its content, not from its name.
- Error messages: "**The file is too large. Maximum 5 MB.**", "**No more than 5 files.**" and "**Only photos (JPEG, PNG) and PDF are allowed.**". A text file renamed to receipt.jpg is rejected.

> **IMPORTANT**
>
> HEIC photos from an iPhone are accepted in Safari only. The hint "**HEIC photos from an iPhone are accepted in Safari.**" under the buttons reminds you. If another browser cannot open such a file, the form says: "**This browser cannot open HEIC photos from an iPhone. Add the photo in Safari, or on the iPhone turn on Settings → Camera → Formats → "Most Compatible"**" That is, add the photo through Safari or change the phone setting once: "Settings → Camera → Formats → Most Compatible". After that the iPhone shoots in JPEG and the photo opens anywhere. While a photo is being converted you see "**Converting the photo...**".

### Saving and errors

For a cash expense the till must have enough money, and the account must have an opening balance. If not, the form shows one of these messages.

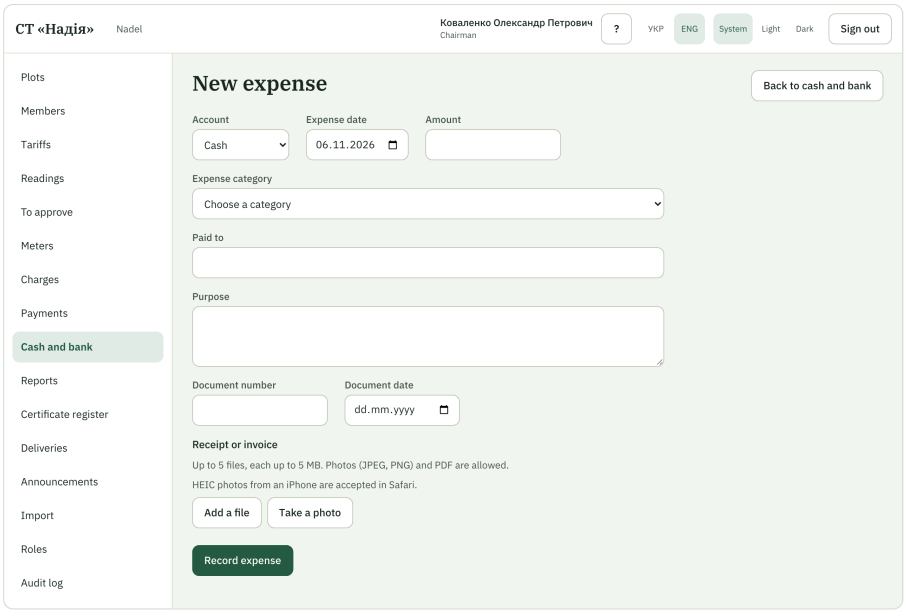
| What is shown | What to do |
|---|---|
| **Not enough cash in the till: balance {balance} on {date}** | A smaller amount, or first put money into the till. A cash expense without money in the till is not recorded. |
| **Enter the opening balance of the account first.** | The chairman enters the opening balance on the Cash and bank screen. |
| **The date is before the opening balance of the account.** | Enter a date not earlier than the opening balance. |

## Worked example

The treasurer bought a cable and terminals in a shop to repair a lamp and paid in cash.

1. On the "Cash and bank" screen press "**Expense**".
2. Account: "**Cash**", amount 860.00, category "Materials". Paid to: Будмаркет «Демо». Purpose: "Cable and terminals for a lamp repair". Document number: ЧК-1187.
3. Press "**Take a photo**" and photograph the receipt.
4. Press "**Record expense**". The cash balance went down by 860.00 ₴, and the expense card has the photo of the receipt.

## Screenshots




The new expense form

## Related pages

- Expenses
- Expense card
- Cash and bank
- Settings: expense and income categories

# Expense card

## What the screen is for

The card shows one expense with all its data and the receipt files. Here the chairman reverses an expense that was recorded with a mistake.

## Who sees it

The card is seen by those who have "**View cash and bank**": the chairman and the treasurer. The button "**Reverse and record again**" is seen only by those who have "**Approve cash book operations**" (the chairman). A treasurer who made a mistake asks the chairman.

## Every element

### Expense data

A table with the fields: **Account**, **Date**, **Amount**, **Category**, **Paid to**, **Purpose**, **Document** (number and date), **Booking month** and **Recorded by**. The accounting month may differ from the date if the date fell in a closed month: then the expense counts in the nearest open one.

The button "**Back to cash and bank**" returns to the cash.

### Files

The section "**Files**". If there are no files, it says "**No files.**".

- Photos are shown as thumbnails. A click opens the photo at full size right on the page. The Esc key closes it, arrows switch between files; the buttons "**Close**", "**Previous photo**" and "**Next photo**" do the same with the mouse.
- A PDF is shown as a file icon. It opens in a new tab ("**Open PDF**").
- The button "**Download**" downloads a file. Files can be downloaded by those who have "**View cash and bank**".
- If a file did not open, you see "**The file could not be loaded.**".

### Reverse and enter again

The button "**Reverse and record again**" opens the window "**Reverse the expense**". Enter a reason (up to 500 characters; otherwise "**Enter the reason for the reversal.**") and press "**Reverse**". "**Cancel**" closes the window.

What happens next:

- The expense gets the mark "**Reversed**", with the reason and the date. The money returns to the balance of the cash or the bank.
- The form of a new expense opens at once with the same values and files. Fix the mistake and record it.
- A reversed expense is not reversed a second time.
- The action goes to the change log.

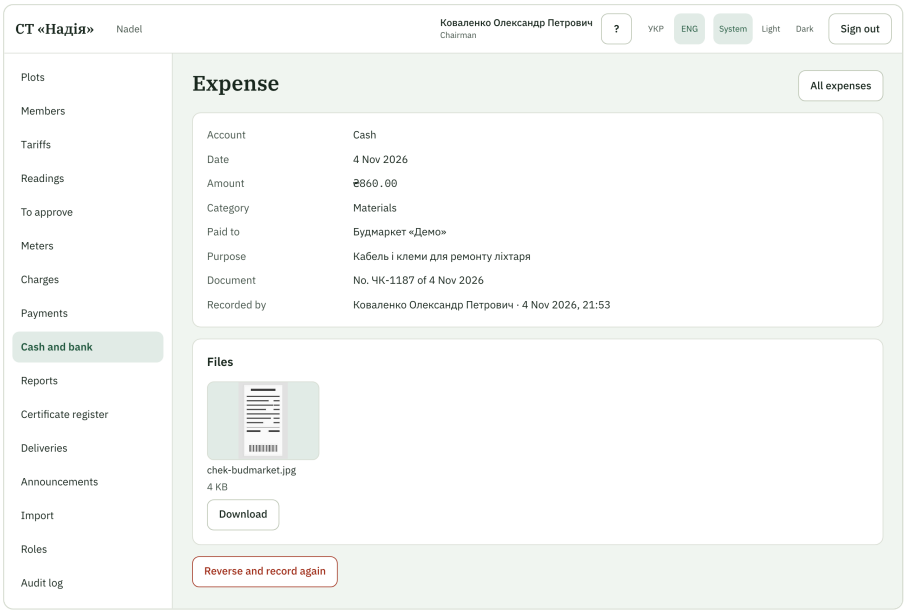
> **IMPORTANT**
> A reversed expense is not deleted and stays in the list. If you close the new expense form without recording it, the money has already returned to the balance. Enter the expense again through "**Expense**".

## Worked example

The treasurer entered 860.00 ₴ for materials under the wrong category: it should have been "Other".

1. The chairman opens the card of this expense.
2. He presses "**Reverse and record again**", writes the reason "Wrong category" and confirms.
3. A form opens with the same 860.00 ₴, document ЧК-1187 and the photo of the receipt. The chairman changes the category and presses "**Record expense**".
4. The old record has the mark "**Reversed**", the new record has the right category, and the cash balance is the same as it was.

## Screenshots




An expense card with a photo of a receipt

## Related pages

- Expenses
- Enter an expense
- Cash and bank
- Change log

# Cash count

## What the screen is for

From time to time the treasurer counts the cash in the till and compares it with the book. This screen shows how much there should be according to the book, takes the actual amount and at once shows the difference. The chairman also settles a surplus or a shortage here. The page has two views: a **new count** and a **saved count**.

## Who sees it

Saved counts are seen by those who have "**View cash and bank**". A new count is recorded by a person with "**Enter cash book entries**". The button "**Book the difference**" is seen only by a person with "**Approve cash book operations**" (the chairman).

## Every element

A new count is opened with the button "**Cash count**" on the Cash and bank screen. A saved one is opened with the link "**Last count: {date}**" in the cash card. The button "**Back to cash and bank**" returns back.

### A new count

| Field | What to enter |
|---|---|
| **Date and time of the count** | When you counted the money. A future time is not accepted: "**The time of the count cannot be in the future.**". |
| **Cash actually counted** | How much cash is really in the till. Not less than 0.00. |

As soon as you enter the amount, the screen shows "**Book balance at the end of the day**" and "**Difference**" (actual minus book). Below stands "**Cash movements since the previous count**": the cash movement since the previous count, so you can see what could have been lost. If there are many lines, only a part is shown: "**Showing {shown} of {total} lines.**".

- A difference of 0.00: the count is recorded as "**Matches**".
- Any other difference needs an explanation in the field "**Explanation of the difference**" (up to 500 characters). Without it: "**Explain the difference.**".

The button "**Save the count**" saves the count. A saved count keeps the book balance it showed: an entry that later appears with an old date does not change it.

### A saved count

A table: **Date and time of the count**, **Cash actually counted**, **Book balance at the end of the day**, **Result** (matches, or the difference with the explanation), **Status** and **Counted by**.

If there is a difference that is not yet settled, the status is: "**Unresolved difference {amount}**". The same mark hangs on the cash card until the chairman settles the difference.

- **Book the difference** (the chairman only). In the window "**Book the difference**" enter a reason and press "**Book**". A line "Surplus" (receipt) or "Shortage" (expense) appears in the cash journal. A difference is settled once. A shortage after which the cash would go negative on later days is rejected.
- **Cash count act** downloads a PDF: the cooperative's name, date and time, the book balance, the actual balance, the difference (with the explanation and, once settled, the way it was settled), signature lines for the treasurer and the chairman. The act is always in Ukrainian.
- The settlement of a difference can be reversed in the cash journal with the button "**Reverse**" by the difference line; then the count shows "**The booking is reversed**".

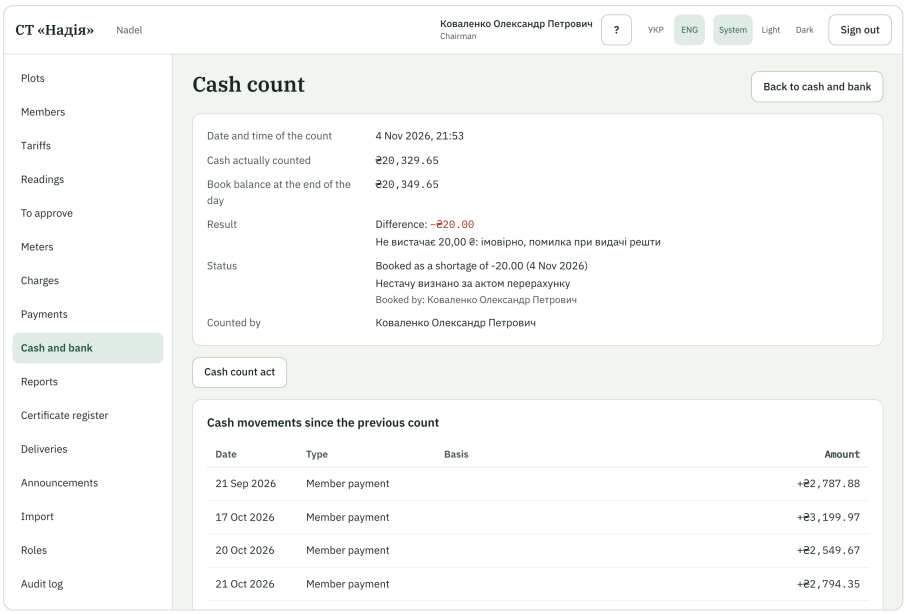
> **IMPORTANT**
>
> A count and its record are not deleted. If you recorded a wrong amount, make a new count: it will also show the movement of money between the two counts.

## Worked example

According to the book the till has 6,520.00 ₴, and the treasurer counted 6,500.00 ₴.

1. The treasurer presses "**Cash count**" and enters 6500.00.
2. The screen shows a difference of −20.00. The treasurer writes the explanation: "20.00 ₴ is missing: probably a mistake when giving change" and presses "**Save the count**".
3. The cash card shows "Unsettled difference −20.00 ₴".
4. The chairman opens the count, presses "**Book the difference**" and gives the reason "Shortage recognised by the count act". A "Shortage" line of −20.00 ₴ appears in the cash journal; the cash balance went down by 20.00 ₴.
5. The chairman downloads the count act, has the treasurer sign it and signs it himself.

## Screenshots




A saved cash count with a difference and an act

## Related pages

- Cash and bank
- Bank reconciliation
- How the system calculates cash, bank and reconciliation

# Bank reconciliation for a month

## What the screen is for

At the end of a month the treasurer compares the bank statement with what is recorded in the book. The screen shows both sets of figures side by side, the difference for each and all the bank lines of the month. When the closing balance agrees, the month is marked "**Reconciled**".

## Who sees it

The screen is seen by those who have "**View cash and bank**". The statement figures are entered and "**Reconciled**" is pressed by a person with "**Enter cash book entries**" (the treasurer, the chairman). The button "**Bank difference**" and the reversal of a bank difference are seen only by a person with "**Approve cash book operations**" (the chairman).

## Every element

A month is opened from the bank card on the Cash and bank screen: press the name of the month. The title is "**Bank reconciliation, {month}**". The button "**Back to cash and bank**" returns back. Next to the title you see the status: "**Not reconciled**", "**Reconciled**" or "**Needs a new check**", and for a reconciled month also who reconciled it and when ("**Reconciled by {name}, {date}**").

### Bank statement and book

The section "**Bank statement and books**". The treasurer enters four figures from the bank statement:

1. **Balance at the start of the month**;
2. **Receipts**;
3. **Spendings**;
4. **Balance at the end of the month**.

They must add up: opening + receipts − debits = closing. Otherwise the system says, for example: "The statement figures do not add up: 100.00 + 50.00 − 20.00 ≠ 200.00" (the text template is "**The statement figures do not agree: {opening} + {incoming} − {outgoing} ≠ {closing}**"). The button "**Save the statement**" saves the statement. While the statement is not entered you see "**The statement of this month is not entered yet.**".

The table has three columns: "**Statement**", "**Books**" and "**Difference**".

### How to find a difference

Under the table there is the section "**Bank movements of the month**": all the bank lines of the month. This makes it easier to find what is missing.

- If a record is missing (the bank charged a fee and you do not have it), enter it: a payment, an expense or other income. The comparison updates by itself.
- If no record explains the difference, the chairman presses "**Bank difference**". In the window "**Bank difference**" enter the amount (negative means a debit) and a reason and press "**Book**". The line goes to the bank journal; it can be reversed ("**Reverse the bank difference**").

### The button "Reconciled"

It works only when the difference of the closing balance (closing by the statement minus closing by the book) is 0.00. Otherwise the hint "**Available when the closing balance difference is 0.00.**" stands next to it, and the action is rejected with the message "**The month cannot be marked reconciled: enter the statement and clear the differences.**".

A difference in receipts or debits does not block the button. The screen shows it as a warning: "**The statement's total in or total out differs from the books. This does not block the reconciliation: it happens when a mistaken line was reversed in another mon**
"

A reconciled month cannot be changed ("**The month is reconciled: its statement cannot be changed.**").

### A reconciled month can become "unreconciled"

If a bank line dated in a reconciled month appears later (for example, on 3 November a bank payment for 20 October is entered), the month goes to "**Needs a new check**". The screen says what was added and when: "The month must be reconciled again: 04.11.2026, 10:15 a member payment dated 20.10.2026 for 500.00 ₴ was entered." The same happens to every following reconciled month, because their opening balance changed. Check the months one by one starting with the earliest and press "**Reconciled**" for each. The change log keeps a record of which line reopened them.

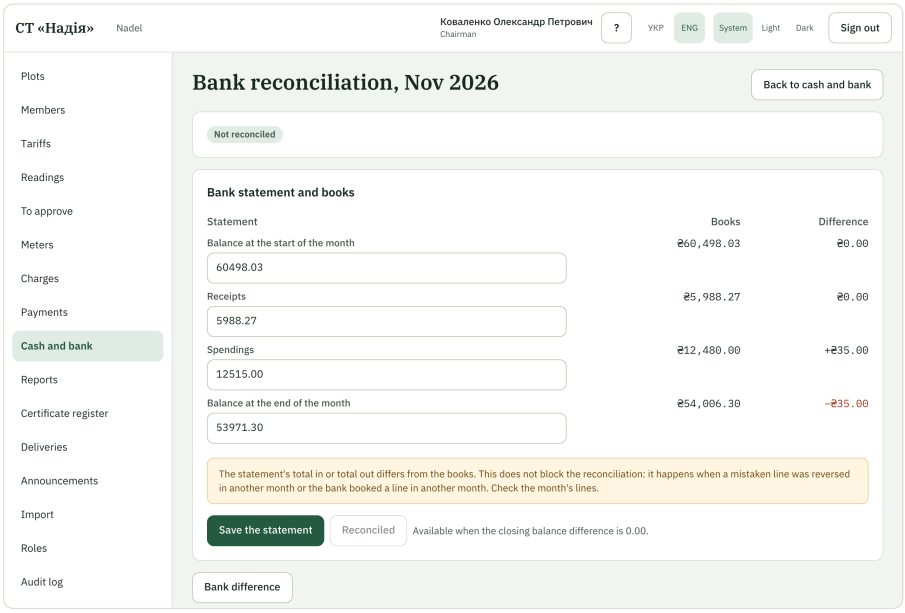
> **IMPORTANT**
>
> A mistake of a past month that you reverse today: the reversal goes into the current month with today's date. In the past month this gives a difference in receipts or debits: it only warns and does not block the reconciliation.

## Worked example

The statement for the month: opening balance 48,200.00, receipts 5,000.00, debits 14,000.00, closing balance 39,200.00. The book shows 35.00 ₴ less in debits, because the bank charged a fee that you do not have.

1. The treasurer enters the four figures and presses "**Save the statement**". The difference of the closing balance is 35.00 ₴, the button "**Reconciled**" does not work.
2. The fee is not among the bank lines. The treasurer enters it as an expense of 35.00 ₴ in the category "Bank fees".
3. The difference became 0.00. The treasurer presses "**Reconciled**", and the month is marked reconciled.

## Screenshots




Bank reconciliation for a month: statement, book and differences

## Related pages

- Cash and bank
- Enter an expense
- Cash count
- How the system calculates cash, bank and reconciliation

# Certificate register

## What the screen is for

A certificate on the state of settlements says how much a plot owes the cooperative on a certain date, or that there is no debt. It is asked for, for example, when a plot is sold. This screen is the register of all issued certificates. Here you find them, open their PDF and annul wrong ones.

The certificate itself is issued on the plot card, on the "Certificate" tab, see Plot card.

## Who sees it

The register is seen by those who have "**View charges**". Only those who have "**Issue certificates**" can issue and annul certificates: by default the chairman. The button "**Annul**" is only theirs. A member does not see certificates.

## Every element

### The filter "Status"

Shows "**All**", "**Issued**" or "**Annulled**". When no certificates match, it says "**No certificates match this filter.**", and if there are none at all, "**There are no certificates yet.**".

### The table

| Column | What is in it |
|---|---|
| **Number** | A number like "Д-2026-1". Numbers run in sequence within the cooperative and the year, with no gaps and no repeats. |
| **Plot** | The plot number; the link leads to its card. |
| **As of** | The date as of which the certificate was made. |
| **Debt** | The total debt on that date. |
| **Issued by** and **Issued** | Who issued it and when. |
| **Status** | "**Issued**" or "**Annulled**". |
| **Actions** | "**Download PDF**" (PDF) and "**Annul**". |

### What is in a certificate

A certificate contains the cooperative's details, the plot with its area and current owners, the debt by type and in total, and the prepayment on the date. These are the same figures as in the report "**Plot balances**". If the total debt is 0.00, it says "No debt"; otherwise "The debt is ...". It has the name and position of the person who issued it, the date of issue and a place for a signature and a stamp. A certificate can be issued for a plot without an owner, but not for common land or for an archived plot. The document is always in Ukrainian, because it is an official document.

### Annulling

The button "**Annul**" opens the window "**Annul certificate {number}?**". The window warns: "**The certificate stays in the register marked as annulled. An annulment cannot be undone.**" Enter the "**Reason for annulment**" (required) and press "**Annul certificate**".

An annulled certificate stays in the register with the reason and the name of the person who annulled it ("**Annulled by {name}, {when}**"). Its PDF carries the mark "ANNULLED". It cannot be annulled a second time ("**The certificate has already been annulled**"), and its number is not reused.

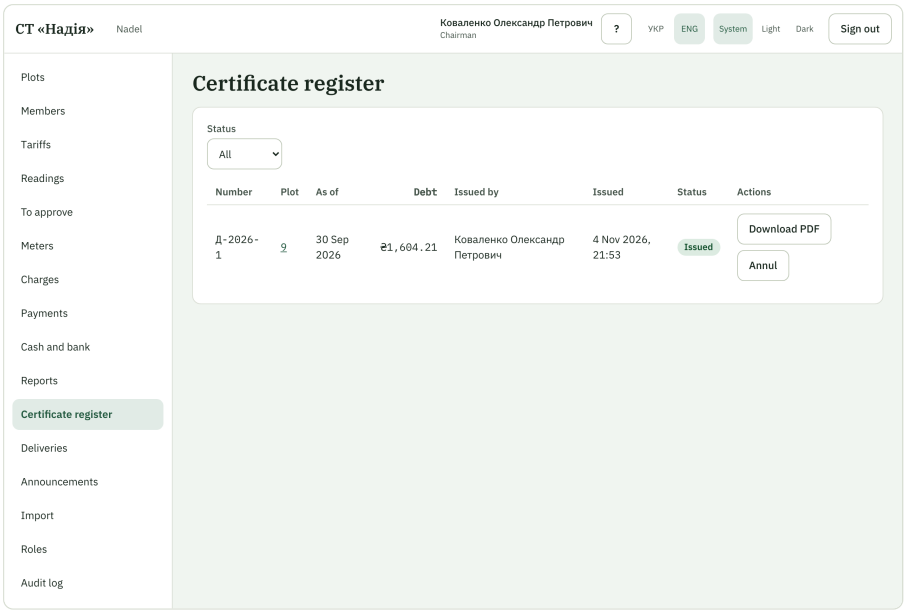
> **IMPORTANT**
> An issued certificate does not change. A mistake is corrected by a new certificate, and the old one is annulled with a reason. An annulment cannot be undone.

## Worked example

The owner of plot 9 asks for a certificate. The chairman opens the card of plot 9, the "Certificate" tab, chooses the date and presses "**Issue certificate**". The system shows "Certificate issued: Д-2026-1" and downloads the PDF. The next day it turns out that the date was wrong.

1. The chairman opens the "Certificate register" and finds certificate Д-2026-1.
2. He presses "**Annul**", writes the reason "Wrong date" and confirms.
3. The certificate has the status "**Annulled**". The chairman issues a new certificate with the right date (its number is Д-2026-2).

## Screenshots




The certificate register with the filter and the PDF and Annul buttons

## Related pages

- Plot card
- Reports
- How the reports work

# Delivery

## What the screen is for

The system sends invoices, payment notifications and board announcements to the cooperative's members through two channels that belong to the cooperative itself: e-mail from your mailbox and Telegram through the cooperative's own bot. This screen shows what was sent in a month and how each sending ended. The treasurer sees to whom an invoice arrived and who needs paper.

## Who sees it

The screen is seen by those who have "**Issue invoices**": the chairman and the treasurer. The link to an announcement from the rows is seen by those who also have "**Send announcements**". The channels are set up by the chairman on the Settings page, in the section "Delivery".

## Every element

### The summary for the month

At the top there is the table "**Month summary**" by channel: **Queued**, **Sent**, **Failed** and **To print / printed**.

### Filters

| Filter | What it does |
|---|---|
| **Month** | The month the delivery is shown for. The current one by default. |
| **Kind** | "**Invoice**", "**Payment**", "**Payment reversal**", "**Announcement**" or "**All**". |
| **Status** | The state of a sending (below). |
| **Channel** | Telegram, e-mail, paper. |

When nothing was sent in the month you see "**Nothing has been sent for this month yet.**", and if the filters leave nothing, "**Nothing matches these conditions.**".

### The table

Columns: **Plot**, **Member**, **Channel**, **Status**, **Reason**, **Attempts**, **Sent at** and **Actions**.

| Status | What it means |
|---|---|
| **Queued** | Waits for its turn or the next attempt. |
| **Sent** | The mail server or Telegram accepted the message. This does not yet mean the person read it. |
| **Failed** | It did not work. The "**Reason**" column says why. |
| **To print** / **Printed** | An invoice for paper, until the board presses "**Printed**". |

For announcement rows, if you have the right to send announcements, a link to the announcement card is added.

For a member who chose Viber, the row has the note "**Viber is not supported yet**": the invoice goes to them on paper.

### Error reasons and what to do

| Reason | What to do |
|---|---|
| **The mail server refused the letter** | The mailbox refused the letter (a limit, speed, the address not accepted). The system will try twice more. If it does not pass, reduce "Letters per second" in the settings, wait until the next day and send again. |
| **Wrong mailbox login or password** | Correct the app password in the "Delivery" settings and send a test letter. After that press "**Send again**". |
| **Could not connect** | No connection to the mail server or Telegram. This is temporary, the system repeats. Check the server address and port. |
| **The bot is blocked** | The member blocked the bot. The connection is cancelled, invoices go through the fallback channel until they connect Telegram again. |
| **The channel is not configured** | There is no mailbox or bot (or Telegram rejected the token). Set the channel up in "Delivery". |
| **No contact** | The member has no e-mail address or Telegram chat. Contact them personally or enter the address in the member card. |
| **This mail server is not allowed** | The system connects only to mail services on the internet on ports 25, 465 or 587. Enter another server. |

**IMPORTANT**

Gmail accepts only an app password from programs, not the ordinary account password. If mail does not work, first turn on two-step verification in the Google account, create an app password and paste it in the "Delivery" settings.

**Buttons**

- "**Send again**" in a row with the status "**Sent**" or "**Failed**". After fixing the reason, send the message again. For paper and for messages still in the queue it is not there.
- "**Printed**" removes the "to print" mark from all paper invoices of the chosen month. The system asks for confirmation: "**Mark the paper invoices as printed?**" (the number of invoices is stated in the window) and afterwards says how many were marked ("**Marked as printed: {count}.**").

"**Sent**" and "**Printed**" do not change any amounts. They are only a record of delivery.

**Who gets which channel**

The decision depends on the member's preferred way (visible in the member card and changed in their account) and on what is set up.

| Preferred way | Situation | Receives |
|---|---|---|
| Telegram | connected, the bot is set up | in Telegram |
| Telegram | not connected or no bot | by e-mail if there is an address and a mailbox is set up; otherwise on paper |
| E-mail | there is an address, the mailbox is set up | by e-mail |
| E-mail | no address or no mailbox set up | on paper |
| Paper | — | on paper (a PDF to print) |
| Viber | — | on paper, with a note |

The invoice of a plot is received by each of its current owners through their own channel. A member with several plots gets a separate message for each.

**Payment notifications**

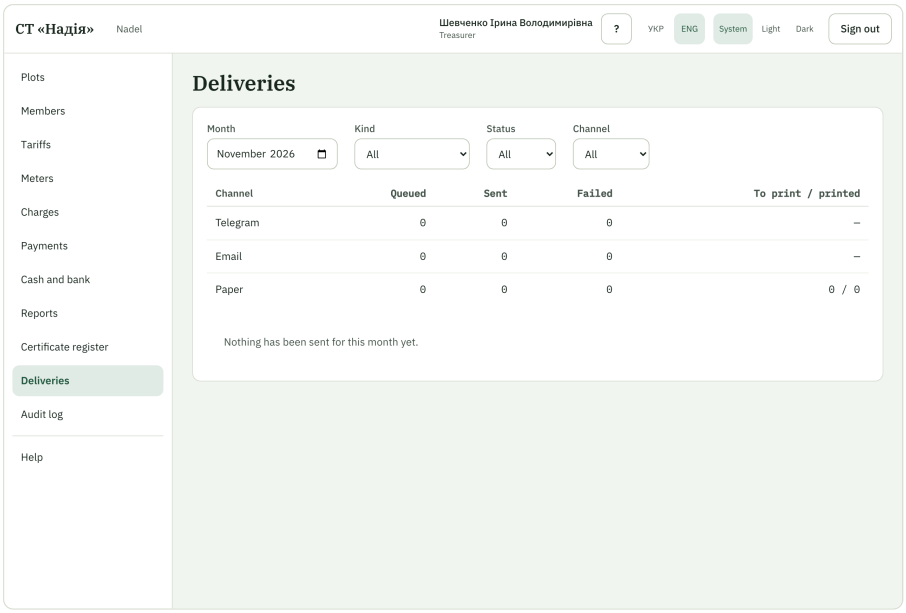
When "Notify about payments" is on in the settings, after a payment is saved each current owner who receives messages in Telegram or by e-mail gets, for example, "Received 500.00 ₴ for plot 9 (28.10.2026). Debt: 1,234.56 ₴" (or "Prepayment: ..." if there is no debt). For cash, the receipt PDF is added. After a reversal comes "The payment of 28.10.2026 for 500.00 ₴ is reversed. Debt: ...". Owners on paper do not get such messages. With the switch off, there are no messages at all.

## Worked example

The treasurer sent the invoices for the month, and for plot 9 the row has the status "**Failed**" with the reason "**No contact**".

1. Open "Delivery", choose the month and the filter "**Status**" → "**Failed**".
2. Open the member card and enter an e-mail address.
3. Return to the delivery and press "**Send again**" by this row. The status changes to "**Queued**", then to "**Sent**".
4. After printing the paper invoices press "**Printed**" and confirm.

## Screenshots




Delivery for a month: summary by channel, filters and a list with error reasons

## Related pages

- Announcements
- Charges of a month: invoices and sending
- Settings: delivery
- How invoice delivery works

# Announcements

## What the screen is for

All the announcements the board has sent to the cooperative's members are collected here: about a meeting, a power cut, payment deadlines. From here you can write a new announcement or open an old one to see to whom it arrived.

## Who sees it

The menu item "Announcements" and the screen are seen by those who have "**Send announcements**". By default that is the chairman only. The right can be given to a custom role on the Roles page.

## Every element

### The button "**New announcement**"

Opens the new announcement form.

### The table

| Column | What is in it |
|---|---|
| **Date** | When the announcement was sent. It is a link to the announcement card. |
| **Author** | Who wrote it. |
| **To** | "**All members**" or a list of streets. |
| **Channels** | Telegram, e-mail or both. |
| **Outcome** | "**Sent: {sent}, failed: {failed}, queued: {queued}, on paper: {paper}**". |

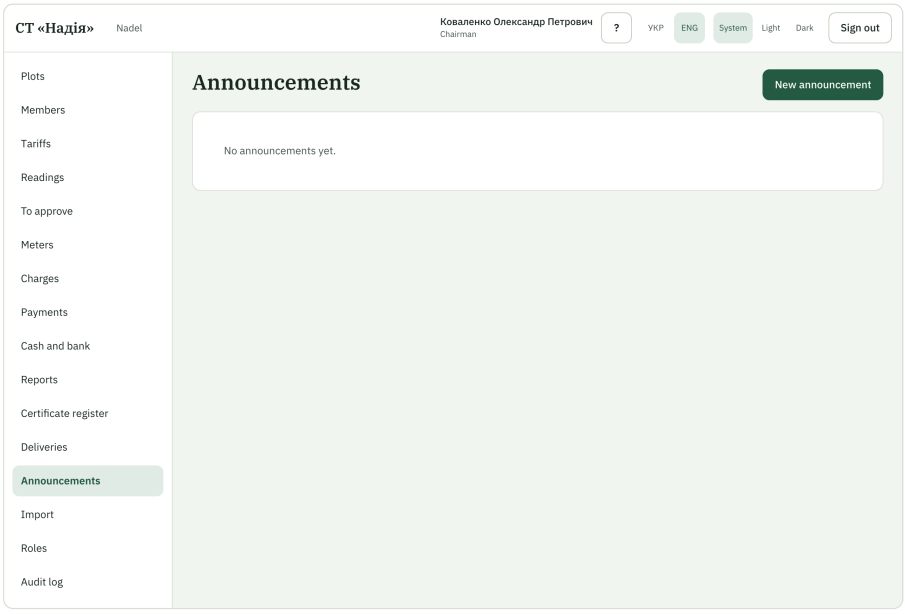
If there were no announcements yet, you see "**No announcements yet.**".

The result in the column reads like this: "Sent": the mail server or Telegram accepted the message; "errors": it failed; "queued": still waiting; "on paper": members who have none of the chosen channels and must be told personally.

## Worked example

The chairman sent an announcement about a meeting to all members. In the list it has a row: "**All members**", channels "Telegram, E-mail", result "Sent: 18, errors: 1, queued: 0, on paper: 6". This means: 18 members received the message, one failed, six have no channels and need a personal notice. The chairman opens the announcement to see who exactly.

## Screenshots




The announcement list with the delivery result

## Related pages

- New announcement
- Announcement card
- Delivery

# New announcement

## What the screen is for

Here the board writes an announcement and sends it to the cooperative's members through Telegram and e-mail. Before sending, the system shows how many members will receive the announcement and whom you must tell personally, because they have no channel.

## Who sees it

The screen is seen by those who have "**Send announcements**" (by default only the chairman).

## Every element

### Fields

| Field | What to enter |
|---|---|
| **Announcement text** | Plain text, up to 2000 characters. The counter "**{count} / {max}**" stands under the field. When there are too many characters, you see "**At most {max} characters**", and the server answers "**The announcement is longer than 2000 characters.**". |
| **To** | "**All members**" or "**Streets**". For streets mark the ones you need in the list "**Streets**" (streets of active plots). If none is marked, you see "**Choose at least one street**". If the cooperative has no streets yet, it says "**The cooperative has no streets yet.**". |
| **Channels** | "**Telegram**", "**E-mail**" or both. With none: "**Choose at least one channel**". |

Telegram works only if the cooperative's bot is connected in the settings. Otherwise the system says "**The cooperative has no Telegram bot set up.**".

### The button "Preview"

It sends nothing. It shows "**Who gets the announcement**": how many members will receive the announcement (by channel: "**Telegram: {telegram}, e-mail: {email}, on paper: {paper}**") and the list "**On paper (inform in person)**". These members have none of the chosen channels: you must notify them yourselves. Print this list and tell each of them. If there is no such list, you see "**Every member gets the announcement in the chosen channels.**".

### The button "Send"

The hint next to it is "**Preview who gets the announcement first.**", that is, first press "**Preview**". The announcement is queued, and you return to the announcement list. The progress of sending is seen on the Delivery screen.

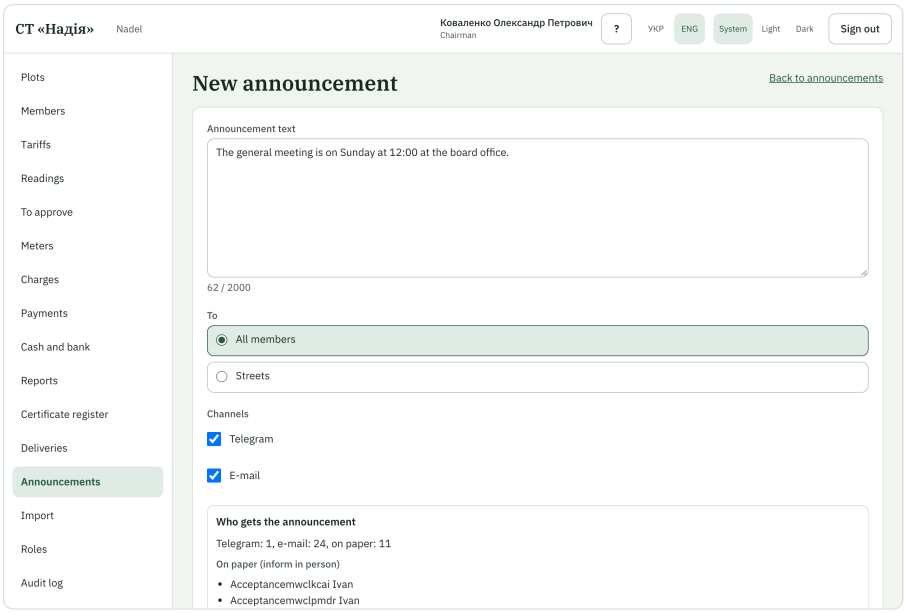
> **IMPORTANT**
> A sent announcement cannot be recalled or changed. Before sending, check the text, the streets and the channels in the preview.

## Worked example

The chairman writes an announcement about a meeting.

1. Text: "The general meeting is on Sunday at 12:00 at the board office."
2. "**To**": "**All members**".
3. "**Channels**": "Telegram" and "E-mail".
4. "**Preview**": the screen shows how many members will receive the announcement in Telegram and by e-mail, and the list of those who stay on paper.
5. The chairman prints this list to phone the people in it. Then he presses "**Send**".

## Screenshots




A new announcement with a preview: who gets it and who stays on paper

## Related pages

- Announcements
- Announcement card
- Delivery
- Settings: delivery

# Announcement card

## What the screen is for

The card shows one announcement: what was sent, to whom and how the delivery ended. You see here exactly to whom a message did not arrive.

## Who sees it

The card is seen by those who have "**Send announcements**" (by default the chairman).

## Every element

The button "**Back to announcements**" returns to the list.

### Announcement data

A table with the rows "**Text**", "**Date**", "**Author**", "**To**" ("**All members**" or a list of streets), "**Channels**" and "**Outcome**" (for example, "Sent: 18, errors: 1, queued: 0, on paper: 6").

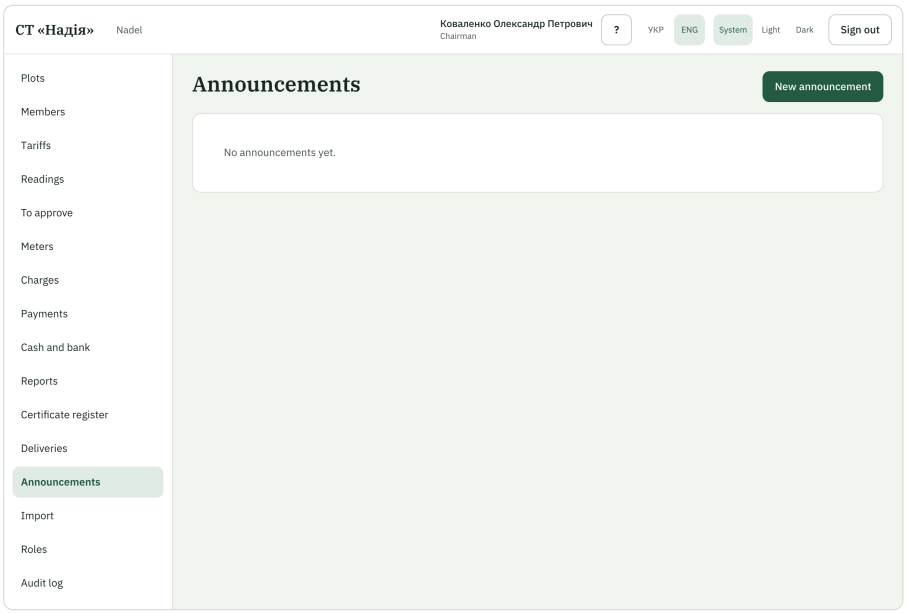
### Delivery

The section "**Delivery**" lists the message to each member: plot, member, channel, status, error reason, number of attempts and time. The statuses and reasons are the same as on the Delivery screen. You can send a message again on the "Delivery" screen with the button "**Send again**", once you have fixed the reason. If the announcement is not found, you see "**Announcement not found.**".

## Worked example

The chairman opens the announcement about the meeting. One row has the status "**Failed**" and the reason "**The bot is blocked**": the member blocked the bot. The chairman tells them personally, and the member still has e-mail as a fallback channel.

## Screenshots




An announcement card with the text and the delivery to each member

## Related pages

- Announcements
- New announcement
- Delivery

# Reports

## What the screen is for

Reports gather into one table what the system already knows: charges, payments, debts, electricity, water, corrections, the cash. Each report is shown on the screen and can be downloaded as an Excel file. Here you choose which report to open, and also the templates a report can be written into.

A report does not calculate anything "its own way": it only selects, groups and adds up the sums already posted. So any sum in a report can be checked against the plot account or the charges month screen to the kopeck. Sums in reports are always effective sums: charges plus adjustments minus reversals.

## Who sees it

Reports are opened and downloaded, and files by templates are generated, by those who have "**View reports**": the chairman and the treasurer. The four cash reports need only "**View cash and bank**". For example, a custom role "Audit committee" with just this right sees exactly these four reports and no others.

| Report | The right needed |
|---|---|
| Most reports | "**View reports**" |
| "**Debtors**" | also "**View members**" (the report has names and phones) |
| "**Payments**" | also "**View payments**" (the payer note is seen only by a person with "**View members**") |
| "**Cash book**", "**Bank movements**", "**Income and expense**", "**Expenses**" | "**View cash and bank**" |
| Uploading, replacing and archiving templates | "**Manage report templates**", by default the chairman only |

A role with "**View reports**" but without "**View members**" sees all reports except "**Debtors**". A member and an electrician do not see the "**Reports**" menu item and get a message that there is no access at a direct address.

## Every element

### Report cards

At the top are the cards of the reports available to you. A card has the report name and one sentence about it; a click opens the report. If nothing is available, it says "**No reports are available.**".

The list of reports:

- "**Charges by month**": what was charged in each month by charge type.
- "**Plot balances**": the debt of each active plot on a date.
- "**Debtors**": plots with a debt, with names and contacts of the owners.
- "**Electricity by meter tree**": electricity from the substation to the plots.
- "**Water**": plots connected to water and charges for water.
- "**Missing readings**": meters without a confirmed reading for a period.
- "**Land tax**": expected and charged land tax for a year.
- "**Corrections**": all reversals and adjustments of the chosen months.
- "**Payments**": each payment, with totals by source.
- "**Cash book**", "**Bank movements**", "**Income and expense**", "**Expenses**": the four cash reports.

For details of each report, its filters and columns, see the Report page.

### Templates

- Those who have "**View reports**" but cannot manage templates see the section "**Report templates**" with ready templates: the button "**Generate**" by a template generates a file for the chosen period.
- The chairman sees the card "**Report templates**", a link to the templates page, where they are uploaded and archived.

### Useful links from other screens

On a charges month the link "**Report**" opens "**Charges by month**" for that very month. On a plot account there is a link to the debt report for the street of that plot.

**IMPORTANT**

The report "**Charges by month**" has two sheets: "**By billing month**" and "**By service period**". The first groups the money by the month in which the charge was booked into the ledger. The second groups the same money by the month the service was provided for. These are different things: a reading taken in October may carry electricity for August and September. In the first sheet this money stands in October, in the second in August and September.

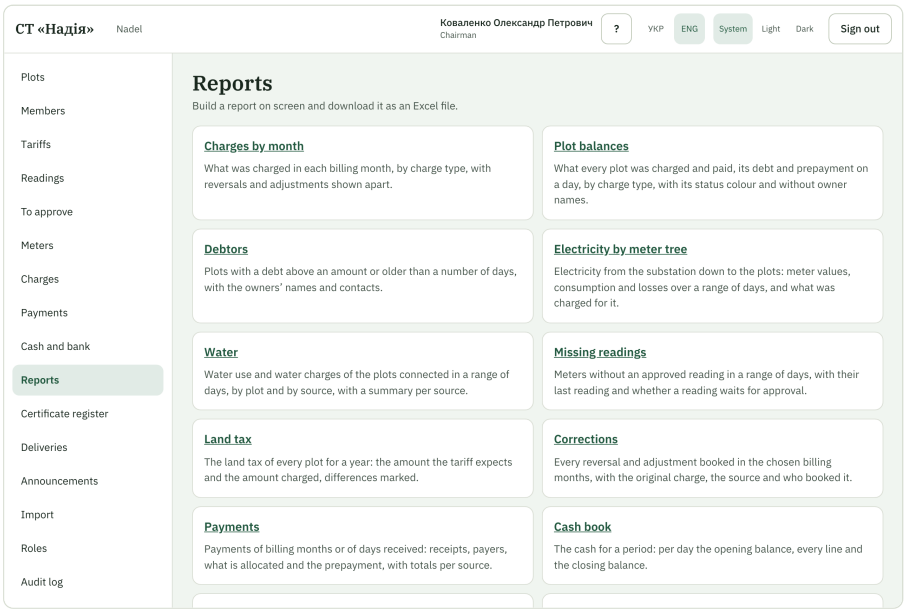
## Worked example

The chairman prepares an electricity report for the general meeting.

1. He opens "**Reports**" and presses the card "**Electricity by meter tree**".
2. On the report page he chooses the period and presses "**Show**".
3. He presses "**Download Excel**" and gets an Excel file.

For a report on the sum charged for August the chairman opens "**Charges by month**" and looks at the second sheet: "**By service period**" will also hold the electricity whose reading was taken in October but which belongs to August.

## Screenshots




The Reports screen: report cards and templates

## Related pages

- Report: filters, preview and Excel
- Report templates
- Cash and bank
- Charges of a month
- How the reports work

# Report: filters, preview and Excel

## What the screen is for

This is one report: filters at the top, a table below and a button to download an Excel file. All reports are built the same way; the filters and columns differ.

## Who sees it

Reports are seen by those who have "**View reports**". The four cash reports ("**Cash book**", "**Bank movements**", "**Income and expense**", "**Expenses**") need only "**View cash and bank**". The report "**Debtors**" also needs "**View members**", because it has the names, phones and addresses of owners. The report "**Payments**" also needs "**View payments**". If a report is not shown to you, a message that there is no access is shown instead.

## Every element

### The title and buttons

- "**All reports**" returns to the list of reports.
- "**Download Excel**" downloads an Excel file with all rows and the same totals. While the file is being prepared the button says "**Downloading...**". If something failed, you see "**Could not download the file. Please try again.**".

The Excel file has one sheet per report (some reports have two), a header (the cooperative, the report name, the filters, the date and time in Kyiv, who generated it), a frozen heading row; numbers are stored as numbers, dates as dates. The language of the file is the interface language. The file name is the report code and the period in Latin letters, for example land-tax_2026-01-01_2026-12-31.xlsx.

### Filters

The filters are above the table. Press "**Show**" to update the table. Which filters there are depends on the report: "**From month**" and "**To month**", "**From**" and "**To**", "**As of**", "**Year**", "**Charge types**", "**Street**" and others. Your choice stays in the page address.

Under the filters it is written what the period is counted by.

| Text | What it is |
|---|---|
| **The period is counted in billing months.** | The charging month: the month a charge belongs to in the ledger. |
| **The period is counted in service days.** | The days the service was provided. |
| **The period is counted by reading dates.** | The dates of meter readings. |
| **The period is counted by the days payments were received.** | The days payments arrived. |
| **The period is counted by the dates of money movements.** | The dates of cash and bank movements. |

These are different things: an adjustment posted in March for January stands in March in charging months, and in January in service days.

### The table

The screen shows the first 200 rows and the totals. The text "**Showing first {shown} of {total} rows. The Excel file has all of them.**" says how many rows the report has in all. If there are no rows, you see "**No rows for these filters.**". A value that is unknown (for example, a reading at the start of a period) is shown empty with a reason, not as zero. In the reports "**Plot balances**" and "**Charges by month**" there are columns "Paid" (the allocation of payments), and the balances also have "Prepayment".

The limit: a report with more than 20,000 rows on a sheet is not generated: "**The report has more than 20 000 rows on a sheet. Narrow the filters and try again.**" Every generation of a report is written to the change log: who, which report, the filters, the number of rows. The content is not recorded.

### Twelve reports

**Charges by month.** What was charged in each month by charge type. Filters: "**From month**", "**To month**", charge types, street. A row is a month and a type: "Plots", "Charged", "Opening balances", "Reversals", "Adjustments" and "Total charged" (the sum from the month screen). The first sheet is "**By billing month**", the second is "**By service period**": the same money by the month of the service. A reading taken in October may carry electricity for August and September: in the first sheet it is in October, in the second in August and September.

**Plot balances.** The debt of each active plot by charge type as of a date ("**As of**", today by default), with the oldest unpaid charge, days overdue and the status (red, yellow, green, grey). There are no owner names. For a past date the status is computed with today's thresholds, as stated in the header.

**Debtors.** Plots with a debt from a sum ("**Debt from**", by default the cooperative's red threshold) or older than a number of days ("**Older than, days**"). Here there are names, phones and e-mail addresses of owners, so the report also needs "**View members**". Do not pass the file to outsiders.

**Electricity by meter tree.** Electricity from the substation to the plots for a period (by default last month, service days). The sheet "Consumption": a tree node per row, readings at the start and end with the type ("Actual", "Estimated", "Unknown"), consumption, "Below it, kWh", losses in kWh and %, and "Plots fed". If a value is unknown, everything that depends on it in the row is empty with a reason. The sheet "Charged" shows what was posted in the ledger for this period:

plot consumption, losses, calculations of common consumers, adjustments and reversals in separate rows. The header states the current loss mode.

**Water.** Plots connected to water on the chosen days (filters "**From**", "**To**", "**Water source**"). A row: source, days connected, meter, m³ only for the connected days and charges by type: "By meter", "Subscription", "Well pump share", with a separate column for reversals and adjustments. The sheet "By source" sums plots, plot-days, m³ and the amount. Plots that are not connected are not shown.

**Missing readings.** Meters that have no confirmed reading for the period. Filters: dates, resource, meter kind (by default only plot meters), street. A row: resource, kind, plot or name, serial number, last reading, days since, and the mark "Reading awaits confirmation". Under the table there is the link "**Open “Not charged”**".

**Land tax.** The filter "**Year**". A row is a plot the tax applies to: area in m² and in sotkas, the rate per sotka, "Expected" and "Charged for the year". Plots where these sums differ are marked in the column "Discrepancy", and under the table is a hint that a recalculation may be needed (the report posts nothing itself). Example: a rate of 35.00 ₴ per sotka per year, a plot of 600 m² = 6 sotkas, 6 × 35.00 = 210.00 ₴ per year.

**Corrections.** All reversals and adjustments of the chosen charging months. Filters: months, charge types, "Reversal" or "Adjustment". A row: posting date, month, plot, type, original charge, amount, source, reason, basis and who posted it. This is a ready list of changes to the past for the general meeting.

**Payments.** Each payment, with totals by source (cash, bank).

**Four cash reports.** They work the same way but need only "**View cash and bank**".

- "**Cash book**": the period; for each day the opening balance, every line (time, kind, receipt or document number, counterparty, receipt or expense) and the closing balance; totals for the period. The closing balance of each day equals the cash balance of that day on the Cash and bank screen.
- "**Bank movements**": the same for the bank account, with the reconciliation status of each month.
- "**Income and expense**": the report for the general meeting for a period. Income: member payments by charge type, "Prepayments" (the part of payments not allocated to any charge), reversals of earlier accepted payments as a separate negative row, other income by category. Expenses by category. Settled differences of counts and reconciliations as a separate row. Result = income − expenses + differences, and it equals the change of cash plus bank for the period. Also the cash and bank balances at the start and end.
- "**Expenses**": a list of expenses with the filters "Period", "Category", "Account". On the screen each expense has a link to its files.

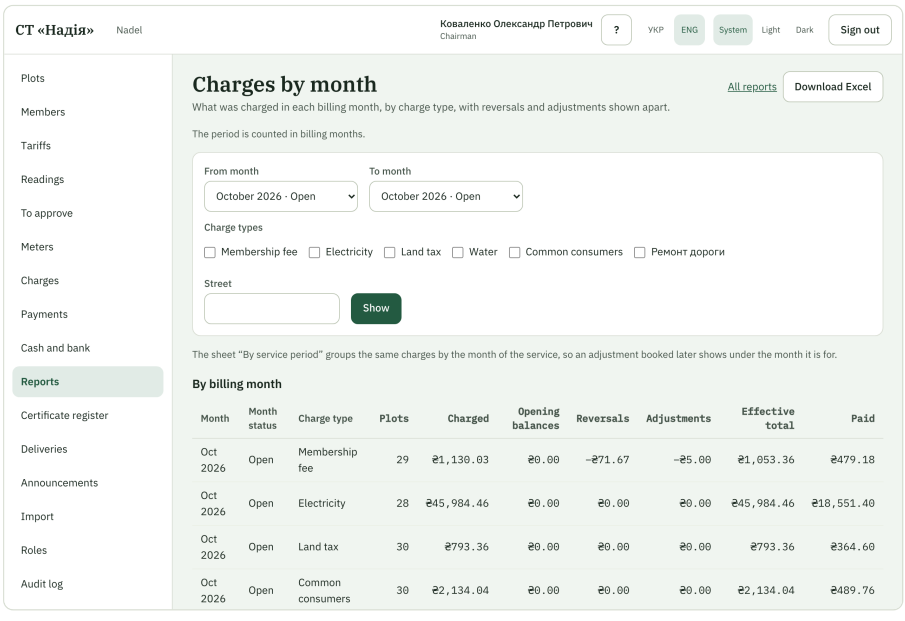
> **IMPORTANT**
> The light of the well pump is split between its plots as a row of a common consumer. This row is on the sheet "Charged" of the electricity report and in the column "Well pump share" of the report "**Water**". It is the same sum shown from two sides: do not add these two reports to each other.

## Worked example

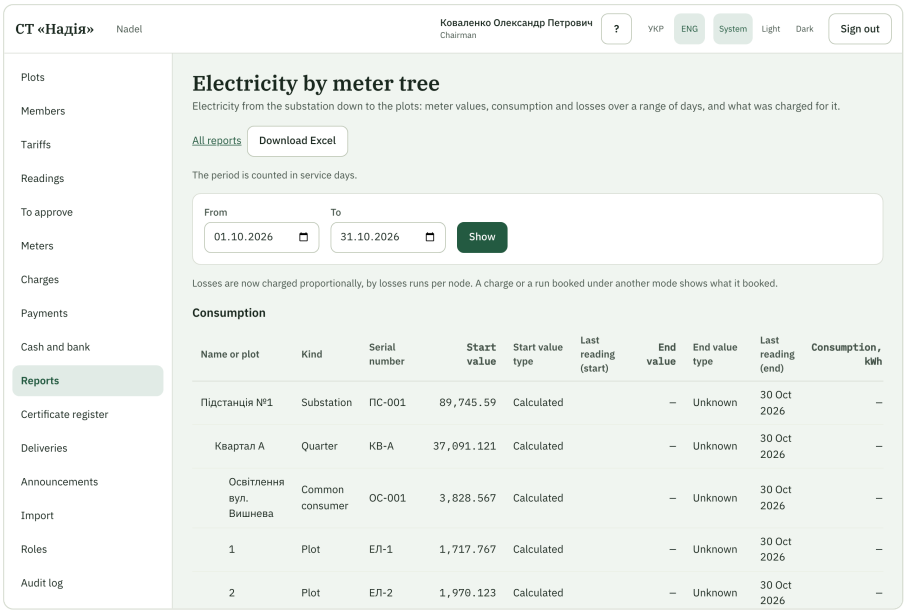
The chairman opens "**Land tax**" for 2026. The rate is 35.00 ₴ per sotka per year, three plots: 600, 800 and 1000 m².

1. The filter "**Year**": 2026. He presses "**Show**".
2. In the table the plots are 6, 8 and 10 sotkas: expected 210.00, 280.00 and 350.00 ₴, 840.00 ₴ in total.
3. The column "Discrepancy" is empty, so exactly as much was charged as expected.
4. "**Download Excel**": the file land-tax_2026-01-01_2026-12-31.xlsx with all rows.

## Screenshots




The report "Charges by month": filters and table




The report "Electricity by meter tree"

## Related pages

- Reports
- Report templates
- Plot account and charges month
- How the reports work

# Report templates

## What the screen is for

If you need your own form (a statement for a meeting, a report for the tax office), prepare it in Excel as usual, and where a value from the system should stand, write a field like `${name.of.field}`. The system replaces the fields with data and keeps your formats, merged cells, column widths and formulas. Here templates are uploaded, updated and archived, and files are generated from them.

## Who sees it

The page is seen by those who have "**Manage report templates**": by default the chairman only. The treasurer generates files by templates from the Reports screen, the section "**Report templates**", and needs "**View reports**" for it.

## Every element

The button "**All reports**" returns to the reports.

### The tax report

A separate card "**Tax report**" with the state "**The template is not added yet**" until the chairman uploads the form with the button "**Upload the template**". The official form is added as an ordinary template, without changes to the program. When the form changes, upload a new version ("**Upload new version**"). A second tax report template cannot be added ("**The tax report template is already added. Upload a new version of it instead.**").

### Other templates

The section "**Other templates**". If there are no templates, you see "**No other templates yet.**".

The button "**Add a template**" opens a form: "**Name**", "**Description**", "**Excel file (.xlsx, up to 2 MB)**" ("**Choose file**"). The button "**Upload**" uploads the template. A valid file is an ordinary .xlsx workbook up to 2 MB. If another template already has this name, you see "**Another active template already has this name.**".

By each template: the status ("**Active**" or "**Archived**"), the version number, who uploaded it and when, and these buttons.

| Button | What it does |
|---|---|
| **Generate** | Opens the window "**Generate a file**" with the fields "**From**" and "**To**" (both days are in the period, the period is no longer than a year). "**Generate and download**" creates the file and downloads it. The system does not store generated files. |
| **Upload new version** | Creates version 2, 3... The latest version is always the one generated. |
| **Versions** | Shows all versions. Each has "**Download**", because old versions can be downloaded. |
| **Archive** | The template stops being offered for generating and stops accepting versions, its name is freed, and its versions stay available for download (see the window "**Archive the template?**"). |

The name of the file you generate contains the template name. The template and its version are recorded in the file properties and in the change log.

### Substitution fields

The block "**Placeholders**": a list of fields with a type, a description and an example. The search is "**Search placeholders**", the button "**Copy**" copies a field. Groups: single values and sets of repeating rows (plots, charging months, electricity meters, water, payments).

- **A field alone in a cell** (`${rows.plots.landTax.expected}` and nothing more) is filled with a number or a date, not text, so formulas over it work. For a date give the cell a date format.
- **A field inside text** ("Land tax for `${period.year}`") is filled as text in the Ukrainian format of numbers and dates.
- **Single values:** `${cooperative.name}`, `${cooperative.officialName}`, `${cooperative.iban}` (the details from the settings), `${period.from}`, `${period.to}`, `${period.year}` (the year of the last day of the period), `${totals.debt}`, `${totals.landTax.expected}` and other totals.
- **A repeating row:** if a row has fields of one set (`rows.plots.*`, `rows.months.*`, `rows.meters.*` or `rows.water.*`), that row is repeated as many times as there are rows in the set. One such row per sheet. Two sets in one row, or two repeating rows on a sheet, are rejected by the system.

There are no names or contacts of people in templates: only data of the cooperative, plots, meters and sums.

### What the system does with a template

- An unknown field (a spelling mistake) does not disappear: it stays in the cell as written, and a sheet "**Warnings**" with the cell and the reason appears at the bottom of the file. The same goes for a field whose value is unknown (no reading, a detail not entered): the cell is empty, not zero. After generating you will see "**The file was downloaded. Warnings in the file: {count}.**".
- The system does not calculate formulas: it writes the fields and leaves formulas to you. Excel or LibreOffice recalculates them when you open the file; a viewer that does not calculate formulas shows them empty.
- Files with macros (.xlsm), active content and embedded objects, damaged ones or ones too large after unpacking are rejected, and nothing is saved. Links to other files are removed from the template with a warning. Charts that refer to a repeating row are not shifted (a warning).

### If a file does not fit

| What is shown | What to do |
|---|---|
| **This is not an Excel file. Choose a file with the .xlsx extension.** | Save the form as an Excel workbook (.xlsx). |
| **The file is too large: the maximum is 2 MB.** | Make the file smaller. |
| **The file contains macros. Save it as a plain .xlsx without macros.** | Save as an ordinary .xlsx without macros. |
| **A sheet has more than one repeating row. Keep one repeating row per sheet.** | Leave one repeating row per sheet. |
| **The template is archived and takes no new versions.** | Create a new template: an archived one accepts no new versions. |

**IMPORTANT**

An archived template does not come back to work, but its versions can be downloaded. So if the form must change, upload a new version rather than archiving the template.

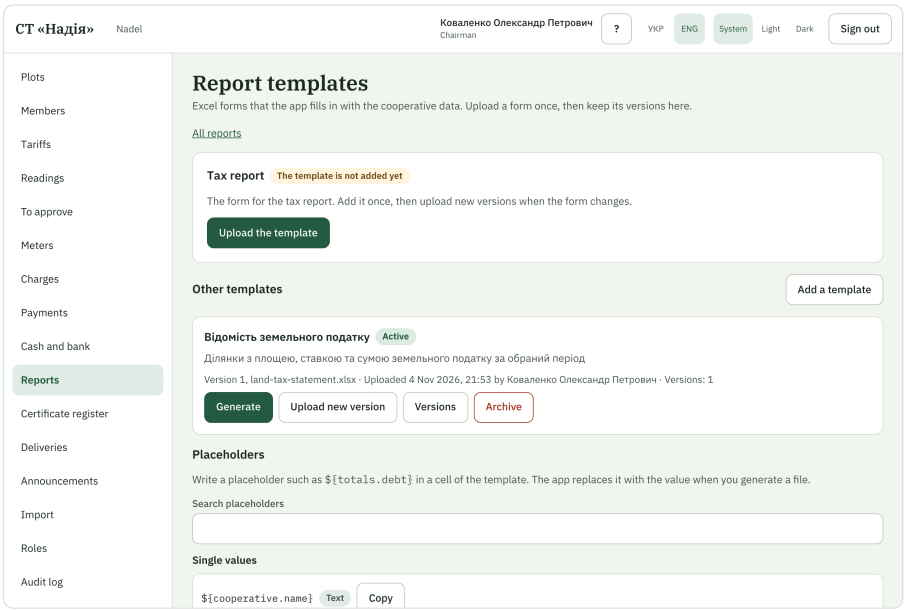
## Worked example

The template is a land tax statement (a rate of 35.00 ₴ per sotka per year; plots of 600, 800 and 1000 m²). In Excel you write:

| Cell | A | B | C |
|---|---|---|---|
| 1 | `${cooperative.officialName}` | | |
| 2 | Land tax for `${period.year}` | | |
| 4 | Plot | Area, m² | Tax for the year, ₴ |
| 5 | `${rows.plots.number}` | `${rows.plots.area}` | `${rows.plots.landTax.expected}` |
| 6 | Total | | `=SUM(C5:C5)` |

1. The chairman presses "**Add a template**", writes a name, chooses the file and presses "**Upload**".
2. He presses "**Generate**", enters the period 01.01.2026 – 31.12.2026 and "**Generate and download**".
3. Row 5 is repeated for each plot: 600 m² gives 6 × 35.00 = 210.00; 800 m² gives 280.00; 1000 m² gives 350.00. The system turned the formula `=SUM(C5:C5)` into `=SUM(C5:C7)`, so "Total" equals 840.00.
4. The heading in row 1 is the full name of the cooperative from its details, in row 2 "Land tax for 2026".

## Screenshots




The "Report templates" page: the tax report, other templates and substitution fields

## Related pages

- Reports

- Report: filters, preview and Excel
- Settings: the cooperative's details

# Cooperative settings

## What the screen is for

Here the chairman sets the rules by which the cooperative works in the system: who sees plot colours, when a plot is red, how losses and common consumers are charged, whether a prepayment goes to new charges by itself, which requisites are printed on documents, which mailbox and which bot send the messages, which water is connected to whom, and which categories the cash book uses. Every panel is saved separately with its own button.

## Who sees it

The screen shows the panels you have a permission for.

| Panel | Permission |
|---|---|
| Plot colours, threshold, losses, payments, common consumers, requisites, delivery | "**Manage cooperative settings**" |
| "**Water supply**" | "**View plots**"; the buttons in it need "**Manage meters**" |
| "**Expense and income categories**" | "**Approve cash book operations**" |
| The link "**Meter tree**" | "**Manage meters**" |

By default the chairman has all of this. The treasurer does not see the settings screen. The menu item "**Cooperative settings**" appears if there is a permission for settings or for managing meters.

Every change of settings is written to the change log. If saving succeeded, "**Settings saved.**" appears above the button.

## Every element

### Who sees plot colours

A panel with the choice "**Who sees plot colours**".

- "**Own plots only**" (the default). **A member sees the colour only of plots they own; other plots are neutral.**
- "**All plots**". **Every member sees the colour of every plot, but never amounts, owners or personal data of other plots.**

The **Save** button is active only when the choice has changed.

> **IMPORTANT**
> Neighbours know whose plot is whose, so the colour effectively shows the debtor. Turn on "**All plots**" only by a decision of the general meeting. Under the choice the system reminds you: "**Show the debt colour of all plots to all members only by decision of the general meeting**". Amounts, owners and personal data of other people's plots are never shown to a member under any setting.

The setting applies to the plot map that will come later. Today members do not see the plot list, so they do not see other people's plots at all, whatever is chosen here.

### The "Red status threshold" panel

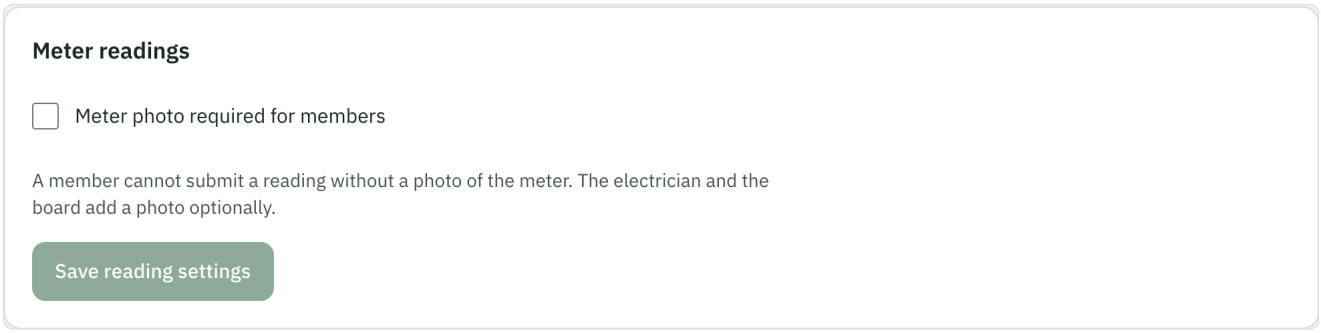
A plot turns red when its debt reaches a certain amount or its oldest debt has lasted a certain number of days. Any other debt is yellow. The fields:

- "**Red from the debt amount, ₴**". An amount with kopecks, for example 1500.00.
- "**Red from the debt age, days**". A whole number of days.

The **Save the threshold** button. Errors: "**Enter an amount such as 1500.00**" and "**Enter a whole number of days**". The new condition applies at once to the plot colours in the list and on the map.

### The "Meter readings" panel

One switch: "**Meter photo required for members**". The hint under it: "**A member cannot submit a reading without a photo of the meter. The electrician and the board add a photo optionally.**" When the rule is on, an owner cannot submit a reading from their account without a photo of the meter and sees the message "**Add a photo of the meter: the board asks for a photo with every reading.**". A reading in the Telegram bot is not changed by this rule. The electrician and the board add a photo if they wish. The **Save reading settings** button saves the choice. How the owner sees it: How to submit a meter reading.




The "Meter readings" panel

**The "Losses" panel**

Losses of a node of the meter tree (see Meter tree). There are two modes in the "**How losses are charged**" field:

- "**Fixed percentage of the consumption**": the percentage from the "**Losses percentage, %**" field is added to the consumption charge of every plot;
- "**Proportional to the consumption**": the node's loss calculation divides the losses between the plots of the node by itself. The percentage does not apply in this mode, the field is disabled, and the system explains: "**In this mode the losses are charged by the losses run of the node, the percentage is not used.**"

The **Save the losses** button. The percentage must be from 0 to 100, for example 5 or 5.25, otherwise the system answers: "**Enter a percentage from 0 to 100, for example 5 or 5.25**". More about the calculation: Losses.

**The "Payments" panel**

The switch "**Allocate prepayment automatically**" (on by default).
**A plot's prepayment is placed on new charges as soon as they are booked. When off, the treasurer allocates it by hand.** The **Save the payment settings** button.

> **IMPORTANT**
> When the switch is on, the free part of a payment goes straight to the oldest debts of the plot and does not stay as a prepayment. A prepayment stays only when the plot has no open debts. You can see this in the payment form.

**The "Common consumers" panel**

How the consumption of every common meter (the well pump, street lighting) is divided between plots.
**How the consumption of each common meter (the well pump, street lighting) is shared among the plots. A consumer without its own split follows the cooperativ**

- The choice "**Default split of the cooperative**": "**Among connected plots**", "**Equally per plot**", "**Among service users**" or "**Not charged**". The **Save the default** button.
- Below, for every common meter, the "**Split**" field: keep the default or choose its own way. If "**Among service users**" is chosen, the "**Service users**" list appears: add the plots that use this consumer with the search "**Add a plot**". The **Save** button. If there are no common consumer meters yet, it says: "**There are no common consumer meters yet**".

More: Common consumers.

**The "Requisites" panel**

**The details printed on invoices and reports.** The fields: "**Official name**", "**Legal address**", "**Bank name**", "**IBAN**". The "**EDRPOU code**" code is read-only. The **Save the requisites** button. If the IBAN is wrong, the system says: "**The IBAN is not valid**".

Requisites are printed in invoices, certificates and reports. Until the IBAN and the registration code are filled in, invoices cannot be built: "**Invoices cannot be built until the IBAN and the ЄДРПОУ of the cooperative are filled in**"

**The "Delivery" panel**

**The cooperative mailbox and the Telegram bot the system uses to send invoices and notices to members.** Next to each channel is a mark "**configured**" or "**not configured**". Until a channel is set up, nothing is sent through it. Who gets an invoice by which channel and how to read the result: see Delivery.

#### E-mail

The simplest way is to create a free mailbox for the cooperative (Gmail or Ukr.net) and use an app password: this is a separate password for programs. It is created in the mailbox settings and does not open the mailbox itself. Most services do not accept the ordinary mail password for such programs.

| Service | What to do in the mailbox | Server | Port and encryption | User and password |
|---|---|---|---|---|
| Gmail | Turn on two-step verification, then create an "app password" | smtp.gmail.com | 587, STARTTLS | the full mailbox address; a 16-character app password |

| Service | What to do in the mailbox | Server | Port and encryption | User and password |
|---|---|---|---|---|
| Ukr.net | In the mailbox settings allow access from mail programs and create a password for them | smtp.ukr.net | 465, SSL | the full mailbox address; the password for programs |

The values are typical: check them in the help of your mail service, they may change.

The fields: " **Mail server (SMTP)** ", " **Port** ", " **Encryption** " (STARTTLS, SSL or " **no encryption** "), " **User** ", " **Password** ", " **Sender address** " and " **Sender name** " (for example "CT Надія": this is how members see the sender).

- The **Save the mailbox** button saves the data. After saving the password is never shown again: the field says " **saved** ". To keep the password unchanged, leave the field empty. The change log shows only that the password was changed.
- The **Send a test letter** button sends a test letter to your own address from your member card (without one the button says there is nowhere to send: " **Your member card has no e-mail address, so there is nowhere to send a test letter.** "). The button works with the saved data, so the system reminds you: " **Save your changes first, then send a test letter.** " The result is " **The mail server accepted the test letter.** " or a refusal by the server in plain words, for example " **Wrong mailbox login or password** ".

**IMPORTANT**

Gmail accepts only an app password from programs, not the ordinary account password. If the " **Password** " field holds the ordinary password, the system says: " **Gmail requires an app password, not the account's usual password.** " First turn on two-step verification in the Google account (an app password cannot be created without it). Then create the password on the page myaccount.google.com/apppasswords (the screen has the link " **Create an app password** ") and paste it into the " **Password** " field. Gmail settings: server smtp.gmail.com, port 587, encryption STARTTLS, user is the full mailbox address, for example st.nadiya@gmail.com. The server's reply, in English, is under " **Details** "; you can show it to whoever helps with the setup.

The system connects only to mail services on the internet and only on ports 25, 465 or 587. An address like localhost or an internal network address (192.168...., 10....) cannot be saved: " **This mail server cannot be used. Enter the server of a mail service (for example smtp.gmail.com) and the port 25, 465 or 587.** "

#### Telegram bot

The chairman creates the bot in Telegram itself: it is free and takes a minute. Above the token field the screen has a short instruction.

1. In Telegram open @BotFather (the official bot for creating bots).
2. Send the command /newbot.
3. When asked for a name, write the name of the cooperative, for example "Надія — рахунки": members see it.
4. Choose a unique user name ending in bot, for example nadiya_rakhunky_bot.
5. BotFather replies with a message containing the token: a long string like 123456:ABC.... Copy it.
6. Paste the token into the " **Bot token** " field and press **Save the token** .
7. Press **Check the bot** . The system asks Telegram and shows the name of the bot or " **The token is not valid** ". The button works with the saved token, so save it first.

The token is the key to the bot: do not give it to anyone. After saving the screen shows only the last 4 characters. If the token is changed, the cooperative's bot restarts in about a minute. If Telegram rejects the token (the bot was deleted or the token reissued), the bot stops until the token is replaced. When Telegram is not available, the system says: " **Telegram is unavailable** " How a member connects Telegram is described in My account.

#### General parameters

- " **Letters per second** ". **From 1 to 10. Fewer letters per second is slower but gentler on the mail server.** Free mailboxes limit the number of letters per day and block sending that is too fast. That is why the default is 1: 120 letters go out in a couple of minutes. Raise the number only for paid mail. If the cooperative is larger than the daily limit, some letters get the refusal " **The mail server refused the letter** ": send them again the next day.
- The switch " **Notify about payments** " (on by default). **A member is told about their payments and reversals.**
- The **Save the options** button saves both the speed and the switch.

### The " **Water supply** " panel

**Which plots take water from the cooperative supply or from a common consumer meter.** For every source (the cooperative's water supply or a common consumer's meter) you see the connected plots with dates. The **Connect plots** and **Disconnect plots** buttons open a form: choose a date, find plots with the search " **Find a plot to add** " and press **Apply** . The action is "all or nothing": if even one plot cannot be changed, none changes. A single plot is connected on its card.

### The " **Expense and income categories** " panel

**Categories for the expenses and other receipts of the cash book. An archived category is not offered in a new entry but stays on old ones.** Two lists: " **Expense categories** " and " **Income categories** ". Next to a category are the buttons **Rename** , **Archive** (or **Restore** for an archived one), **Up** and **Down** , and at the bottom of the list **Add a category** . In the form enter a name and press **Save** . Categories are not deleted, only archived. Errors: " **Enter a name.** " and " **This category already exists.** "

**The "Meter tree" link**

**Substations, quarter meters, common consumers and the plots they feed.** The link opens the meter tree.

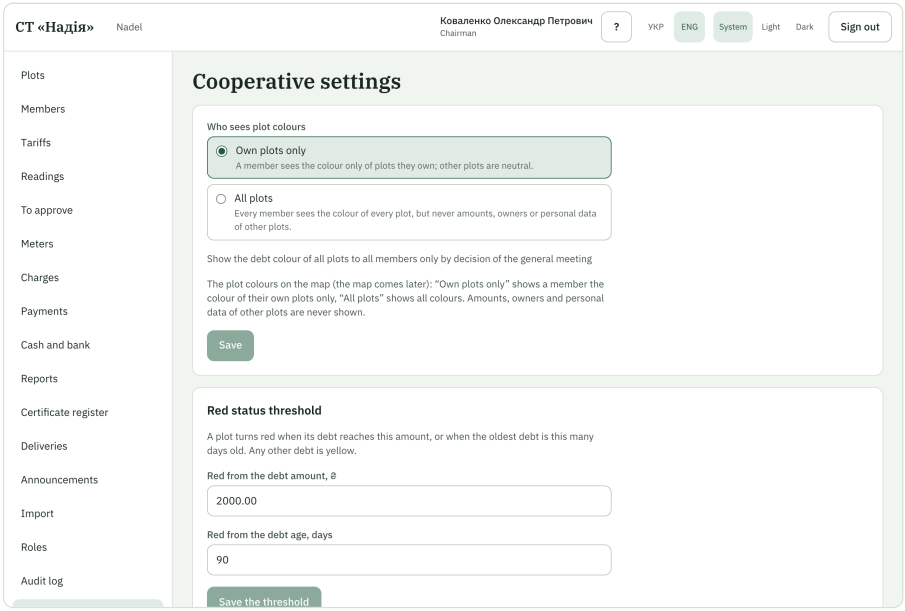
## Worked example

The chairman sets up the cooperative "Nadiya" before the first sending of invoices.

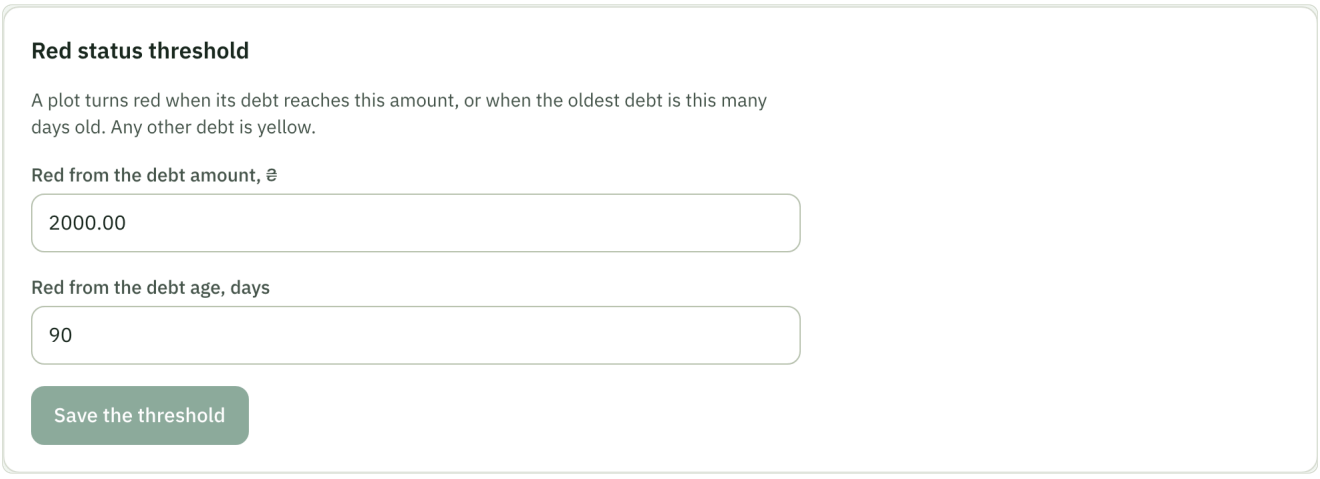
1. In the "**Red status threshold**" panel keeps 1000.00 ₴ and 90 days. A plot with a debt of 1,200.00 ₴ is now red, and a plot with a debt of 400.00 ₴ whose first debt is a week old is yellow.
2. In "**Requisites**" enters the full name, the legal address, the bank and the IBAN. Without the IBAN invoices will not be built.
3. In "**Delivery**" enters the data of the mailbox st.nadiya@gmail.com: server smtp.gmail.com, port 587, STARTTLS, the app password. Presses **Save the mailbox**, then **Send a test letter**. A letter arrives at the chairman's address, and the system says "**The mail server accepted the test letter.**".
4. Creates the bot with @BotFather, pastes the token and presses **Save the token** and **Check the bot**. The mark changes to "**configured**".
5. Leaves the switch in the "**Payments**" panel on. Now, if a plot paid 1,000.00 ₴ against a debt of 750.00 ₴, the remaining 250.00 ₴ becomes a prepayment and goes to new charges by itself as soon as they are booked. If the switch were off, the treasurer would allocate the prepayment by hand.

Plot colours for members stay at "**Own plots only**" until the general meeting decides otherwise.

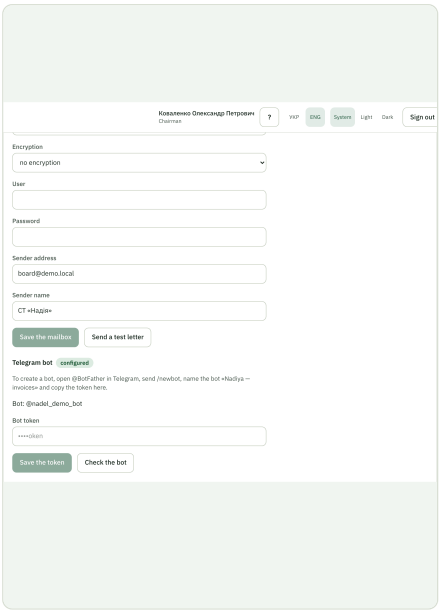
## Screenshots




Cooperative settings




The red status threshold panel




The delivery panel: mail, Telegram bot, parameters

## Related pages

- Roles and permissions
- Delivery
- My account
- Enter a payment
- Losses
- Common consumers
- Meter tree
- Charge types
- Change log

# Roles and permissions

## What the screen is for

Here the chairman sets who can do what. A role is a set of permissions, and a permission is a separate action, for example "enter payments". A member is given one or several roles and sees in the system only what their roles allow.

## Who sees it

The screen is seen by whoever has the permission "**Manage roles**". By default this is the chairman. Roles are given to members on the member card with the "**Edit roles**" button.

## Every element

### The list of roles

Every role is a row with a name, the mark "**Base role**" or "**Custom role**", a description and the number of members with this role. On the right are the **Edit** button and, for custom roles, **Delete**. If there are no roles yet, it says: "**No roles yet.**".

- **Base roles** are "**Chairman**", "**Treasurer**" and "**Member**". They cannot be deleted. They can be renamed: the name and description change for all languages. The permissions of the treasurer and the member can be changed, those of the chairman cannot.
- **Custom roles** are created with the "**Add role**" button. They have a name, a description and any set of permissions. A person can have several roles; their permissions are then added together.

### The "**Add role**" button and the role form

The "**New role**" form (for an existing role, its name) has these fields.

- "**Name**". For a base role there is a hint about the standard name under the field. The **Restore standard name** button cancels the renaming. If another role already has this name, the system warns: "**Another role is already shown under this name.**".
- "**Description**". For base roles the **Restore standard description** button returns the standard description.
- "**Permissions**". A list of tick boxes: next to each permission are its name and an explanation. Tick the ones you need. For the chairman all boxes are always ticked, and the system says: "**The chairman always has every permission.**"
- The **Save** and **Cancel** buttons. After saving: "**Saved**". Everything is recorded in the change log.

### The "**Delete**" button

The system asks "**Delete this role?**" and waits for confirmation. After deleting: "**Role deleted**".

### Error messages

| What is shown | What to do |
|---|---|
| **A role with this name already exists.** | Choose another name. |
| **This role is assigned to members and cannot be deleted.** | First remove the role from all members on their cards. |
| **Base roles cannot be deleted, and the chairman's permissions cannot be changed.** | The chairman's permissions cannot be changed. Adjust the other roles instead. |
| **The cooperative must keep at least one chairman.** | Give the chairman role to another member before taking it from the last one. |

### Permissions

Here are all the permissions that can be given to a role. The right column shows who has them by default. The chairman always has all permissions. Custom roles and the role "**Member**" have nothing until you tick it.

| Permission | What it gives | By default |
|---|---|---|
| **View members** | See all members, their contacts and notes | chairman, treasurer |
| **Edit members** | Add, change, archive and restore members | chairman |
| **Invite to the system** | Invite members to sign in, resend an invitation and switch access off | chairman |
| **Manage roles** | Create, change and delete custom roles, change the permissions of base roles, give roles | chairman |
| **View audit log** | Read the change log | chairman, treasurer |
| **View plots** | See all plots with details | chairman, treasurer |
| **Edit plots** | Add, change and archive plots | chairman |

| Permission | What it gives | By default |
|---|---|---|
| **Edit ownership** | Add, change and correct owners | chairman |
| **Run import** | Load data from Excel | chairman |
| **Manage cooperative settings** | Change the settings of the whole cooperative | chairman |
| **View tariffs** | See charge types, tariffs and versions | chairman, treasurer |
| **Manage tariffs** | Create and change charge types, tariffs and versions | chairman |
| **View meters** | See the meter register and tree and their readings | chairman, treasurer |
| **Manage meters** | Register, change, replace and remove meters; build the tree; connect plots to water | chairman |
| **Enter readings** | Enter, correct and delete readings, for example on a round | chairman |
| **Approve readings** | Approve or reject readings passed by members | chairman |
| **View charges** | See the account of every plot: charges, debt by type, the calculation of every charge | chairman, treasurer |
| **Run charges** | Enter opening balances and run charges | chairman |
| **Reverse charges** | Reverse a charge with a reason | chairman |
| **Close periods** | Close charge months | chairman |
| **Recalculate charges** | View and run recalculation of past charges | chairman |
| **View reports** | Open and download reports, build files from templates (the debtors report also needs "**View members**") | chairman, treasurer |
| **Manage report templates** | Upload, replace, download and archive report templates | chairman |
| **View payments** | See payments, their allocation and the payments report | chairman, treasurer |
| **Enter payments** | Enter cash and bank payments, split them between plots, reallocate, allocate prepayment | chairman, treasurer |
| **Reverse payments** | Reverse a payment with a reason | chairman |
| **Issue invoices** | Build, reissue, print and send invoices with a QR code | chairman, treasurer |
| **Issue certificates** | Issue and annul certificates of the state of settlements | chairman |
| **Send announcements** | Send announcements by e-mail and in Telegram and print them | chairman |
| **View cash and bank** | See the cash desk and the bank, journals, expenses, cash counts, reconciliations and cash reports | chairman, treasurer |
| **Enter cash book entries** | Enter expenses, bank handovers, other income, cash counts, statement figures | chairman, treasurer |
| **Approve cash book operations** | Enter opening balances, book differences, reverse cash entries, keep expense and income categories | chairman |

**IMPORTANT**

A change of permissions takes effect immediately after saving: if you remove a permission, the person loses both the button and the access to the data. The check is done by the system, not only by the screen, so hiding a button is not enough. A change of a role's permissions is written to the change log, which shows the added ("+") and removed ("−") permissions.

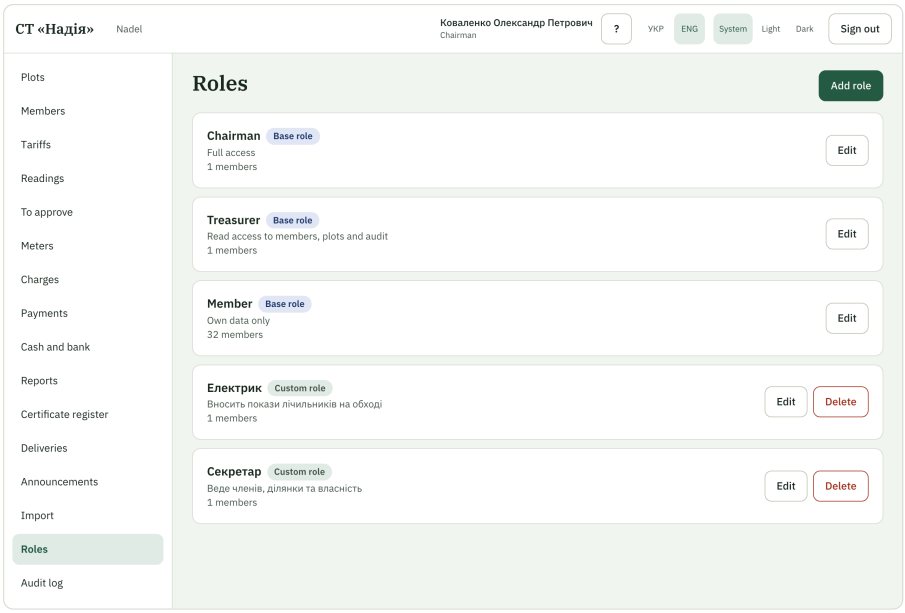
## Worked example

The cooperative has an electrician who only takes readings.

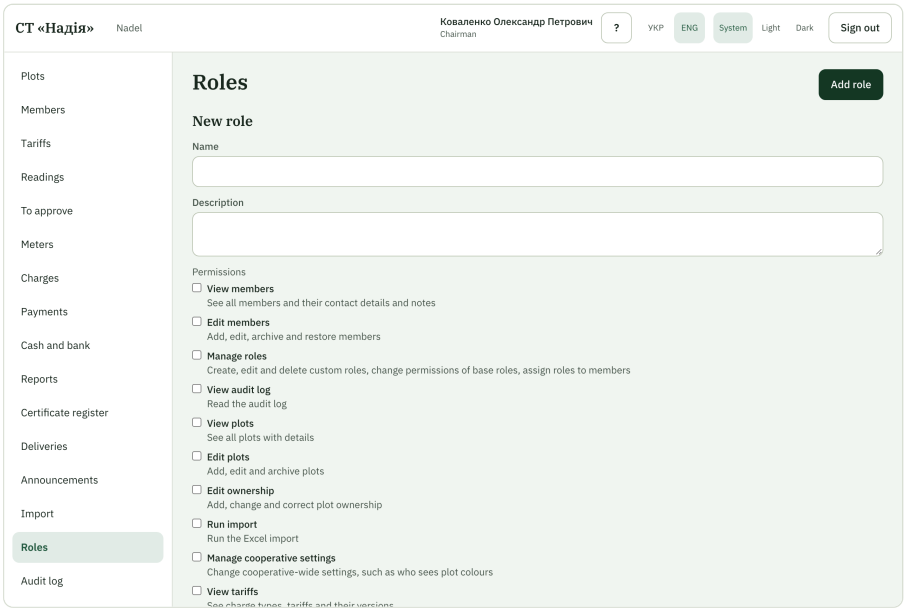
1. Open "**Roles**" and press **Add role**.
2. In the "**Name**" field enter "Electrician", in "**Description**" briefly what the role is for.
3. In "**Permissions**" tick only "**View meters**" and "**Enter readings**". Press **Save**.
4. Open the member card, press **Edit roles** and add the new role to "**Member**".
5. The member signs in again: in the menu they see "**Readings**" and "**Meters**", and the other sections are not shown. The row of this role in the list shows "Members: 1".

In the demo cooperative this is the role "Електрик" (Electrician; held by Tkachenko Serhii Ivanovych, currently without permissions) and also the custom role "Секретар" (Secretary) with permissions to view and edit members, plots and ownership.

## Screenshots




The list of the cooperative's roles




The new role form with the list of permissions

## Related pages

- Member card: giving roles
- Members
- Change log
- Getting started

# Change log

## What the screen is for

Every action of the board is written to the log: who, when and what changed, with old and new values. The log helps to find out where a change came from without relying on memory. Log records are never changed or deleted. Data entered by demonstration scripts is not shown in the log: only the actions of people in the system.

## Who sees it

The log is read by whoever has the permission "**View audit log**": the chairman and the treasurer. The old and new values of phones, e-mail, address, Telegram, Viber and notes of a member are seen only by those who also have "**View members**". Everyone else sees that the field changed, and the word "**hidden**" stands instead of the values. Other fields, such as the name, are shown as usual. The mailbox password and the Telegram bot token are shown to nobody: the log says only "**changed**".

## Every element

### Filters

- "**Kind of record**": "**All**" or one record kind: **Member**, **Role**, **Plot**, **Settings**, **Ownership**, **Charge type**, **Tariff**, **Meter**, **Meter reading**, **Charge**, **Losses run**, **Common consumers run**, **Recalculation** and **Payment**.
- "**From**" and "**To**": the limits by date. If a date is wrong, the system shows it next to the field.
- The marks "**Only this record**" and "**Only this author**" appear when the log is filtered to one record or one author. The cross on a mark ("**Remove filter**") cancels that filter. The log filtered to one member is opened by the "**Change history**" button on the member card.

Filters are kept in the page address, so a link to a filtered log can be shared. When there are no records yet, it says: "**No entries yet**", and when a filter found nothing: "**Nothing found**". A page holds fifty records, with a page switch at the bottom.

### Columns

| Column | What is in it |
|---|---|
| **Time** | The date and hour of the action. |
| **Author** | Who did it. Clicking the name leaves only the records of this author. Actions done by the system itself are marked "**System**". |
| **Action** | What was done, for example "**Owners changed**", "**Plot updated**", "**Payment reversed**" or "**Settings changed**". |
| **Record** | The kind of record and the record itself. For members, plots, tariffs, meters, recalculations and others it is a link to their card. If the record was deleted, it says "**Deleted or unnamed record**". |
| **Changes** | The old and new value of every changed field, joined by an arrow. Only fields that really changed are shown. For a role you see the added ("+") and removed ("−") permissions. |

The log records almost every action that changes data: adding and changing members, roles, plots, owners, settings, import, charge types and tariffs, meters and readings, charges, reversals, month closing, recalculations, payments, reports, invoices, certificates, delivery, the cash desk and the bank. In particular the log holds:

- invoices: building the invoices of a month (the month, how many were built, how many skipped) and reissuing an invoice; certificates: an issued certificate and an annulled certificate with the reason. Downloads of documents are not recorded;
- delivery: a change of the settings (which ones, without secrets), "Invoices sent" (the month and the counts by channel), "Announcement sent" (to whom and how many), connecting and disconnecting a member's Telegram. Single sends do not enter the log: they are on the Delivery screen;
- cash and bank: the opening balance, entering and reversing expenses, bank deposits and other income, cash counts and the booking of their differences, the statement figures, bank differences, "Reconciled" and the reopening of a month (by the system, with a line showing who reopened it), adding, renaming, moving and archiving items. The counterparty and the purpose are visible there only to those who see members' personal data.

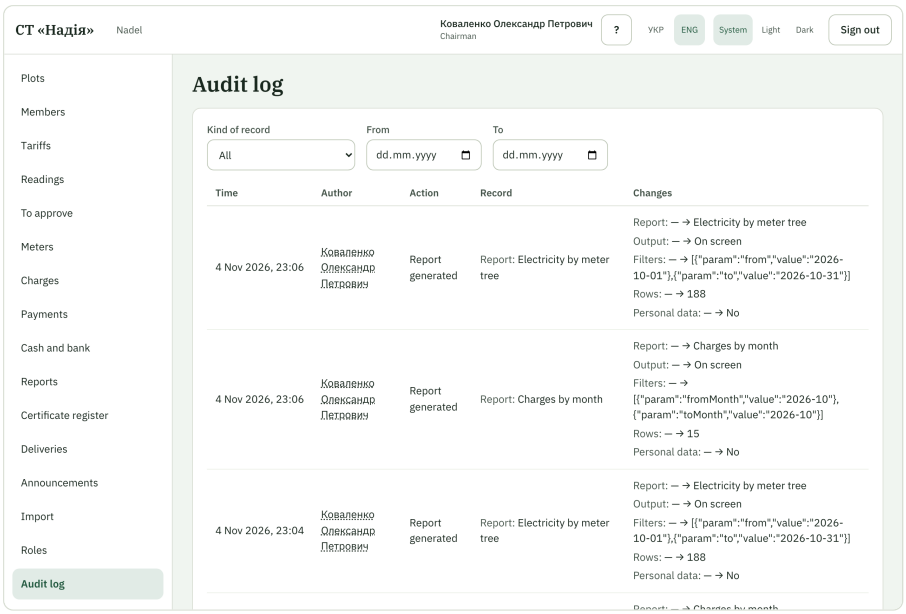
## Worked example

The chairman wants to find out who changed the owner of plot 9.

1. Open "**Audit log**" and in "**Kind of record**" choose "**Ownership**".
2. Find the record with the action "**Owners changed**", the record "Plot 9" and the changes: the owners, the "**Owners from**" date, the basis.
3. Click the author's name: only their actions stay in the log.
4. Click "Plot 9" in the record column to open the plot card.

Suppose the chairman created a custom role that reads the log but does not have the permission "**View members**". Such a person will see that the "**Phones**" field of a member changed, but will read "**hidden**" instead of the numbers. The chairman sees both the old and the new number.

## Screenshots




The change log with filters and records

## Related pages

- Member card
- Plot card
- Roles and permissions
- Cooperative settings

# Core principles

- **Only the server counts money** — The web page and the app only show amounts, debts and statuses; they never calculate them. One formula, in one place.
- **Kopecks** — Hryvnia with two decimals. Intermediate calculations are exact, never rounded. An amount is rounded to kopecks **once**, half up, only when a charge is produced.
- **A rate is not money** — A tariff rate has up to four decimals (1.4532 ₴ per m²) and is never rounded. A fifth decimal is rejected, not cut.
- **Nothing is erased** — Charges, owners and readings are never edited in place. A mistake is fixed by a new record (a reversal, a new version, a new reading); the old one stays in the history.
- **Reports calculate nothing again** — A report only selects, groups and adds up amounts already booked under these same rules, so every amount in a report can be reconciled to the kopeck with a plot account or the month screen (the page Reports and templates).
- **Audit log** — Every board action is recorded with old and new values.
- **One language** — The interface shows only the selected language, never a mix.
- **Minimal personal data** — No tax numbers, passports or dates of birth; contacts are stored encrypted; personal data never goes to the server logs.

> **Why round once**
>
> A rate of 0.05 ₴ for 15 + 15 days of a 30-day month is exactly 0.05 ₴. Rounding each half separately gives 0.03 + 0.03 = 0.06 ₴, a kopeck more. Over thousands of plots and months such kopecks add up.

# Tariffs and the expected amount

A charge type (membership fee, electricity, land tax, water, special-purpose fees) has one or more tariffs. A tariff has **versions**: each applies from its date until the day before the next version. Two versions never start on the same day.

| | |
|---|---|
| `units` | `per plot → 1 · per sotka → area m² / 100` |
| `period share` | `monthly 1 · quarterly 1/3 · yearly 1/12` |
| `month amount` | `rate × units × period share × (days the version applies in the month) / (days in the month)` |
| `one-time fee` | `the whole rate in the month of its due date, 0.00 in other months` |

## Worked examples

| Case | Given | Result |
|---|---|---|
| Membership fee | 50 ₴/sotka/year, plot 600 m² | 50 × 6 / 12 = 25.00 |
| Odd area | 50 ₴/sotka/year, 612 m² | 50 × 6.12 / 12 = 25.50 |
| Rounded at the end | 33.3333 ₴/sotka/year, 700 m² | 19.444425 → 19.44 |
| Per plot, quarterly | 300 ₴ per quarter | 100.00 per month |
| Version change mid-month | 50 ₴ until the 15th, 60 ₴ from the 16th, 600 m², 30 days | 12.50 + 15.00 = 27.50 |
| Special-purpose fee | 1,200 ₴, due 01.05.2026 | 1,200.00 in May, 0.00 otherwise |
| Land tax | 35 ₴/sotka/year, 600 m² | 17.50 per month |

## Which plots

- **All plots** minus exclusions (each with a reason), or **Selected plots**. Never archived plots or common land.
- Per-kWh and per-m³ tariffs only for plots with an active meter of that resource. **No meter, no charge** (confirmed by the cooperative's chairman).

## Correcting a version

A version can be corrected or deleted **while no charge uses it**. After that a correction goes through recalculation (the page Corrections and recalculation): charges already issued to people never change silently.

# Meters and the tree

A meter measures one resource: electricity (kWh) or water (m³). Meter values are not money: up to three decimals.

The meter tree: substation → quarters → plots; common consumers are nodes of their own. There may be any number of substations, quarters and wells.

- The chairman builds the tree; nothing about it is hard-coded. Moving a meter under another node takes effect from a date; the tree's history is kept, so a past range is calculated with the tree as it was then.
- Any number of substations, quarter meters and common consumers (wells included); every quarter meter and every well has its own list of plots.
- A plot has at most one active meter per resource.

## Replacing a meter

A replacement is one action: the final reading of the old meter, the serial number and the initial reading of the new one. Consumption of a period with a replacement is the sum over both meters.

> **Example**
>
> Old meter: 8,140 at the start, removed at 8,205. New one: installed at 3, reads 69 at the end. Consumption: (8,205 − 8,140) + (69 − 3) = 65 + 66 = **131 kWh**.

A counter wrapping past zero (99,999 → 00,012) is "a replacement of the meter by itself": same serial number, final 99,999, initial 0. The same rule then applies.

# Readings and consumption

**A reading is a meter's value on a date**, not "the reading for a month". The electrician comes when they can: after a few days or after a few months. The person enters the date; it defaults to today, with an unobtrusive hint.

## Reading statuses

| Status | Meaning | Counted |
|---|---|---|
| Approved | Entered by the electrician or the board, imported, or a member's reading approved | yes |
| Pending | Submitted by a member, waits for approval | no |
| Rejected | With a reason the owner sees | no |
| Superseded | Replaced by a new reading for the same date or a correction; kept in the history | no |

## Intervals and the value on a date

- **Interval consumption** is the difference between a reading and the previous reading of the same meter.
- **The value on any date**: the real reading if there is one on that date; otherwise a linear interpolation by days between the neighbouring readings, marked `calculated`; with no reading after the date yet, unknown.
- **Consumption for a range "from … to …"** = the value at the end − the value at the start.

Real readings (green) and values calculated at the range boundaries (dashed)

> **Where 1,150 comes from**
>
> Between 01.03 (1,000) and 01.04 (1,310) there are 31 days and 310 kWh. 16.03 is day 15: 1,000 + 310 × 15 / 31 = 1,150. Calculated values are always marked so that nobody takes them for a real reading.

## Checks

- A reading lower than the previous one (or higher than the next one, when entered back-dated) is refused. A lower value is valid only as the initial reading of a new meter.
- A date in the future or outside the meter's working period is refused.
- Daily consumption three times the average of the last 180 days: the reading is accepted with the warning "Check the reading".
- A reading can be corrected or deleted while no charge uses it; after that, through recalculation.

## Electricity by reading

The days of an interval run from the day after the previous reading to the day of the new one, both included. When the tariff changes inside the interval, the kWh are split by days, each part times its rate, and the total is rounded once.

> **Examples**
>
> Readings 112.4 (22.09) and 205.7 (04.10): 93.3 kWh × 4.32 = **403.06 ₴** for 23.09–04.10 (12 days).
>
> Readings 1,000 (01.10) and 1,300 (31.10), tariff 4.32 until 16.10 and 5.00 from 17.10: 15 days each, 150 × 4.32 + 150 × 5.00 = 648.00 + 750.00 = **1,398.00 ₴**.
>
> A meter replacement gives two charges: 65 kWh = 280.80 and 66 kWh = 285.12.

# Charges and closing months

A charge is an entry in the plot's account: charge type, amount, month, the days it pays for (service period) and how the amount was obtained (rates, units, kWh, the readings and tariff versions used).

## Where charges come from

| Source | How it arises |
|---|---|
| Recurring tariffs | The board presses **Charge the month**: preview → "Post". A rerun adds only what is missing, never duplicates. |
| Electricity | Every approved reading at once gives a charge for the interval since the previous reading. |
| Network losses | Depending on the chairman's setting: a percentage on every electricity charge, or a board run for a range (the page Network losses and common consumers). |
| Common consumers | A separate board run for a range, a separate line in the account (the page Network losses and common consumers). |
| Opening balance | A plot's debt on the day the cooperative starts in the system — by hand or with an Excel sheet. |

## Months and closing

- A month is open or closed. Months are closed only in order; a closed month never reopens.
- Before closing the board sees a check list: is the month posted, what was not charged, which meters have not been read for long. None of it blocks closing — a person decides with everything in view.
- A charge for an already closed month (a late reading, say) goes to the earliest open month, marked "for September, charged in October".

## Reversals and the **not charged** queue

- Charges are never changed or deleted — the database itself enforces it. A wrong charge is cancelled by a **reversal** (the same amount with a minus and a reason). A whole losses or common-consumer run can be reversed together.
- What the system could not charge goes to the **not charged** queue with the reason: no tariff for some days; two tariffs of one type on a plot on the same day; no node readings. Once the cause is fixed, **Retry**.

## Debt

```
open amount of a charge = the charge with its adjustments (reversed: 0.00) − Σ allocated to it
debt per charge type    = Σ open amounts of the charges of that type
plot debt               = Σ debts per charge type
prepayment              = Σ payments (net of reversals) − Σ allocated
```

A payment pays specific charges, so the debt is what is still unpaid. How a payment is allocated and what happens when a charge is reversed is on the page How the system allocates payments.

## On the screen: **Charges**

The **Charges** menu item (permission **View charges**) has four tabs: Months, Not charged, Losses and Common consumers.

1. Open a month. The page shows what was charged per charge type and the check list (**Before closing**): is the month run done, how many items wait in **Not charged**, which meters have no reading for more than 60 days.
2. **Charge the month** (permission **Run charges**) first shows a **preview**: the number of plots, the new charges per charge type and the items that cannot be charged. **Book** books exactly what was shown. If anything changed meanwhile the system asks to preview again.
3. A rerun creates only what is missing: a plot or a tariff added later gets its charge, nothing is duplicated. A month after the current one cannot be charged.

## **Not charged**

One list of everything the system could not charge, with the reason and a link to fix it: no tariff for every day, two tariffs on the same day, a plot outside every tariff, an unknown node or consumer value. **Retry** tries one item again; the next month run retries the tariff and reading items. An item disappears when the retry succeeds. Items never block closing a month.

## Closing a month

**Close the month** (permission **Close periods**) shows the check list once more and asks to confirm. Only the earliest open month can be closed (**Close the earlier months first** otherwise). A closed month never changes; late entries go to the earliest open month.

## The plot account, reversals and opening balances

On the plot card the **Account** panel shows the debt per charge type and the charges, newest first, with a date filter. **Details** opens the calculation of a charge: kWh, rate, readings, days, the losses percentage. A charge booked later than the days it pays for reads "for 13.09–28.09, charged in October 2026".

- **Reverse** (permission **Reverse charges**) asks for a reason and creates an entry for the same amount with a minus. A charge is reversed once; a reversal cannot be reversed; a charge in a closed month is corrected by an adjustment (a later phase), the button is not offered.
- **Enter an opening balance** (permission **Run charges**): charge type, amount, "debt counts from" (not later than today) and an optional reason. Many plots at once: the **Opening balances** sheet of the Excel import (the page Excel import). One opening balance per plot and charge type; to change it, reverse it first.
- Reversing a whole losses or common consumers run is done on its page (below).

## Plot colours on the map and in the list

| Colour | When |
|---|---|
| Red | The debt is above the "red from amount" or the oldest debt is more than "red from days" old (strictly greater) |
| Yellow | Any other debt |
| Green | No debt |
| Grey | No owner, or common land |

The thresholds are set in **Cooperative settings** (defaults 1,000.00 ₴ and 90 days; the demo cooperative uses 2,000.00 ₴ and 90 days).

## The treasurer and the member

The treasurer sees months, accounts and the lists but has no buttons. A member sees only the account of their own plots under **My charges** (the page My account), read-only, with the calculation of every charge; other plots' amounts are never shown.

# Network losses and common consumers

## Losses: two modes, the chairman chooses

- **Fixed percentage**: every electricity charge gets, for example, 5 % kWh on top. 65 kWh → losses 3.25 kWh = 14.04 ₴.
- **Proportional by node**: the board chooses a range and runs the calculation. For every node, bottom-up: losses = node consumption − the sum of its children's consumption. The losses are shared among the plots under the node in proportion to their consumption.

  **Example: a quarter and the substation**

  Quarter A: 1,100 kWh, plots P1 = 600 and P2 = 400 below it → losses 100: P1 60 kWh (259.20 ₴), P2 40 (172.80 ₴).

  Quarter B: 1,050, plot P3 = 1,000 → losses 50 → P3.

  Substation: 2,300; children A 1,100 + B 1,050 + well 50 = 2,200 → losses 100 shared 600 / 400 / 1,000 of 2,000: P1 30, P2 20, P3 50 kWh.

- **Negative losses** (the plots together "used" more than the node shows: a wrong reading, different reading dates, a faulty meter): nothing is charged, the board sees a warning with the numbers.
- **An unknown value** (no reading of the node or a child after the range end): that node and the nodes above it are skipped with a warning; the rest of the tree is charged.
- Two runs cannot cover the same days of the same node.

## Common consumers — a separate line

Well pumps, street lighting, the office. Their consumption is a separate charge type **Common consumers** with its own per-kWh tariff and its own board run. The split is chosen per consumer:

| Split | Who pays | Example |
|---|---|---|
| Connected plots | Plots connected to that well, for their connected days | Pump 60 kWh, three connected plots: 20 kWh each = 86.40 ₴; an unconnected plot nothing |
| Equally | Every active plot, with or without a meter | Lighting |
| Service users | The plots listed in the settings | Lighting 30 kWh, 3 plots: 10 each = 43.20 ₴ |
| Not charged | Nobody — covered by the membership fee | Shown in reports |

## On the screen: Losses and Common consumers

Both tabs of **Charges** work the same way: pick the range **From** and **To** (not later than today), press **Preview**, read the table and warnings, then **Book** (permission **Run charges**). The booking is written in the month of the **To** day, or in the earliest open month after it. The history of past runs is below, with **Details** and **Reverse the run** (permission **Reverse charges**, a reason is required).

- The losses mode is chosen in the settings (**Losses**): with a fixed percentage there is nothing to run here and the page says so.
- If some nodes were already charged for these days the whole run cannot be booked; **Preview without the charged nodes** books the rest.
- A node with an unknown value goes to **Not charged**; when readings arrive, **Retry** charges it alone.
- Common consumers need a tariff of the **Common consumers** charge type; without it the preview says **No tariff** and nothing can be booked. The split of each consumer, its service users and the default are in the settings.

# Water and connections

Water works like electricity: **a plot not connected to the well pays nothing for water**.

- There may be several wells: each plot is connected to a particular well, and each well's pump is shared only among its own plots.
- A plot has a history of water connections: the source (a well or "the cooperative's water supply"), from a date, until a date.
- Any water tariff — by water meter (m³) or fixed (per plot per season, say) — is charged only for the connected days. A per-m³ tariff also needs a water meter.
- A connection starting or ending mid-month splits the amount by days: 300 ₴ per year, connected from 16.10 (31 days) → 300 / 12 × 16 / 31 = **12.90 ₴**.
- The well pump's electricity is shared only among the plots connected to it, by plot-days: P1 and P2 the whole month, P3 from 16.10, pump 60 kWh → 30 + 30 + 15 = 75 plot-days → P1 24, P2 24, P3 12 kWh.
- Connections are entered on the plot card, in bulk on the well's page, or with the `Water` column of the import.

## On the screen

A plot's connection is on its card, panel `Water` : `Connect` (source and the first connected day) and `Disconnect` (the last connected day); the history is listed below. Many plots at once are connected or disconnected on the `Water supply` section of `Cooperative settings`, per source. Without a connection a plot is never queued in `Not charged` for water and pays nothing for it.

IMPORTANT

**Two things to know**

- If a well meter has been removed, its lasting connections are no longer offered in the settings: end them on each plot's card.
- A plot with a lasting water connection cannot be archived or made common land: end the connection on the plot card first.

# Corrections and recalculation

A back-dated tariff version, a correction of a used version or of a used reading — one mechanism:

- **Open month**: the old charge is reversed and a new one created.
- **Closed month**: nothing in it changes; the current open month gets an adjustment for the difference, referring to the original month.
- A reason is required. Before posting, a preview: how many plots and how much.
- If a debt goes down after it was already paid, the excess becomes a prepayment, never a "negative debt".

## On screen

**Recalculations** is a tab of **Charges** (viewing needs **View charges**). It is a log: when, the cause (manual, tariff version, reading), who booked it, the explanation and the **Total** difference. **Open** shows one recalculation: the scope, the totals by charge type and the lines "was charged → should be → difference", the action (**Reversal and new charge**, **Adjustment**) and the entries created, with a link to the plot account. A recalculation is never changed or undone: a mistake is corrected by the next recalculation.

**New recalculation** (**Recalculate charges**, the chairman only): choose **From month** and **To month** (not later than the current month, at most 24 months), the charge types and the plots (**All** or "Chosen"), and press **Preview**. The preview step shows what would change: the plots, the totals by charge type and the lines with their reasons (**A tariff version changed**, **The plot area changed** ...). Nothing is written until you enter a **reason** (required, up to 500 characters) and press **Book the recalculation**; exactly the preview you saw is booked. If the charges changed in the meantime, the preview is refreshed and must be confirmed again. When the charges already match the data, the preview says **No changes** and there is nothing to book.

**Correcting a used tariff version** is done on the tariff card: **Edit** or **Delete** the version. When charges already use it, the same preview step opens instead of **Version corrected**: the change of the version and the recalculation are saved together or not at all. The same goes for a new back-dated version.

**Correcting a charged reading** is done on the meter card: **Correct** or **Delete**. The chairman gets the preview step (**Recalculation for the reading**) with a reason; the electrician (may enter readings, may not recalculate) sees **The reading has already been charged — ask the chairman** and nothing changes.

**On the plot account** the adjustment of a closed month is a separate row "Adjustment for Sep 2026: −₴5.00" with the reason, the links **Original charge** and **Recalculation** and the badge **Adjustment**. The checkbox **Adjustments only** keeps only such rows. A member sees the adjustments of their own plots (the reason, but not the recalculation). The demo data has such a row on plot 2.

Screens that change data affecting past charges (plot area, tariff scope, water connection, moving a meter in the tree) recalculate nothing themselves and remind you: "Past charges have not changed. If needed — Charges → Recalculation". Only those who may recalculate see the hint.

**Example**

From 1 September 60 ₴/sotka/year was entered by mistake instead of 50; a 6-sotka plot. September is closed, October open.

| | Was | After the correction |
|---|---|---|
| September (closed) | 30.00 | Unchanged; in October an adjustment −5.00 "for September" |
| October (open) | 30.00 | Reversal −30.00 and a new charge 25.00 |
| Total | 60.00 | 50.00; if 60.00 was paid, a prepayment of 10.00 |

IMPORTANT

**Known limit**

**To approve** has no recalculation step. If a reading is already approved and charged, correct it on the meter card, which has the preview step.

# Reports and templates

Reports put into one table what the system already knows: charges, payments, debts, electricity, water, corrections. Every report is shown on screen and can be downloaded as an Excel file. Besides the built-in reports, the chairman can upload their own Excel form (a template) and the system fills it in with the cooperative's data.

> **Reports introduce no new rules**
>
> A report calculates nothing "its own way": it only selects, groups and adds up the amounts already booked under the rules of chapters 18-25 and 27. So any amount in a report can be reconciled to the kopeck with a plot account or the billing months screen. Amounts in reports are always **effective** amounts: charges plus adjustments minus reversals.

## Who sees reports

| What | Permission | Who has it by default |
|---|---|---|
| Open and download reports, generate files from templates | View reports | Chairman, treasurer |
| The **Debtors** report (this permission is needed as well) | View members | Chairman, treasurer |
| The **Payments** report (this permission is needed as well; the note about the payer is shown only to someone who also has **View members**) | View payments | Chairman, treasurer |
| Upload, replace, download and archive templates | Manage report templates | Chairman only |
| The four cash book reports: **Cash book**, **Bank movements**, **Income and expense**, **Expenses** (only this permission is needed, not **View reports**) | View cash and bank | Chairman, treasurer |

A member and the electrician have no **Reports** item in the menu, and a direct address shows a message that they have no access. You can give your own role these permissions in **Roles** (the page Roles). A role with **View reports** but without **View members** sees every report except **Debtors**.

## The Reports screen

Menu → **Reports**. At the top are the cards of the reports available to you, below them the templates. A card has the report's name and one sentence about it; a click opens the report.

- **Filters** are at the top of a report page; press **Show** to refresh the table. Under the filters it is written what the period counts by: *billing months* (the month a charge belongs to in the ledger), *service days* (the days a service was provided) or *reading dates*. They differ: an adjustment booked in March for January stands in March in "billing months" and in January in "service days".
- **Preview**: the screen shows the first 200 rows and the totals; the line "Showing first … of … rows. The Excel file has all of them" tells how many rows the report has in all.
- **Download Excel** gives a file with all rows and the same totals. It has one sheet per report (some reports have two), a header block (cooperative, report name, filters, date and time in Kyiv time, who made it) and a frozen header row; numbers are stored as numbers and dates as dates. The file is in the interface language. The file name is the report code and the period in Latin letters, for example `land-tax_2026-01-01_2026-12-31.xlsx`.
- **The limit**: a report with more than 20 000 rows on a sheet is not produced; you see "The report has more than 20 000 rows on a sheet. Narrow the filters and try again".
- A value that is not known (for example a reading at the start of the period) is shown **empty** with the reason, never as zero. **Plot balances** and **Charges by month** have **Paid** columns (the allocation of payments), **Plot balances** also **Prepayment**; the separate **Payments** report lists the payments themselves.
- Every report that is made is recorded in the audit log: who, which report, the filters and the row count; the content is not recorded.

## The eight reports

- **Charges by month** — What was charged in each month, by charge type. Filters: **From month**, **To month**, charge types, street. A row is a month and a type: **Plots**, **Charged**, **Opening balances**, **Reversals**, **Adjustments** and **Effective total** (the amount you see on the month screen). The second sheet, **By service period**, groups the same money by the month of the service.
- **Plot balances** — The debt of every active plot by charge type as of a date (**As of**, today by default), with the oldest unpaid charge, days overdue and status (red, yellow, green, grey). There are no owner names. For a past date the status is worked out with today's thresholds; the header says so.
- **Debtors** — Plots with a debt from an amount (**Debt from**, by default the cooperative's red threshold) *or* older than a number of days (**Older than, days**). Unlike the other reports, this one has the **names, phones and e-mails of the owners**: personal data. So the report needs the **View members** permission as well, and making it is recorded in the audit log. Do not pass the file on to outsiders.
- **Electricity by meter tree** — Electricity from the substation down to the plots for the range **From** - **To** (the previous month by default, service days). The **Consumption** sheet: one tree node per row (indented by level), readings at the start and end with their type (**Real**, **Calculated**, **Unknown**),

consumption, **Children, kWh** (the sum of the consumption of the nodes below it), **losses in kWh and %** (the node's consumption minus the sum below it; only for substations and quarter meters whose values are known at both ends) and **Plots fed**. When a value is unknown, everything that depends on it in the row is **empty** with the reason, not zero. The **Charged** sheet shows what was booked in the ledger for the range: plots' consumption (kWh, price, amount), losses (a fixed percent or runs per node), common-consumer runs with their split, plots and plot-days; adjustments and reversals are separate rows with the original service period. The totals are **Consumption of plots**, **Losses charged** and **Common consumers charged**. The header shows the current losses mode; a run booked under another mode shows what it booked.

- **Water** — Plots connected to water on the chosen days (**From**, **To**, **Water source**: **Cooperative supply** or **Well**). A row has the source, **Connected days**, the meter, cubic metres for connected days only, and the charges by kind: **By meter**, **Fixed fee**, **Well pump share**, with a separate column for reversals and adjustments. The **By source** sheet totals plots, plot-days, cubic metres and amount. Plots that were not connected are not listed.
- **Missing readings** — Meters with no *approved* reading in the range; filters: dates, resource (electricity, water), meter kind (plot meters by default), street. A row has the resource, kind, plot or name, serial number, last reading (date and value), days since it and the mark **Reading waiting for approval** (such a meter stays in the list). Not listed: a water meter of a plot that was not connected on any day of the range; a meter installed after the range or removed before it; and also a meter installed or removed *inside* the range, which is treated as having had a point to read. Under the table is how many open **Not charged** items there are for these meters, with a link to the list.
- **Land tax** — The filter is **Year**. A row is a plot the tax applies to: area in m² and sotkas, rate per sotka, **Expected** (the amount the system expects for the year) and **Charged for the year**. If two rates applied in the year, the plot has one row per rate with its days. A plot whose expected and charged amounts differ is marked in **Differs**, and a hint under the table says that a recalculation may be due (the report books nothing itself). Example: the rate is 35.00 ₴ per sotka per year and the plot is 600 m² = 6 sotkas → 6 × 35.00 = **210.00 ₴** a year (17.50 ₴ a month).
- **Corrections** — Every reversal and adjustment of the chosen billing months (filters: months, charge types, **Reversal** or **Adjustment**). A row: booked on, month, plot, type, the original charge (period and amount), the amount, the source (**Manual reversal**, **Recalculation**, **Run reversal**), the recalculation cause, the reason and who booked it. It is a ready list of changes of the past for a general meeting. When there are no corrections, the report says so.

> IMPORTANT
>
> **The same pump share in two reports**
>
> The electricity of a well pump is split among its plots as a common-consumer row (the page Water and connections). That row is on the **Charged** sheet of the electricity report *and* in the **Well pump share** column of the **Water** report. It is the same amount seen from two sides: do not add the two reports together.

Reports are easy to open from other screens: on a billing month the **Report** link opens **Charges by month** for exactly that month, and a plot account has a link to the balances report for that plot's street.

## Four cash book reports

Besides the eight reports, the **Reports** screen has **Cash book**, **Bank movements**, **Income and expense** and **Expenses**. They work the same way (filters, preview, Excel), but need only the **View cash and bank** permission. They are described on the page Reports, the rules on the page Income and expenses and the reconciliation act.

## Templates: your own Excel form

If you need your own form (a statement for the meeting, a report for the tax office), prepare it in Excel as usual, and where a value from the system should stand, write a **placeholder** such as `${name.of.field}`. The system replaces placeholders with data and keeps your formats, merged cells, column widths and formulas. The full list of placeholders is on the **Report templates** page, in the **Placeholders** block: type, description, example, a search box and a **Copy** button.

- **A placeholder alone in a cell** (`${rows.plots.landTax.expected}` and nothing else) is filled with a number or a date, not text, so formulas over it work. Give the cell a date format for a date. **A placeholder inside text** ("Land tax for ${period.year}") is filled as text in Ukrainian number and date format.
- **Single values**: `${cooperative.name}`, `${cooperative.officialName}`, `${cooperative.iban}` (requisites from the settings), `${period.from}`, `${period.to}`, `${period.year}` (the year of the last day of the period), `${totals.debt}`, `${totals.landTax.expected}` and other totals.
- **A repeating row**: when a row holds placeholders of one set — `rows.plots.*` (plots), `rows.months.*` (billing months), `rows.meters.*` (electricity meters) or `rows.water.*` (water) — the row is repeated once per row of the set. **One repeating row per sheet**; two sets in one row or two repeating rows on a sheet are refused.
- Names and contacts of people are not available in templates: only data of the cooperative, plots and meters, and amounts.

**A small example: the land tax statement**

The template is an Excel table (rate 35.00 ₴ per sotka per year; plots of 600, 800 and 1000 m²). What you write in the template:

| Row | A | B | C |
|---|---|---|---|
| 1 | `${cooperative.officialName}` | | |
| 2 | Land tax for `${period.year}` | | |
| 4 | Plot | Area, m² | Tax for the year, ₴ |

| Row | A | B | C |
|---|---|---|---|
| 5 | `${rows.plots.number}` | `${rows.plots.area}` | `${rows.plots.landTax.expected}` |
| 6 | Total | | `=SUM(C5:C5)` |

What you get for the period 01.01-31.12.2026 (row 5 is repeated for every plot):

| Plot | Area, m² | Tax for the year, ₴ |
|---|---|---|
| 1 | 600 | 210.00 |
| 2 | 800 | 280.00 |
| 3 | 1000 | 350.00 |
| Total | | 840.00 |

An area of 600 m² is 6 sotkas, 6 × 35.00 = 210.00; likewise 8 × 35.00 = 280.00 and 10 × 35.00 = 350.00. The formula `=SUM(C5:C5)` under the repeating row was turned by the system into `=SUM(C5:C7)` : it covers all three copies, so **Total** equals the sum of the rows. The title in row 1 is the full name of the cooperative from the requisites, and row 2 reads "Land tax for 2026".

## What the system does with a template

- **An unknown placeholder** (a typo, or a field that is not in the catalogue) **does not disappear**: it stays in the cell as written, and a **Warnings** sheet at the end of the file names the cell and the reason. The same goes for a placeholder whose value is unknown (no reading, a requisite not entered): the cell is empty, not zero, and it is listed in the warnings.
- The system does **not evaluate** formulas: it writes the placeholders and leaves the formulas to you. Excel or LibreOffice recalculates them when you open the file; a viewer that does not calculate formulas shows them empty.
- A valid file is an ordinary `.xlsx` workbook **up to 2 MB**. Files with macros ( `.xlsm` ), active content or embedded objects, damaged files and files that are "too large when unpacked" are refused and nothing is stored. Links to other files are removed from a template with a warning. Charts that refer to the repeating row are not shifted (a warning).

## Versions, the archive and the tax report

- **Add a template**: a name, a description, a file. Uploading a new file into an existing template ( **Upload new version** ) makes **version 2, 3…**; the latest version is always the one used, and older ones can be downloaded under **Versions** ( **Download** ). The generated file's name contains the template's name; the file properties and the audit log record the template and its version.
- **Archive**: the template is no longer offered for generation and takes no new versions, its name becomes free, and its versions stay downloadable.
- **Tax report** is a separate card with the status **The template is not added yet** until the chairman uploads the form with **Upload the template** . The official form will be added as an ordinary template, without changing the program; when the form changes, upload a new version. Fields the official form may be missing will be added by a later feature.
- **Generate** next to an active template opens the **Generate a file** window: **From** and **To** (both days are included, the period is up to one year). **Generate and download** makes the file and downloads it. The system **does not keep** generated files. If there are warnings you see "The file was downloaded. Warnings in the file: …" with the first of them; the full list is on the **Warnings** sheet. The treasurer generates from the **Reports** section, the chairman from the templates page.

## Cooperative requisites

**Settings** has a **Requisites** block (changing it needs the **Manage cooperative settings** permission): **Official name** , **Legal address** , **Bank name** , **IBAN** and **EDRPOU code** . The IBAN must be a valid Ukrainian IBAN (otherwise **The IBAN is not valid** and nothing is saved). The EDRPOU code is read-only — **Changed by the platform operator** . These are public data of the legal entity, not personal data; changes are recorded in the audit log. These are the data that go into `${cooperative.officialName}` , `${cooperative.iban}` and so on.

# How the system allocates payments

A payment is a row in the ledger that erases nothing: a mistake is corrected by a reversal, and the allocation to charges changes only by new rows. How to work with the screens is the page Payments; here are the rules and the examples.

- **The allocation order** — The oldest debt first (by the day it counts from); for the same day, in the order of the cooperative's charge types (membership, electricity, land tax - as on the charge types page); then by the start of the service period and the booking order. Each charge receives at most its open amount; the rest is prepayment if the plot has no open debt (while automatic allocation is on, the free remainder goes at once to the oldest debts).
- **The treasurer's change** — The proposal can be changed before saving: the lines may not add up to more than the payment, and a line may not exceed the charge's open amount. At the bottom you see "Allocated X of Y"; if a line exceeds the balance, the line "More than the remaining {amount}" with the balance stands above the amount field. With automatic allocation on, the unallocated remainder goes to the oldest debts; with it off, or when no debt is open, it becomes prepayment.
- **Taking money back from a charge** — When money is taken off a charge (it is reversed, an adjustment lowers it, or the payment is reversed), this is a separate "release" record. The released sum becomes the plot's prepayment and, if automatic allocation is on, goes to the other open charges.
- **Reversing a payment** — The chairman only, with a reason. Counted in the payment's own month while it is open, otherwise in the earliest open month. A payment is reversed once; a part of a group on its own.
- **A payment of a closed month** — Counted in the earliest open month and keeps its date received; the closed month does not change.
- **Splitting between plots** — In proportion to the debts, kopecks rounded down and the leftover kopecks to the largest remainders (a tie goes to the plot listed first). This is the only place of rounding in allocation; the parts always add up to the payment.
- **One at a time** — Every change of a plot (payment, allocation, release) first locks the plot's ledger, so two treasurers cannot allocate the same money twice; at the end the database checks it too and refuses to allocate more than a payment holds or a charge is worth.

## Worked examples

Order of charge types: membership, electricity, land tax.

> **Example 1. A payment smaller than the debt**
>
> Debts: January membership 600.00, February electricity 250.00. Payment 700.00. The oldest first: membership 600.00, then electricity 100.00. Electricity has 150.00 open, prepayment 0.00. The plot's debt is 150.00.
>
> **Example 2. A payment larger than the debt**
>
> The same debts, payment 1,000.00: membership 600.00 and electricity 250.00 are paid in full, prepayment 150.00. There is no debt, so the plot is green although there is a prepayment.
>
> **Example 4. Prepayment goes to a new charge**
>
> Example 2, then a land tax charge of 120.00 is booked. Automatic allocation is on: 120.00 goes from the prepayment to the land tax and the prepayment becomes 30.00. With the switch off, the land tax would stay a debt of 120.00 and the prepayment 150.00 until the treasurer presses `Allocate prepayment`.
>
> **Example 19. A payment with a remainder while debts are open**
>
> Debts: membership 600.00 (January), electricity 250.00 (February). Payment 700.00; the treasurer allocates only the electricity 250.00. Automatic allocation is on: the remaining 450.00 is placed at once on the membership, 150.00 of membership stays open, the prepayment is 0.00.
>
> **Example 20. The same with automatic allocation off**
>
> The same case with the switch off: electricity 250.00, the remaining 450.00 stays as prepayment and the membership stays open at 600.00. The treasurer can press `Allocate prepayment`.
>
> **Example 6. A charge is reversed**
>
> Example 1 (payment 700.00: membership 600.00 and electricity 100.00), then the chairman reverses the January membership. 600.00 is taken back from it - that is a prepayment of 600.00. Automatic allocation moves 150.00 to the open electricity; the prepayment stays 450.00 and there is no debt.
>
> **Example 12. One payment for two plots**
>
> A member has plot 7 with a debt of 300.00 and plot 15 with a debt of 100.00 and pays 200.00. The parts in proportion to the debts: 200 × 300 / 400 = 150.00 for plot 7 and 50.00 for plot 15. These are two payments of one group; cash has one receipt number. If the member paid 500.00, plot 7 would get 300.00 and another 100.00 as prepayment (the excess goes to the first plot), and plot 15 would get 100.00.

## Debt and colour after payments

- **The plot's debt** is the sum of the open amounts of its charges, that is charged minus allocated. A payment lowers the debt only by what is allocated to charges.

- **The oldest unpaid** is the oldest debt day among charges with an open amount above zero; a charge paid in full does not count, a partly paid one counts with its open part.
- **The colour** uses the same thresholds as before, but over the debt after payments: red if the debt exceeds the threshold or the oldest unpaid day is older than the threshold in days; yellow for any other debt; green when there is no debt, **also when there is a prepayment**.
- The colour is refreshed after every payment, reversal, allocation and release - within seconds. Example: a plot with a debt of 1,200.00 (red with a threshold of 1,000.00) after a payment of 900.00 has a debt of 300.00 and is yellow.
- An adjustment upward (for example +50.00 to a paid membership) leaves 50.00 open, so the plot is no longer green. An adjustment downward (−100.00 to a paid membership) takes 100.00 back from the latest allocation and it becomes prepayment, not a "negative debt".

# Invoice numbers

An invoice does not recalculate anything: it reads the same ledger as the plot account and the reports, and arranges it by billing month. How to work with the screens is the page The charges month and invoices; here are the rules and the examples.

- **The booking-month rule** — Every row of the ledger belongs to the billing month it is *booked* in. For an invoice of October, **Charged** is everything booked in October (regular charges, adjustments and reversals), and **Paid** is the payments placed in October. "Debt at the start" is what was open before October, "debt at the end" what is open after it.
- **A closed month never changes** — A correction made after October was closed is booked in the month it is made in. So the October invoice and its re-issues keep the same numbers, and the correction shows up in the invoice of the later month, with the month it corrects.
- **Amount due** — Debt at the end of the month, minus the money received after the month and before the issue, never below 0.00. Those payments are listed under the table as "Received after October". When the amount due is 0.00 there is no QR code; a prepayment is shown as **Prepayment**.
- **The invoice number** — Year-month/plot number, for example `2026-10/9`. A re-issue keeps the number and raises the version.

## Worked examples

Plot 9, charge types in the order Membership, Electricity. The invoice for October 2026, issued on 05.11.2026.

| # | Situation | The October invoice |
|---|---|---|
| 1 | No debt before. October: membership 75.00, electricity 338.26. No payments | Start 0.00; charged 413.26; paid 0.00; end 413.26; due 413.26; QR amount 413.26 |
| 2 | Electricity debt from September 1,135.08. In October a payment of 1,500.00 went to 1,135.08 (September electricity) + 75.00 (membership) + 289.92 (October electricity) | Start 1,135.08; charged 413.26; paid 1,500.00; end 48.34 (electricity); due 48.34 |
| 3 | Example 1 and a payment of 413.26 on 03.11 | End of October 413.26; "Received after October" 413.26; due 0.00; no QR code |
| 4 | Overpaid: no debt at the end, prepayment 100.00 | Due 0.00; "Prepayment 100.00"; no QR code |
| 5 | October was closed on 03.11. On 06.11 a recalculation lowers an October charge by 20.00 (booked in November) | The October invoice does not change, and a re-issue gives the same numbers. The November invoice shows the -20.00 adjustment "for October 2026" |

# Which channel, and how sending is retried

Delivery calculates nothing and changes no amounts: it only delivers documents that are already built. How to work with the screens is on the page Delivery; here are the rules for choosing a channel and for retries, with examples.

- **The channel is decided for each owner separately** — For every *current* owner of a plot, the owner's preferred way (Telegram, e-mail, paper or Viber) is taken and checked for being possible: Telegram - the member is connected and the bot exists; e-mail - there is an address and a mailbox is set up. If not, the fallback of the table on the page Delivery applies: from Telegram to e-mail, then paper; from e-mail to paper; Viber is always paper.
- **One message per plot and owner** — A member who owns two plots gets two messages, each with its own PDF. Two people who own a plot together get one message each, each through their own channel.
- **The same is never sent twice** — For one invoice version, one member and one channel there is exactly one delivery record. Pressing "Send invoices" again adds nothing; a new invoice version is sent as a new message.

## Worked examples

October 2026 invoices; the cooperative's mailbox and bot are set up.

| No. | Situation | What happens |
|---|---|---|
| 1 | Plot 9, one owner Olena: prefers Telegram, connected | Olena gets the invoice in Telegram: the text, the PDF and the "Оплатити" (Pay) button |
| 2 | Plot 7 is co-owned. Ivan: prefers Telegram, has not connected yet, has an e-mail address. Maria: prefers paper | Ivan gets a letter with the PDF; Maria is on the "to print" list, her copy is printed with its own QR code for connecting Telegram |
| 3 | Peter: prefers Viber (not supported yet). Hanna: prefers Telegram, has not connected, no e-mail address | Both are "to print"; Peter has the note **Viber is not supported yet** on the Deliveries screen. When Hanna connects Telegram, the next invoices go there |
| 4 | A member owns plots 9 and 10, prefers e-mail | Two letters: one for plot 9, one for plot 10, each with its own PDF |
| 5 | The invoices are sent already and the treasurer presses "Send invoices" again | Nothing new: the system says everything has been sent |

## Sending attempts

Sending runs in the background at a limited pace: at most one message a second to one member, up to 20 messages a second per bot in Telegram, and e-mail at your **Letters per second**. Every message has **three attempts**: now, after 1 minute and after 5 minutes. After the third failure the status is **Failed** with the reason of the last attempt.

| Reason | Retried? | What it means |
|---|---|---|
| The mail server refused the letter | yes, up to three attempts | the mailbox did not accept the letter: the daily limit, too fast, an invalid recipient address |
| Could not connect | yes, up to three attempts | no connection to the mail server or Telegram |
| Wrong mailbox login or password | no, **Failed** at once | the password does not fit; a retry changes nothing until it is corrected |
| The bot is blocked | no | the member blocked the bot; the connection is cancelled |
| No contact | no | no e-mail address or connected chat |
| The channel is not configured | no | no mailbox or bot, or Telegram rejects the token |

If Telegram asks to wait (too many requests), the message is simply moved to the time Telegram names, and that pause does *not* use up an attempt. A message that became **Failed** does not disappear: it stays on the Deliveries screen, and after the cause is fixed it is sent with **Send again**.

# Cash, bank and reconciliation

The cash and the bank are kept as a **journal of lines** that erase nothing: a line is neither edited nor deleted, a mistake is corrected by a reversal. How to work with the screens is the page Cash and bank; here are the rules and examples. The server calculates all amounts.

- **The balance on a day** — The opening balance plus every receipt and minus every spending from the opening date up to and including that day. The opening balance is the balance *at the start* of its date; it is entered once. An entry other than a member payment dated before the opening balance is refused.
- **Member payments in the journal** — A payment from the page Payments becomes a line by itself: a cash payment in the cash, a bank payment in the bank. Reversing a payment adds a line of the opposite amount dated on *the day of the reversal*. The member's debt view (the account, the reconciliation act) keeps the dating from the page How the system allocates payments. When the feature is first installed, every payment already entered and every reversal of a payment gets its line exactly once.
- **Cash never goes negative** — A cash expense, a handover, the reversal of a cash income and a booked shortage are refused if they would make the cash negative *on their date or on any later day*. The message names the balance: "Not enough cash in the till: balance 200.00 on 07.10.2026". A wrong balance is fixed by a cash count, not by going negative. Two entries saved at the same moment can never together push the cash below zero.
- **The only exception** — Reversing a *cash member payment*: the money is already gone, so the reversal is allowed. If the cash is negative because of it, the card says "The cash balance is negative - do a cash count".
- **The bank may be negative** — An overdraft is the bank's business: the screen warns **The bank balance is negative** but does not refuse.
- **Closed months** — A line dated in a closed billing month keeps its date and falls into the earliest open month. Reports by day use the date, reports by billing month use the month it belongs to.
- **Cash count** — Difference = counted − book. Zero - **Matches**. Otherwise an explanation is required, and the difference hangs as **Unresolved** until the chairman books it as a "Surplus" or a "Shortage". A saved count keeps the book balance it showed.
- **Bank reconciliation** — The statement figures must agree: opening + in − out = closing. Four pairs of figures are compared: the opening balance, the total in, the total out and the closing balance (from the books: the bank balance at the start of the month, the sum of receipts, the sum of spendings, the balance at the end of the last day). **Reconciled** is available only when the **closing balance** difference is 0.00; a difference in the total in or out only warns. Months are calendar months by the date of the line. Any bank line dated in a reconciled month (a payment too) sends it to **Needs a new check**.

## Worked examples

Cash opening balance on 01.10.2026 is 1,000.00, the bank's is 20,000.00.

**Example 21. Cash balance by day**

05.10 a cash payment of 500.00 - cash 1,500.00. 06.10 a cash expense of 300.00 **Materials** - cash 1,200.00. 07.10 a handover of 1,000.00 to the bank - cash 200.00, bank 21,000.00.

**Example 22. Not enough cash**

Then on 07.10 the treasurer enters a cash expense of 250.00. The cash is 200.00, so the entry is refused: "Not enough cash in the till: balance 200.00 on 07.10.2026".

**Example 23. A back-dated entry**

On 08.10 the treasurer enters an expense of 400.00 dated 06.10. On 06.10 there was 1,200.00 and after it there would be 800.00 - fine. But 07.10 after the handover would be 200.00 − 400.00 = −200.00, so the entry is refused and the message names that day and its balance.

**Example 24. Reversing a cash payment**

On 12.10 the chairman reverses the cash payment of 500.00 from 05.10 when the cash holds only 200.00. The cash journal gets a line of −500.00 dated 12.10; the cash becomes −300.00. The reversal is allowed (the money is already given out), and the cash card asks for a cash count.

**Example 25. A shortage found at a count**

On 10.10 the book says 200.00, the treasurer counted 180.00: the difference is −20.00, an explanation is required, "Unresolved difference −20.00". The chairman books a "Shortage" of −20.00: the cash is 180.00 and no difference hangs.

**Example 26. October reconciliation: a bank fee**

The statement: opening 20,000.00, in 3,000.00, out 150.00, closing 22,850.00 (20,000.00 + 3,000.00 − 150.00 = 22,850.00, the figures agree). The books: in 3,000.00 (2,000.00 member bank payments plus the 1,000.00 handover), out 0.00. The "out" differs by 150.00 (and the closing balance too). It is a bank fee nobody entered: the treasurer enters an expense of 150.00 **Bank fees** in October, the differences become 0.00 and October can be marked **Reconciled**.

**Example 27. A reconciled month is reopened**

October is **Reconciled**. On 03.11 a bank payment dated 20.10 is entered. October goes to **Needs a new check** and the screen says: "Reconcile this month again: a member payment of 20.10.2026 for 500.00 ₴ was entered on 04.11.2026, 10:15." November, if it is also **Reconciled**, is reopened with the same notice: its opening balance changed. Check October first, then November.

**Example 28. The statement figures do not agree**

Opening 100.00, in 50.00, out 20.00, closing 200.00. 100.00 + 50.00 − 20.00 = 130.00, not 200.00: the statement is not saved - "The statement figures do not agree: 100.00 + 50.00 − 20.00 ≠ 200.00".

**Example 29. A closed month**

September is closed. On 3 October the treasurer enters an expense dated 30.09: the journal keeps the date 30.09 and the billing month is October.

IMPORTANT

**Limits of reconciliation**

**Reconciled** needs only a zero closing balance difference. A mistake made in a past month and reversed today cannot be accounted for in that month (the reversal is dated today), but this is only a warning about a difference in the total in or out and does not block. A back-dated entry changes the opening balance of all later months, so the month of the entry and every later **Reconciled** month are reopened; check them one by one, starting with the earliest.

# Income and expenses, and the reconciliation act

How to work with the report and the act is the page Reports; here is what exactly they calculate.

- **Member payments in** `Income and expense` — The payments received in the period are taken by charge type - *as the payments are allocated at the end of the period*. The part not allocated to any charge is the line "Prepayments". Reversals made in the period of payments received earlier are a separate negative line.
- **The identity — Income − expenses + differences = the change of (cash + bank) over the period.** A handover of cash to the bank is neither income nor expense: it is a move between two accounts and cancels out.
- **The reconciliation act** — The act reads the same ledger as the plot account, so the balances at the start and the end of the period equal the balances in the `Plot balances` report on those days. Every charge, adjustment, reversal and payment stands by date in the columns `Charged` and `Paid`. The period runs from the plot's first charge up to today.

  **Example 30. Reconciliation act, plot 9, 01.09-31.10**

  The balance on 31.08: a debt of 200.00. September is charged 300.00, October 250.00; on 15.10 a payment of 300.00 is entered. Turnover: charged 550.00, paid 300.00. The balance on 31.10: 200.00 + 550.00 − 300.00 = 450.00, and the `Plot balances` report shows the same debt on 31.10. The act ends with "According to the cooperative's records the debt as of 31.10.2026 is 450.00".

  **Example 31. Income and expenses for October**

  Cash 1,000.00 and bank 20,000.00 on 01.10. During October: member cash payments of 500.00 and bank payments of 2,000.00 (2,500.00 together - 1,800.00 of it membership fee, 500.00 electricity, 200.00 prepayments); bank interest 35.00; a cash expense of 300.00 and a bank fee of 150.00; a shortage of −20.00; a handover of 1,000.00 from cash to the bank. Income 2,500.00 + 35.00 = 2,535.00; expenses 300.00 + 150.00 = 450.00; differences −20.00. The result: 2,535.00 − 450.00 − 20.00 = 2,065.00. The change of cash: 1,000.00 + 500.00 − 300.00 − 1,000.00 − 20.00 = 180.00, that is −820.00 for the month. The change of the bank: 20,000.00 + 2,000.00 + 1,000.00 + 35.00 − 150.00 = 22,885.00, that is +2,885.00. Together −820.00 + 2,885.00 = 2,065.00 - equal to the result. If a payment of 100.00 from September were reversed in October, the income would get a line of −100.00 and both the result and the change would fall by the same amount.

# My account

## What the screen is for

**My account** is the personal page of everyone who signs in, and the home screen of a cooperative member: after signing in they land here, it is the first item of their menu, and the logo in the header leads back here. A plot owner sees everything about their plots in one place: what will be charged, what is already charged and paid, their meters and invoices. From here you can submit a meter reading, download an invoice, connect Telegram and get a reconciliation act.

## Who sees it

Every signed-in user sees the account, and everyone sees only their own data. Other people's plots, amounts, owners and personal data are not here. Members are not shown the list of all plots. The setting "Who sees plot colours" applies to the plot map that will come later, and the amounts of other plots are never shown.

If you have no plot, the parts about plots do not appear. Then you see only your own data and roles.

## Every element

The screen is made of separate blocks, one under another, in this order.

### Installing the app

At the top of the account there may be a card "**Install the app**" with the button **Install the app**, and on an iPhone in Safari a card "**Install on iPhone**" saying "**Share → Add to Home Screen**". It offers to add Nadel to the home screen. The button "**Do not show again**" hides the card on this device. When Nadel is already installed, the card is not shown. Step by step: Install on your phone.

### My data, roles and rights

- The **My details** block shows your contacts, your preferred channel for invoices, whether Telegram is connected, and your roles. Only the board can change the data: write or tell them.
- The **Roles** block shows your roles and the name of the cooperative (**Cooperative**).
- The **What you can do** block lists what you are allowed to do, with a short explanation of each right. If your roles give no extra rights, it says "**Your roles give no extra permissions. You can see your own data.**"

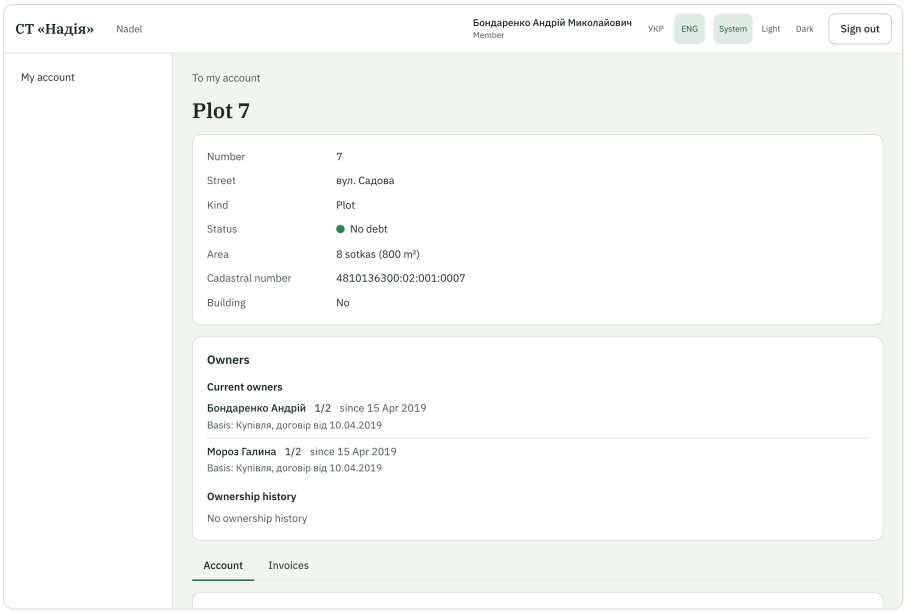
### My plots

The **My plots** block has a card for each of your plots: the number (it leads to the plot card), the street and your share. Below are the co-owners and the history of owners. If you have no plots, the system says "**You do not own any plots**".

Next to each plot there is the **Reconciliation act** button. It opens the **Reconciliation act** form: choose the dates in the **From** and **To** fields and press **Download the reconciliation act**. You get a PDF with the charges, payments and the debt or overpayment at the start and the end of the period. The **Close** button folds the form. More: Documents for a member.

### The card of my plot

The plot number in the **My plots** card is a link: press it and the plot card opens, with a "**To my account**" link. The card shows the plot data (area, street, debt status), the owners with the ownership history, your account with charges and payments, the expected charges by tariffs, the plot's meters and its invoices. You have no "**Notes**" field and no buttons that change data, and the system does not show the card of someone else's plot. You do not need an overall view of all plots: you open everything of yours from here.




Card of plot 7 as its owner sees it

#### My meters

The **My meters** block shows the meters of your plots. For each meter you see its name, the mark **In service** or **Removed**, the last reading with its date, and the history of readings, each with a status: **counted**, **pending**, **rejected** or **superseded**. A rejected reading shows its reason.

The **Submit a reading** button is next to a working meter. How to use it: How to submit a meter reading. If your plots have no meters, the system says "**There are no meters on your plots**".

#### Invoices

The **Invoices** block first shows the latest invoice of each of your plots. For each one you see the month, the number, the version, the "**Due**" amount and the date it was made. The **Download PDF** button downloads the PDF. Older months are hidden under "**Earlier months**". If there are no invoices yet, the system says "**There are no invoices yet.**". How to read an invoice: How to read my invoice.

Under an invoice that has an amount to pay there are buttons: **Pay in the bank app** opens the transfer in the banking app, **Copy IBAN** and **Copy payment purpose** copy those values so that you can paste them into the bank. After a press the button says "**Copied**" for a moment. If copying fails, the system says "**Could not copy. Select the value below and copy it manually.**" and you can select and copy the value by hand. There are no buttons when there is nothing to pay.

#### Notifications and Telegram

In the **Notifications** block you choose, in the "**How to receive invoices**" field, one way: e-mail, Telegram or paper. Only the ways available to you are shown. The **Save** button saves the choice and the system answers "**Saved.**".

Below, if your cooperative has a bot, is the **Telegram** block. The **Connect Telegram** button opens the bot; after that the block says "**Telegram connected**" and has the **Disconnect Telegram** button. Step by step: How to connect Telegram.

The same block has the switch "**Notifications on this device**": it turns on short notifications on this phone or computer and applies to this device only. If notifications are not set up in the system, there is no switch. If they are blocked in the browser, on an iPhone the page is not opened in the installed app, or the browser cannot show them, an explanation stands in place of the switch. More: Notifications on your phone.

#### My charges and payments

The **My charges** block shows the account of each of your plots read-only: the debt by charge types and every charge. The **Details** button next to a charge opens the calculation: the rate, units, readings and days. The payments of your plot are there too: the date, the receipt or bank document number, the amount, what exactly it paid, reversed payments with a mark, the prepayment and the debt after payments. You do not see who made a payment or the note about the payer.

#### Expected charges

The **Expected charges** block shows, for each plot, what the system will charge this month by the current tariffs, and the amount in the "**Total**" line. If the amount depends on meter readings, the line says "**by readings**" instead of an amount; "**no meter**" means the plot has no meter, and without a meter nothing is charged for that resource. How the amount comes out: Tariffs and the expected amount.

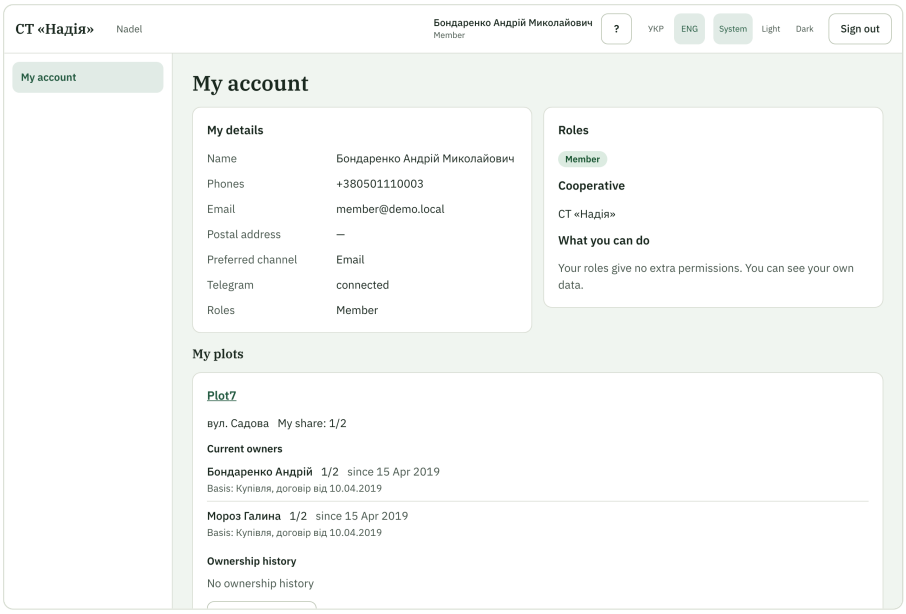
## Worked example

Olena owns plot 9. She opens her account after the board has closed October.

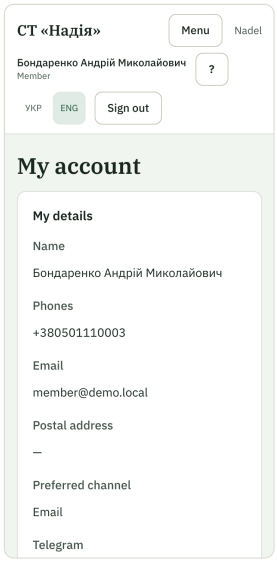
1. In the **Invoices** block she sees the October invoice. Its "**Due**" is 413.26. She presses **Download PDF** and pays by the QR code in her banking app.

2. A few days later the **My charges** block shows her payment with the bank document number and zero debt.
3. She reads her electricity meter, presses **Submit a reading** in the **My meters** block and enters the value and the date. The system answers "**The reading is submitted and waits for approval**". After the board approves it, the reading becomes **counted**.
4. To show the state of payments for the year to a bank or relatives, she presses **Reconciliation act**, sets the period and downloads the PDF.

## Screenshots



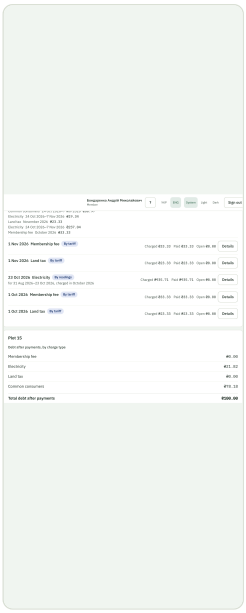

The member's account



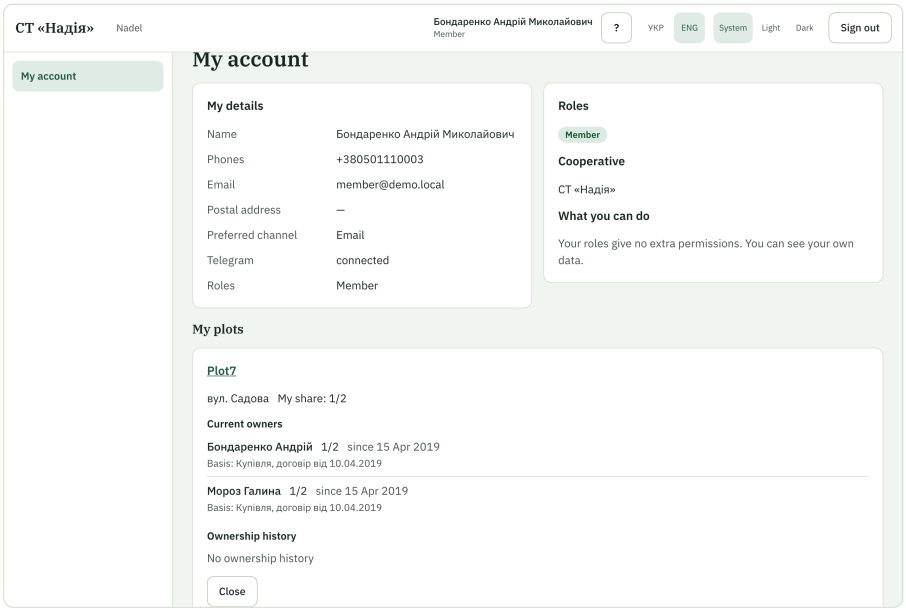

The account on a phone




The invoices block



The "My charges" block




The reconciliation act form

## Related pages

- How to read my invoice
- How to submit a meter reading
- Documents for a member
- How to connect Telegram
- Install on your phone
- Notifications on your phone
- Plot card
- Getting started as a member
- Getting started

# How to read my invoice

The invoice for a month reaches you by e-mail, in Telegram or on paper. You can also download it in your account, in the **Invoices** block. The invoice invents nothing: it takes the amounts from the account of your plot, and only the system calculates them.




Invoices in the account

## What is in an invoice

- The name of the cooperative and its requisites.
- The invoice number, for example `2026-10/9`: year, month and plot number. If the board issued the invoice again, "(version 2)" is added to the number. The number stays the same.
- Your plot, the street and the owners with their shares.
- A table with one line for each charge type. It has the debt at the start of the month, charged, paid and the debt at the end.
- Under "**Charged**" are the details people ask about most. Every electricity interval is written as one span of reading dates with both readings, the kWh and the price, for example "readings 10.08.2026 → 03.10.2026: 3 435,900 → 3 705,500 = 269,600 kWh × 4.32 ₴". Losses and common consumers stand separately, as do the basis of a fixed tariff and, for a correction, the month it corrects.
- The payments of the month (the block "Payments for October 2026"): the date, the receipt number or the bank reference and the amount. You can check them against your bank. A reversed payment is shown as the payment itself and a line "payment reversal" with a negative amount.
- Payments received after the month but before the invoice was built ("Received after October"). They reduce the "**Due**" amount.
- The total: the debt at the end of the month, the prepayment and the "**Due**" amount. Next to it is a QR code if "**Due**" is above zero, with the line "Pay in your bank app": on a phone, a tap opens the transfer in the bank without scanning.

Every invoice fits one A4 page. A very long invoice continues on a second page, whose header repeats the invoice number and the plot. An invoice for a month that is not closed yet carries the mark "as of" a date. How the figures come about: Invoice numbers.

## How to read the amounts

| Line | What it means |
|---|---|
| Debt at the start | What was unpaid before the month began |
| **Charged** | Everything the system charged for the month |
| **Paid** | The payments counted in this month |
| Debt at the end | What is unpaid after the month |
| **Due** | The debt at the end of the month minus the money received after the month and before the invoice was made. It is never below zero |
| **Prepayment** | Money paid in advance above the debt |

A payment clears the oldest charges first, so the lines of the table can hold debt that is not from the current month. More: How the system allocates payments.

## Worked example

Plot 9. The invoice for October 2026.

1. There is no earlier debt. For October the membership fee is 75.00 and electricity is 338.26. There are no payments. Debt at the start 0.00, charged 413.26, paid 0.00, debt at the end 413.26, amount due 413.26. The QR code is also for 413.26.
2. If on 03.11 you paid 413.26, the invoice would still show the debt at the end of October as 413.26. Under the table there would be the line "Received after October" with 413.26, and "**Due**" would be 0.00. There would be no QR code.
3. If you had paid 100.00 too much earlier and there is no debt at the end, the invoice would show "**Prepayment** 100.00", "**Due**" 0.00 and have no QR code.

More cases with numbers: The numbers of the invoice.

## The pay buttons in the account

In My account, in the **Invoices** block, every invoice with an amount to pay has buttons. **Pay in the bank app** opens the transfer in the banking app (on a phone without scanning the QR code). **Copy IBAN** and **Copy payment purpose** copy the IBAN and the payment purpose so that you can paste them into the bank if the app did not open. After copying, the button says "**Copied**" for a moment. If it failed, the system says "**Could not copy. Select the value below and copy it manually.**": select the value and copy it by hand. Check the recipient, the IBAN, the purpose and the amount just as with the QR code.

## The QR code

The code follows the rules of the National Bank of Ukraine. A banking app reads the QR code and opens a filled-in transfer. Before you pay, check in the app:

1. The recipient is the name of your cooperative.
2. The IBAN is the one in the invoice.
3. The payment purpose starts with the word "**Plot**", then your plot number, the month and the invoice number.
4. The amount equals "**Due**". You may change it: you can pay a part.

The other fields (recipient, account, recipient code, purpose) are locked. Close the form without paying if something does not match. If something does not match, do not pay and ask the board.

## If the numbers look wrong

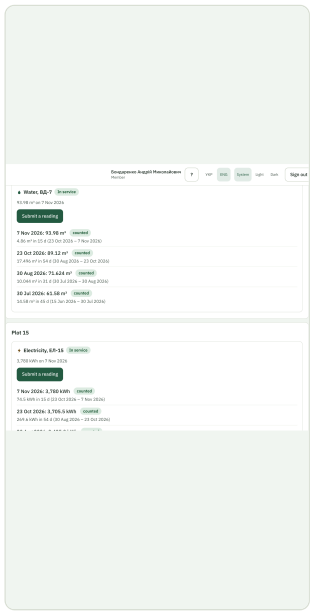
- The invoice of a month that is already closed does not change. If the board corrected a charge afterwards, the correction appears in the invoice of a later month, with the month it corrects named.
- Open the **My charges** block in the account: the **Details** button shows how each amount was obtained.
- If your plot has no meter, nothing is charged for electricity or water on that plot.
- Ask the board about anything else.

## Related pages

- My account
- Documents for a member
- The numbers of the invoice

# How to submit a meter reading

You can submit a meter reading yourself, without waiting for the electrician. You need to be signed in and to have a plot with a working meter.



My meters in the account

## How to submit a reading

1. Open your account and find the **My meters** block.
2. Next to the right meter press **Submit a reading**. The button is not there for a meter that has been removed.
3. Enter the date when you read the meter. Today is there by default.
4. Enter the meter value as it is written on the meter. The value may have up to three decimals.
5. If you wish, add a photo of the meter: press **Add a photo of the meter**. On a phone the camera opens, on a computer a file picker. A HEIC photo from an iPhone is converted to an ordinary one by the system itself. You see a thumbnail above the button. The **Replace** button changes the photo and **Remove** removes it.
6. Press **Submit a reading**. The system answers: "**The reading is submitted and waits for approval**".

Nothing else is needed.

## The photo of a reading

A photo helps the board quickly make sure the number is right. Only photos (JPEG, PNG) up to 5 MB are allowed; otherwise the system says "**Only photos (JPEG, PNG) are allowed.**" or "**The photo is too large. Maximum 5 MB.**". While the photo is being processed, the button says "**Processing...**".

If the board turned on the rule "**Meter photo required for members**" in the settings, the photo is required: without it the reading is not sent and the system says: "**Add a photo of the meter: the board asks for a photo with every reading.**" Add the photo and press **Submit a reading** again. A submitted reading with a photo shows as a thumbnail in the meter history: press it to open the photo full screen. A photo cannot be deleted after submission, it stays with the reading.

## What happens next

The reading is not counted at once. The window says: "**The reading will be counted after the board approves it.**" Until the board answers, the reading has the status "**pending**" and is not in any calculation.

| Status in the list | What it means |
|---|---|
| **pending** | The board has not checked the reading yet |
| **counted** | The reading is accepted. The system charges electricity by it |
| **rejected** | The board rejected the reading and gave a reason, shown under the entry |
| **superseded** | This reading is replaced by a newer or corrected one. It stays in the history |

Under an entry you also see how much was used between this reading and the previous one.

## Why a reading can be rejected

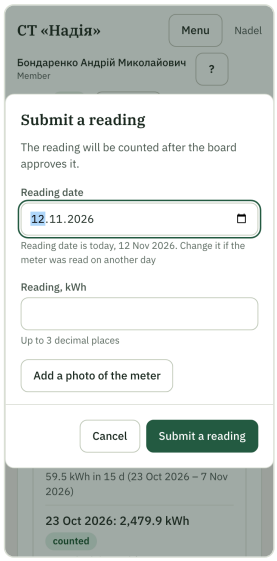
- The value is lower than the previous one. A meter reading cannot go down.
- The date is in the future or outside the time the meter worked.
- The value is far above what is usual for your plot. The system lets it through with a warning, and the board checks it.

If you made a mistake, just submit the correct reading again: the board rejects the wrong one and accepts the right one.

## Worked example

Meter EL-9 on plot 9 had the last reading 3,705.600 from 03.10. You read 3,800.000 and submit it on 31.10. A day later the board accepts it. The system counts the consumption: 3,800.000 – 3,705.600 = 94.400 kWh, and multiplies it by the tariff rate. How exactly: Readings and consumption and Tariffs and the expected amount.

This is the window on a phone with the photo step:




The reading window with the photo button on a phone

You can also submit a reading in the Telegram bot: How to connect Telegram.

## Related pages

- My account
- Readings and consumption
- How to read my invoice

# Documents for a member

A plot owner sees and downloads only the documents of their own plots. A document of someone else's plot is not found by the system.

## The invoice for a month

The board makes the invoice. You can get it by e-mail, in Telegram, on paper, or download it yourself.

1. Open your account, the **Invoices** block.
2. At the top is the latest invoice of each of your plots. Older months are hidden under "**Earlier months**".
3. Press **Download PDF** and the PDF is saved on your device.

If the board issued the invoice again, you will see a new version with the same number. How to read an invoice: How to read my invoice.

## The cash receipt

When you hand cash over at the cooperative's till, the treasurer writes a receipt. It has a running number without gaps. The receipt is internal, it is not a fiscal check. Next to such a payment in the account of the plot there is a "Receipt" link.

## The reconciliation act

The reconciliation act is a PDF for your plot and the chosen period. It holds:

- the cooperative, the plot and its owners;
- the balance at the start of the period: debt, overpayment or nothing;
- all charges, corrections, reversals and payments by dates;
- the balance at the end and a sentence about the state of payments on the last date;
- two places for signatures: from the cooperative and from the member.

How to download the act:

1. Open your account, the **My plots** block.
2. Next to the right plot press **Reconciliation act**.
3. In the **From** and **To** fields set the period. The start cannot be earlier than the first charge of the plot, and the end cannot be later than today.
4. Press **Download the reconciliation act**.

The act is not stored on the server: it is built on request, and the same period always gives the same act. If the requisites of the cooperative are not entered yet, the act cannot be built: ask the board.

## The certificate of payment status

The certificate of debt or no debt is issued by the board. An owner does not download it alone: ask the board to issue one.

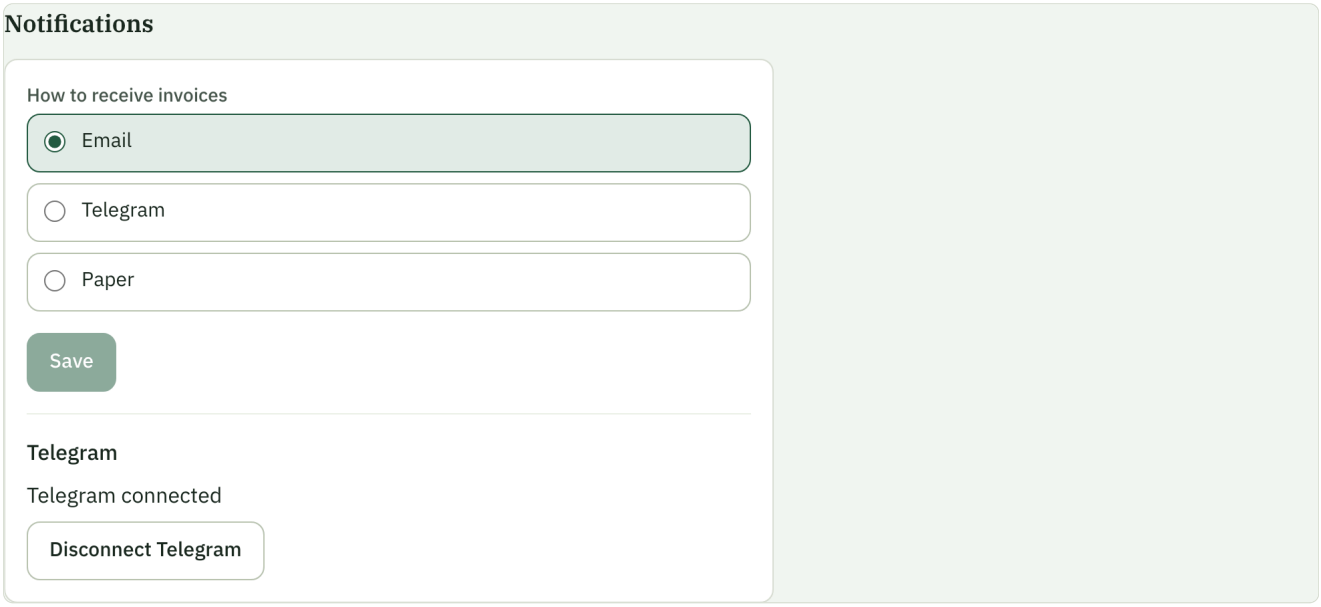
## Related pages

- My account
- How to read my invoice
- How to connect Telegram

# How to connect Telegram

Telegram lets invoices come to you straight in the messenger. A bot cannot write first to a person by phone number or nickname, so once you must open the bot by your personal link. There are two ways.

The connect button is there only when your cooperative has a bot set up. If it is missing, ask the board.




The "Notifications" block in the account

## Way 1. From the account

1. Open your account, the **Notifications** block, and below it the **Telegram** block.
2. Press **Connect Telegram**.
3. The cooperative's bot opens with a one-time code. **The link is valid for 30 minutes. Open it in Telegram and press Start.**
4. The bot greets you by name and shows menu buttons.
5. Go back to the account. It now says "**Telegram connected**" and has the **Disconnect Telegram** button.

## Way 2. From a paper invoice

Every paper copy of an invoice has a small QR code "Receive invoices in Telegram" with a personal code of exactly your copy. It works for 90 days.

1. Point your phone camera at the QR code.
2. Open the bot.
3. Press "Start".

Co-owners who get paper each have their own copy with their own code.

## If it did not work

The code works only once. If the link is used, expired or unknown, the bot answers "The link is not valid, get a new one in your account or from the board". Just press **Connect Telegram** again.

If this Telegram is already connected to another member, the bot asks whether it is you. The connection moves only after the answer "Yes, it is me".

## How to choose what you receive

In the **Notifications** block, in the "**How to receive invoices**" field, choose Telegram, e-mail or paper and press **Save**. E-mail is available when your card has your address. Paper is always available. If the preferred channel is not available, the system sends the invoice another way, for example by e-mail or on paper. Viber cannot be chosen yet: the system says "**Viber is not supported yet — invoices are printed on paper**".

## What the bot can do

The bot shows only your own plots. The menu has buttons.

| Button | What it does |
|---|---|
| "Debt" | For each of your plots: the debt by charge types, the total and the prepayment, as in the account |
| "Invoice" | The invoice for today: a PDF and a "Pay" button. If you have several plots, choose a plot first |
| "Reading" | Choose a plot, then a meter, and send the number. The reading goes to the board for approval, dated today |
| "Disconnect" | Stop receiving invoices in Telegram. The command `/stop` does the same |

When the board approves or rejects a reading, the bot tells you. The commands `/start` and `/help` show the menu and a short help. To any other text the bot answers with the help. Do not send more than 20 messages a minute: the bot will ask you to wait.

If the bot is opened in a chat that is not connected yet, it shows only the help and an invitation to connect, with no amounts.

## How to disconnect Telegram

In the account press **Disconnect Telegram** and confirm in the "**Disconnect Telegram?**" window.
**Invoices will no longer arrive in Telegram until you connect it again.**

## Which channel the invoice comes by

The decision depends on your preferred way.

| Preferred way | Situation | You receive |
|---|---|---|
| Telegram | Connected and the bot is set up | In Telegram |
| Telegram | Not connected or no bot | By e-mail if there is an address and a mailbox is set up, otherwise on paper |
| E-mail | There is an address and a mailbox is set up | By e-mail |
| E-mail | No address or no mailbox set up | On paper |
| Paper | — | On paper |

If a plot has co-owners, each of them gets the invoice by their own channel. An owner of several plots gets a separate message for each.

## Related pages

- My account
- Documents for a member
- How to read my invoice

# Install on your phone

Nadel can be installed like an ordinary app: it gets its own icon on the home screen, opens full screen and can show notifications. You do not download anything from an app store: it is the same page you open in the browser.

A secure connection is needed: the address of the system must start with `https://`. The board gives you exactly such an address. If the browser does not offer installation, check this first.

## Android, Chrome

1. Open My account. When the browser allows installation, there is a card "**Install the app**" at the top with the button **Install the app**.
2. Press **Install the app** and confirm in the phone's window. The phone draws that window itself, so there is no screenshot of it here.
3. The Nadel icon appears on the home screen. Open the system from it from now on.

If the card is not there, open the browser menu (three dots) and choose the item that installs the app, for example "Install app" or "Add to Home screen".

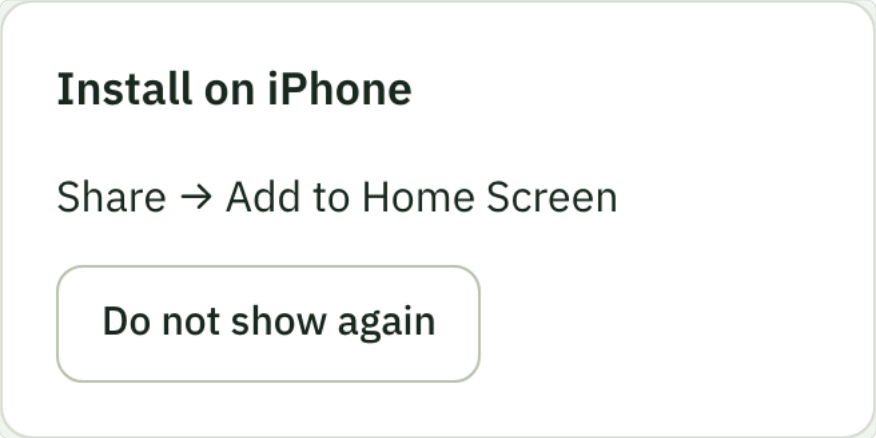
## Desktop, Chrome or Edge

1. Open My account and press **Install the app** in the card "**Install the app**".
2. Or press the install icon at the end of the address bar and confirm.
3. Nadel opens in its own window, and its icon is among your programs.

## iPhone and iPad, Safari

On an iPhone the system can be added to the home screen only from the Safari browser. Chrome and other browsers on an iPhone do not have this item.

1. Open the system in Safari and go to My account. There is a card "**Install on iPhone**" saying: "**Share → Add to Home Screen**".
2. Press the Share button at the bottom of the screen (a square with an arrow pointing up).
3. Scroll the list and choose "Add to Home Screen".
4. Press "Add". The icon appears on the home screen.




The card with the iPhone steps in My account

**IMPORTANT**

Notifications on an iPhone work only in the app added to the home screen, and only on iOS 16.4 or newer. A link opened in Safari does not show notifications. More: Notifications on your phone.

## I do not want this card

The button "**Do not show again**" hides the card on this device. It does not come back. You can still install the app later from the browser menu. When Nadel is already installed, the card is not shown.

## Updates

When a new version is out, a bar "**A new version is available**" with the button "**Update**" appears at the top of the screen. Press it, and the page reloads in the new version. The update never happens by itself, so it does not interrupt you.

If a form with unsaved changes is open, the button does not act and the bar says: "**Finish or cancel your changes in the open form to update.**" Save or close the form and press the button again. During sign-in the bar says: "**Finish signing in to update.**"

## Without internet

When there is no connection, a bar appears at the top: "`No internet connection. Data will appear when the connection is back.`" The app itself opens, but it does not show your data without internet: invoices, amounts and readings are not stored on the device, so that nobody else can see them. When the connection is back, the bar disappears and the data appears by itself. Nothing you enter without a connection is sent: repeat the action when the internet is back.

## Related pages

- Notifications on your phone
- My account
- Getting started for members

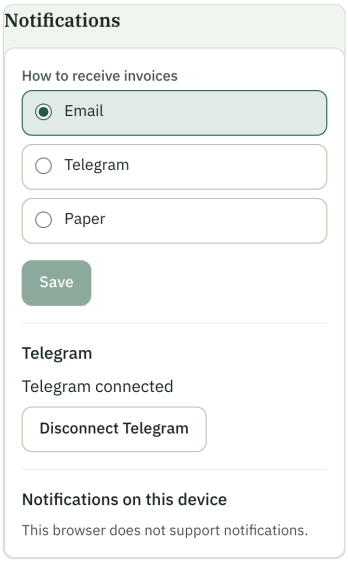
# Notifications on your phone

Notifications are short messages on the screen of a phone or computer, even when Nadel is not open. They work separately from Telegram and e-mail and are set for each device on its own.

## How to turn them on

1. Open My account, the **Notifications** block.
2. Turn on the switch "**Notifications on this device**". Below it is the hint: "**Short messages about invoices, payments and readings will arrive on this phone or computer.**"
3. The browser asks whether to allow notifications. Choose "Allow". The browser asks this itself, so there is no screenshot of it in this help.

The switch is shown only when notifications are set up in the system. If it is not there, ask the board.




The Notifications block on a phone

In this screenshot an explanation, "**This browser does not support notifications.**", stands in place of the switch: the browser it was taken with does not support notifications. On your phone there is a switch in this place.

## How to turn them off

Turn the same switch off. The device stops receiving notifications at once. On another device, for example a computer, they stay on until you turn them off there too.

When you sign out of the system, notifications on this device turn off by themselves. After the next sign-in turn them on again. This way a borrowed phone or a shared computer does not receive your notifications.

## What they tell you

| Event | Notification text |
|---|---|
| An invoice is issued | "New invoice for October" (the month of the invoice stands in place of "October") |
| A payment is recorded | "Payment recorded" |
| A meter reading is accepted | "Meter reading accepted" |
| A meter reading is rejected | "Meter reading rejected" |
| The board publishes an announcement | "New announcement from the board" |

The board and the electrician who review readings get one more: "New readings to review".

Tap a notification and the matching Nadel screen opens.

## Why there are no amounts or names

A notification goes through the notification service of your browser (Google, Mozilla or Apple), and anyone can see it on a locked screen. So the system sends only a short general phrase. Amounts, names and plot numbers never get into a notification. See the details in your account after you sign in.

## If notifications are blocked

If you once pressed "Block", the browser does not ask again, and the switch is off and labelled: "`Notifications are blocked in the browser settings. Allow them for this site and turn them on again.`"

1. Open the site settings in the browser (the icon next to the address, or the app settings in the phone).
2. Change the permission for notifications to "Allow".
3. Go back to My account and turn the switch on again.

## iPhone

On an iPhone notifications work only in the app added to the home screen, and only on iOS 16.4 or newer. While Nadel is open just in Safari, the block says: "`On iPhone notifications work only in the installed app (iOS 16.4 or newer).`" Add the app first: Install on your phone.

## Other messages

- "`This browser does not support notifications.`" means the browser cannot show notifications. Try Chrome, Firefox, Edge or Safari.
- "`Could not change the notification setting. Please try again.`" means a temporary error. Try again.

## Related pages

- Install on your phone
- My account
- How to connect Telegram